

# Cultural Exchange and the Birth of International Development \*

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## Abstract

How did America embrace international development? I study the role of American Protestant missionaries abroad. I use the travel routes of recruiters from the Student Volunteer Movement, the largest missionary movement at the turn of the twentieth century, to generate plausibly exogenous variation in the number of missionaries. Congresspeople representing constituencies with one standard deviation greater missionary exposure were 29 percentage points more likely to support foreign aid bills that established modern development assistance. Missionaries played a fundamental role in shaping foreign policy preferences and perspectives: Denominations with greater missionary exposure were more likely to endorse foreign aid in congressional testimony, after which congresspeople adopted moral universalism in their rhetoric. By the mid-twentieth century, experts on non-European regions disproportionately had a missionary background, and their expertise was especially valued by the government.

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## 1. Introduction

The idea that wealthy nations should support the development of low-income countries from a humanitarian perspective gained prominence after World War II, leading to a shift in global consciousness and sparking a wave of international cooperation aimed at global development (Myrdal, 1970; Sen, 1999; Duflo & Banerjee, 2011). Since the recipients of foreign aid are outside national borders, support for such aid can be fragile, with limited incentives for consistent backing. This lack of interest could potentially be a ‘key obstacle to reforming aid’ (Collier, 2008, p.183), as policy preferences of the public in donor countries can be an important determinant of global development efforts (Mosley, 1985; Lumsdaine, 1993; McDonnell *et al.*, 2003; Chong & Gradstein, 2008; Tingley, 2010; Pritchett, 2015). What, then, were the origins of rich countries’ commitment to international development?

This paper explores how intergroup contact by Protestant foreign missions in the United States shaped American attitudes and commitment to international development. By engaging with individuals from low-income countries, the missionaries reshaped public perceptions about non-European countries. This paper examines the impact of such contact on the establishment of foreign aid legislation, public support for global development efforts, and contributions to social scientific knowledge about developing countries.

This is a suitable setting to test this idea. While the United States became the world’s largest foreign aid donor in the latter half of the twentieth century (Qian, 2015), it was largely isolated from non-European nations in the early 1900s, with minimal immigration from these regions and limited international travel. Missionaries were among the first Americans to venture into distant lands, preceding diplomatic or commercial representatives. In their efforts to spread their faith, they became key sources of information about these regions in the pre-World War II era (Conroy-Krutz, 2017). Historical accounts suggest that these experiences made them more sympathetic toward non-European populations and fostered an interest in the economic development of low-income countries (Brown, 1907; Hocking, 1932; Gallagher, 1946; Gold, 2002; Bays & Widmer, 2009; Hollinger, 2018). Their influence extended beyond their immediate circles as they addressed church congregations and advocated for expanding foreign aid through the media and in Congress, shaping the foreign policy preferences of Americans.

This paper begins by investigating the effect of exposure to missionaries on congressional votes for landmark foreign aid legislation that established the modern development assistance—the Marshall Plan, the Point Four Program, the Mutual Security Act, funding for the International Development Association (a branch of the World Bank), and the Foreign

Assistance Act. To investigate this link, I gather a novel dataset from the archival records of the Student Volunteer Movement (1886–1964), which accounted for over 75 percent of all American missionaries during this period (Hodges, 2017, p. 48). I construct a shift-share variable that measures how missionaries influenced the public through church networks. In this variable, the “shift” is defined as the share of missionaries within each denomination, while the “share” represents the proportion of denomination members in each congressional district. The interaction between these two components captures the missionary influence on local communities by mapping denominational-level variation into geographic areas.

To address the endogeneity in the distribution of missionaries across denominations, I employ an instrumental variable strategy based on denominational exposure to travel secretaries, who were early promoters of the missionary movement during 1886–1887. The exposure to travel secretaries across denominations serves as a suitable instrumental variable for several reasons. First, this exposure is unrelated to interests in global development, which emerged only after prolonged interactions with developing countries. In the 1880s, the primary motivation for foreign missions was urgent religious conversion, driven by eschatological beliefs about the imminent end of the world, rather than concerns for development projects like building schools or hospitals. Second, the proposed instrument captures exposure to travel secretaries at the denominational level, excluding those denominations directly affiliated with the colleges visited by the secretaries. Historical records indicate that travel secretaries primarily visited denominational colleges and universities to recruit missionaries. For denominations outside these directly targeted groups, the visits of travel secretaries to local areas can be considered exogenous. Third, the results remain robust when controlling for denominational exposure from counties with Young Men’s Christian Association (YMCA) chapters, the primary determinant of travel secretaries’ visits. This addresses concerns that the effect is driven by proximity to denominational colleges. Lastly, I provide evidence that the proposed instrument is not associated with increased interest in non-European countries before the Student Volunteer Movement, as indicated by the lack of a pre-existing trend in the birth of authors writing about non-European regions.

Using this instrumental variable approach, I find that congressmen representing denominations with greater exposure to missionaries were more likely to support major foreign aid legislation that established the modern development assistance. The effects are economically significant: a one standard deviation decrease in the exposure to missionaries can account for 29 percent points decrease in the probability of voting for foreign aid.

I provide further evidence from congressional hearings to document the influence of missionaries. By the mid-1950s, foreign aid programs encountered increasing resistance.

Congressional hearings during this period debated over the Mutual Security Act, revealing Congress's attentiveness to the perspectives of religious organizations. I identify a list of denominations that actively advocated for foreign aid during these hearings. My analysis shows that denominations with higher missionary exposure were more likely to appear on this list. Moreover, I found that following the participation of these denominations, members of Congress began to incorporate moral universalism more prominently in their rhetoric. This finding aligns with the literature suggesting that the discourse surrounding foreign aid advocacy increasingly emphasized principles of moral universalism during this era to appeal to religious constituents (Turek, 2024).

The sources of moral universalism can be linked to an understanding of fundamentally different cultures in non-European countries. I demonstrate the essential role missionaries played in shaping American perspectives on these unfamiliar regions. To investigate this idea, I combine individual-level data on missionaries with a bibliometric database, comparing missionaries with a control group of runner-up applicants who volunteered but ultimately did not serve. Available slots to serve as missionaries were limited, and a key factor determining treatment assignment was religiosity rather than interest in foreign countries. The missions board prioritized individuals with strong religious dedication over those primarily motivated by curiosity about foreign cultures. If selection had been based primarily on interest in foreign countries, it would be challenging to estimate the exposure effect on foreign cultures even compared to the runners-up. However, with religiosity as the primary criterion, a comparison of missionaries and runners-up, based on matching them to census data, shows that they share many socioeconomic characteristics. I address any remaining imbalance by including county fixed effects or by using inverse probability weighting estimates. This comparison reveals that missionary experience tripled the likelihood of individuals authoring works on foreign subjects compared to the control group. The content of the publication were primarily social science rather than religion.

The knowledge brought back by missionaries had important policy implications. World War II and the subsequent waves of decolonization increased the demand for understanding non-European regions. In the mid-twentieth century, missionaries and their children rose to positions of importance due to their extensive knowledge of the non-European world because they were uniquely positioned to provide Americans with insights into these regions about local politics, culture, geography, climate, and diseases during this critical period. Using a dataset measuring regional expertise during WWII, I show that the government valued missionary experience for understanding the Asia-Pacific theaters. Further analysis from additional data reveals that missionaries played a significant role in foreign intelligence,

even compared with other prominent experts on foreign countries.

Lastly, as a long-term measure of preference for global development efforts, I investigate the effects on Peace Corps volunteers across counties between 1961 and 2000. I find that missionary exposure also predicts a higher number of Peace Corps volunteers, suggesting a shift from religious to secular forms of international development.

**Related literature.** This paper contributes to several strands of literature. First, it investigates the effects of intergroup contact on attitudes toward outgroup members (Allport, 1954; Boisjoly *et al.*, 2006; Hollinger, 2018; Bazzi *et al.*, 2019; Carrell *et al.*, 2019; Finseraas *et al.*, 2019; Indacochea, 2019; Scacco & Warren, 2018; Rao, 2019; Mousa, 2020; Alan *et al.*, 2021; Billings *et al.*, 2021; Dahl *et al.*, 2021; Lowe, 2021; Steinmayr, 2021; Bailey *et al.*, 2022; Corno *et al.*, 2022; Agte & Bernhardt, 2023; Bagues & Roth, 2023; Okunogbe, 2024; Bursztyn *et al.*, 2024). I examine whether extended intergroup contact—knowing that a member of one’s own group has a positive relationship with outgroup members—can foster favorable attitudes toward the outgroup (Wright *et al.*, 1997; Miles & Crisp, 2014). Recent studies highlight two pathways: on the one hand, indirect contact through media can scale up more easily than direct contact, promoting social integration even when direct intergroup interaction is limited (Marble *et al.*, 2021; Grady *et al.*, 2023; Siddique *et al.*, 2024). Meanwhile, evidence from Mormon missionaries suggests that direct intergroup contact during missionary work can shape attitudes toward undocumented immigrants and support for global development (Crawfurd, 2021; Berinsky *et al.*, 2023). This study advances the literature by combining these insights, showing how intergroup contact through missionaries influenced the rise of policy initiatives aimed at improving conditions in foreign countries—not only by changing the attitudes of those directly exposed to outgroups but also by impacting those reached indirectly.

This paper adds to the growing body of work on the determinants of moral universalism, a concept that spans various policy issues (Tabellini, 2008; Graham *et al.*, 2009; Hruschka & Henrich, 2013; Henrich, 2020; Enke, 2020; Enke *et al.*, 2023; Figueroa & Fouka, 2023; Enke *et al.*, 2024). Previous research highlights determinants such as ancestral livelihood (Le Rossignol & Lowes, 2022), effects of missionaries on developing countries (Bergeron, 2019), the history of American westward expansion (Bazzi *et al.*, 2024), and market access (Henrich *et al.*, 2010; Enke, 2023; Posch & Raz, 2024). I contribute to this literature by studying how cultural connections influence foreign development assistance.

This paper also contributes to the literature on the determinants of foreign aid (Eisensee & Strömberg, 2007; Lumsdaine, 1993; Alesina & Dollar, 2000; Kuziemko & Werker, 2006;

Faye & Niehaus, 2012; Nunn & Qian, 2014). While existing research predominantly focuses on patterns of foreign aid allocation between countries, there is limited evidence on how domestic support for foreign aid is formed in donor countries. Apart from a handful of studies (Milner & Tingley, 2010; Paxton & Knack, 2012; Enke *et al.*, 2023; Mueller, 2024), little is known about how the preferences of non-state actors influence foreign aid allocation. Drawing on a variety of data, I provide new evidence suggesting that the policy preferences of non-state actors in donor countries may play a significant role in shaping foreign aid allocation.

This study also speaks to literature on foreign missions' role in economic development (Woodberry, 2012; Bai & Kung, 2015; Wantchekon *et al.*, 2015; Cagé & Rueda, 2016; Waldinger, 2017; Valencia Caicedo, 2019; Becker & Won, 2021; Calvi *et al.*, 2022; Izumi *et al.*, 2023). While previous research has focused on the developmental effects of foreign missions abroad, I examine their impact on the missionaries themselves and how they shaped support for international development within their home communities. This study suggests that past missionary activities in developing countries may have an additional, indirect effect on economic development through the influence on foreign aid. Relatedly, this paper also contributes to the literature on the domestic consequences of the Protestant foreign missions (Gallagher, 1946; Gold, 2002; Tyrrell, 2010; Hollinger, 2018; Turek, 2020; Conroy-Krutz, 2024).

## 2. Background

### 2.1 Foreign missions and American understanding of non-European countries

The Student Volunteer Movement (1886-1964) was the largest missionary movement in American history, originating from a local YMCA conference in 1886 (Beahm, 1941; Ramsey, 1988). By 1920, it had attracted approximately 48,000 volunteers, with 12,000 ultimately serving abroad, representing about 75 percent of American missionaries overseas at that time (Hodges, 2017, p. 48).

Interactions on the ground with local communities gradually led many missionaries to prioritize educational and medical missions over religious conversion. While the indigenous population often showed indifference to Christianity, building schools and hospitals attracted more attention and approval from them (Ramsey, 1988, p.89). A joint report by several denominations involved in this movement reveals that many of these missionaries began shifting their focus from religious conversion to education and welfare initiatives

(Hocking, 1932). The growing numbers of missionaries were teachers, doctors, nurses, agricultural experts, architect, engineers, rather than priests. While the missionary project was marked by a tension between Anglo-Saxon ethnocentrism and the egalitarian belief in human equality, historical records reveal that contact with people of different races and religions often led many missionaries to reevaluate their own biases, fostering greater respect for foreign cultures and religions (Brown, 1907; Gallagher, 1946; Gold, 2002; Hollinger, 2018).

As the priorities evolved, many missionaries began to engage deeply with local development challenges, moving beyond their original religious goals. Some pursued advanced studies in fields related to these issues, further broadening their expertise. For instance, John Lossing Buck, an agricultural missionary in China, wrote a dissertation on the Chinese rural economy and went on to work as an agricultural economist for the UN Food and Agriculture Organization. These missionaries produced scholarly works on a wide range of topics including agricultural development, nursing practices, medicine, psychology, geography, international relations, and the histories of their host countries (Laubach, 1925; Higginbottom, 1926; Wilson, 1926; Earle, 1929; Buck, 1930; Cressey, 1930; Manchester, 1931; Wright, 1941; Landon, 1943; Haring, 1946; Wiser & Wiser, 1971).

World War II and the subsequent waves of decolonization positioned many missionaries as key authorities when the U.S. government began navigating global affairs. Missionaries were among the few experts on non-European regions, and their insights were highly valued, as demonstrated by their substantial contributions to the war effort. Before the Japanese attack on Pearl Harbor, the U.S. government had underestimated Japan's military capabilities and the likelihood of war. Stanley Hornbeck, a special adviser to the Secretary of State, believed as late as November 1941 that Japan would back down in response to a U.S. trade embargo (Wohlstetter, 1962, pp. 264-65). General Douglas MacArthur even suggested that the pilots involved in the attack must have been white mercenaries, doubting Japan's ability to execute such an operation (Andrew, 2018, p. 636).

In contrast, those with missionary connections in China or Japan recognized the growing threat of Japanese militarism and expressed concerns about potential conflict before WWII. Edwin Reischauer, a missionary's son and later a professor of Asian Studies, proposed a Japanese language training program in 1940, believing war with Japan was likely and the U.S. was unprepared (Japan Society of Boston, 2022). Ernest Price, a former missionary in China and political scientist, warned of conflict after observing Japan's puppet state in China (Price, 1935). Walter Judd, a former medical missionary in China and later a U.S. congressman, delivered 1,400 speeches across 46 states between 1938 and 1940, warning of the growing threat from Japan and the dangers of an isolationist foreign policy (Gao &

Osburn Jr, 2016).

When the need arose for better understanding of Japan and the Asia-Pacific region, very few individuals in the US had relevant knowledge about these regions. It required understanding “the Communist Party of India and the puppet regime in Nanking, inflation in Burma and guerillas in the Philippines, trade routes in the Congo basin and rival cliques in the Japanese Army.” (Katz, 1989) When the U.S. government assembled a small group of individuals with actual knowledge of these areas, many were missionaries or their children. They held key positions in intelligence agencies and the State Department (Reynolds, 2005; Packard, 2010; Hollinger, 2018; Price, 2016; Thomas, 2016; Sutton, 2019). Their knowledge and network was relevant even for people without direct connections with missionaries. John Fairbank, a prominent American historian of Chinese history, extensively relied on missionary networks during his research in China and later while working for the Office of Strategic Services (OSS) during World War II (Fairbank, 1983).

## **2.2 Missionary influence on U.S. foreign aid**

During the latter half of the twentieth century, the U.S. increased its spending on foreign aid. Although U.S. foreign aid as a share of GDP is not higher than that of many other advanced economies, the country remained the largest donor in absolute terms (Qian, 2015). By the 1980s, net foreign aid had surpassed net foreign direct investment from multinational corporations as the primary financial flow from advanced countries to the developing world (Lumsdaine, 1993, p.105). In some recipient countries, foreign aid accounted for over 10 percent of GDP, underscoring its crucial role in development (World Bank, 2005).

Support for foreign aid in advanced economies, particularly in the post-World War II period, was rooted in both political and cultural factors. Lumsdaine (1993) argues that in Europe, especially in Scandinavian countries, the labor unions’ internationalist doctrine, which emphasized political equality, was closely linked to foreign aid policies. Meanwhile, in the United States, Christianity played a more prominent role in shaping foreign aid, as religious leaders emphasized compassion and charity toward those in need worldwide.

The U.S. foreign aid initiative was not entirely novel; by the time the Point Four Program was launched in 1949, American missionaries had long been active in low-income countries. Missionaries had established schools, hospitals, clinics, social service centers, and demonstration farms aimed at improving living standards (Maddox, 1956; Curti, 1963; Bremner, 1988; Curtis, 2018; Turek, 2020).

Missionaries influenced public opinion in the U.S. through their extensive public out-



reach. Hundreds of missionaries who had served in China alone gave an estimated 30,000 speeches annually in colleges, churches, and community gatherings, reaching millions of Americans (Reed, 1983, p. 127). Their efforts significantly influenced young adults, sparking a growing interest in international humanitarian work. By 1920, about 5 percent of American college students were learning about foreign missions, even without intending to become missionaries (Ramsey, 1988, p. 71).

Missionaries' influence extended beyond the church, shaping public opinion in the U.S. through their outreach efforts and engagement with broader society. Henry Luce, the son of Presbyterian missionaries in China and founder of Time magazine, was a leading voice advocating for U.S. global leadership through humanitarian aid in his famous essay "American Century" (Luce, 1941). Similarly, Walter Judd, a former missionary and later a Republican congressman, became a key figure in promoting foreign aid (Gao & Osburn Jr, 2016; Turek, 2024).

By the mid-1950s, critics of foreign aid argued that it wasted taxpayers' money by giving assistance to undeserving recipients. On the other hand, proponents emphasized the moral duty to promote justice and freedom globally and to counter the legacy of colonialism, often invoking religious principles. Religious groups, representing different denominations, rallied public support for foreign aid by appealing to moral and ethical arguments during the congressional hearings on the Mutual Security Act. The rhetoric of congressmen defending foreign aid programs shows that they took into account the importance of religious constituents in garnering public backing for aid expansion (Turek, 2024; US Congress House Committee on Foreign Affairs, 1957).

### 3. Data

**Student Volunteer Movement records.** The building block of this paper's data is manually constructed from a historical archive housed at the Yale Divinity Library. The Student Volunteer Movement tracked all individuals who volunteered for missionary work. For this study, I have scanned and digitized this individual-level data, which is scattered across several repositories in the archive. The dataset includes personal information such as individual names, birth years, states and towns of residence, destination countries, and religious denominations, with varying levels of detail depending on available records. For more details on the digitization process, see Appendix [B.1](#).

The digitization process yielded data on 12,265 missionaries and 24,841 volunteers who did not ultimately serve as missionaries. This dataset includes several notable figures in-

roduced in the background section, such as Henry W. Luce, Walter Judd, or John Lossing Buck.

**Congressional voting.** Data on roll-call voting by U.S. congressmen is sourced from VoteView (Poole & Rosenthal, 2000). I focus on the final roll call votes for five landmark pieces of legislation that expanded foreign aid: the Marshall Plan, Point Four Program, Mutual Security Act, Funding for the International Development Association (World Bank), and the Foreign Assistance Act of 1961, which was followed by the U.S. Agency for International Development (USAID). These key foreign aid bills were identified through Lumsdaine (1993) and Stathis (2014), focusing on legislation that targeted broad groups of countries rather than specific nations. I exclude congressmen who did not participate in the voting from the analysis, as non-participation was most commonly due to travel and scheduling limitations rather than strategic considerations.

**Census of Religious Bodies.** The number of church members of each Protestant denominations in each county is gathered from the Census of Religious Bodies (1890). This data is combined with a standard county level data from National Historical Geographic Information System (NHGIS). I link 47 Protestant denominations to the Census of Religious Bodies (1890) to create a variable that captures the county-level exposure to the missionaries. The 1890 census is regarded as being of exceptionally high quality (Finke, 1989).

**YMCA chapters and travel secretaries.** To construct the instrumental variable, I use the location of local YMCA chapters. To recover the location of all YMCA chapters in 1881, 1884, and 1889, I use *Intercollegian* magazine. Access to the microfilmed records of this magazine is available at the New York Public Library.

The travel secretaries of the Student Volunteer Movement significantly influenced the mobilization patterns of missionaries for decades to come (Beahm, 1941; Ramsey, 1988; Parker, 1994). I use the routes of the travel secretaries of the Student Volunteer Movement in 1886 and 1887. The places that were visited by travel secretaries is available in the *Missionary Review of the World* magazine from February 1887 to June 1887. February 1887 edition includes the locations in the previous months. The denominational affiliations of colleges visited by travel secretaries were obtained from Xiong & Zhao (2023) and cross-referenced with the predominant denominations of college students recorded in the Student Volunteer Movement archives.

**Moral universalism in congressional speech.** To measure moral universalism in congressional speech, I follow Enke (2020), who applied the Moral Foundations Dictionary from

Graham *et al.* (2009) to the congressional speech data in Gentzkow *et al.* (2019). This measures moral universalism embedded in congressional speech based on the word frequency of each congressman’s speech.

**Other geographic controls.** To count the number of college students in each county, I use Xiong & Zhao (2023). To construct standard control variables at the county level, I use county level censuses from National Historical Geographic Information System (NHGIS) and individual level census data (Ruggles *et al.*, 2010). County market access data is taken from Donaldson & Hornbeck (2016) and Hornbeck & Rotemberg (2021). I use market access in 1900. Agricultural land suitability is from Ramankutty *et al.* (2002).

**Individual census.** To measure the socioeconomic background of individual missionaries, I link data on missionaries and volunteers to the US individual level census data from 1900 and 1910. I follow Abramitzky *et al.* (2021) to implement an automated iterative linking procedure to the census data. Details on the linking process is explained in Appendix A.1.

**Publication output.** To estimate the effects on the understanding of foreign cultures, I use two distinct bibliometric datasets to measure knowledge output. First, I collect graduate dissertation data from ProQuest Dissertations Theses. I match individuals to this dataset using first name, middle initial, and last name. Second, I gather data from OpenAlex, which provides publication data on books, academic journals, and conference proceedings. For this dataset, I match on exact names for publications between 1880 and 1945. Based on dissertation or publication titles, I classify output as foreign-related if the title includes foreign country or region names. The list of keywords used for this classification is provided in Appendix B.3.

**Regional expertise during WWII.** I quantify the policy relevance of knowledge output in the context of WWII, when the US government attempted to understand Asia-Pacific regions (Hollinger, 2018; Sutton, 2019).

I use two different data sources to compile a list of individuals involved in foreign intelligence during WWII. First, I use the roster of world specialists from the Ethnographic Board (Bennett, 1947), a branch of the Smithsonian Institution during WWII to serve as a clearinghouse for intelligence agencies in the US military and the State Department. The Ethnographic Board consulted various universities, academic societies, industries, and missionary churches to compile a roster of individuals knowledgeable about foreign countries, especially non-European areas. These rosters were sent to the Office of Strategic Services, the Board of Economic Warfare, the War Department, the Navy Department, the State Department, and many other intelligence agencies. I collect and digitize this data from the

Smithsonian Institution Archives in Washington, DC.

Apart from the Smithsonian roster, I use the Office of Strategic Services (OSS) roster, which is available online.<sup>1</sup> A limitation of this record is that it primarily provides only the individual's name. In some instances, an army serial number is included, but this is limited to a subset of the individuals listed. Some individuals on the roster served in European countries or held roles unrelated to intelligence work, such as typists or drivers. While visiting the National Archives is theoretically an option, it is impractical due to the large number of individuals (23,980) included in this roster. For the purposes of this paper, I restrict my sample to those who can also be identified in the OpenAlex foreign publication data. The intersection of these two datasets provides a list of social scientists specializing in non-European areas within the OSS.

**Wikipedia biographies.** To gain a broader understanding of the role of missionaries in shaping knowledge about foreign countries, I use data from Wikipedia (Laouenan *et al.*, 2022). I first limit the sample to individuals who were active in the United States. I then narrow it down to those born between 1880 and 1925 and who passed away after 1940, focusing on those who might have contributed to intelligence services.

To determine the missionary background of each individual, I use an automated Python script to scan Wikipedia biographies for the term 'missionar\*.' This process identifies individuals who were missionaries, missionary children, or who attended missionary schools during their childhood. I then manually verify each case to confirm whether the individual was indeed a missionary or a missionary child and to identify the country where they served or were born. For the outcome variable, I conduct a similar keyword search for terms such as 'Second World War,' 'World War II,' and keywords related to intelligence or language work. A comprehensive list of these keywords is provided in Appendix B.4.

**Peace Corps volunteers.** As another measure of long-term support for international development other than through monetary contribution, I use the number of Peace Corps volunteers in each county. Through a Freedom of Information Act request, I obtained data on 159,246 Peace Corps volunteers across US zip codes from 1961 to 2000. I map this data to the county level using ZIP code coordinates. 135,255 volunteers have information on zip code.

**Geographical data.** When there is a need to construct a crosswalk between town or city name to counties, I make use of Geographic Names Information System (GNIS) from United States Geological Survey and Berkes *et al.* (2023). I transform county-level data

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<sup>1</sup><https://www.archives.gov/files/iwg/declassified-records/rg-226-oss/personnel-database.pdf>

into congressional district-level data, following the population-weighting method by Ferrara *et al.* (2024).

## 4. Effect on foreign aid

### 4.1 Empirical framework

This section estimates the impact of missionary exposure on the support for foreign aid. The analysis is based on the following equation:

$$aid_c = \beta Exposure_c + \mathbf{X}_c' \gamma + \varepsilon_c \quad (1)$$

where  $c$  denotes a congressional district. The outcome variable  $aid_c$  is an indicator for pro-foreign aid vote. The vector  $\mathbf{X}_c$  includes controls to adjust for confounding variables. The error term  $\varepsilon_c$  is clustered by state.

Missionaries traveled across states to deliver speeches at local churches, and their publications influenced college students studying missions in different states. To capture their influence through churches across space, I construct a shift-share variable to define  $Exposure_c$  at the county or congressional district level. I count the number of missionaries from each denomination in the Student Volunteer Movement records and link each denomination to the 47 denominations in the Census of Religious Bodies (1890).  $Exposure_c$  is defined as follows.

$$Exposure_c = \sum_o \left( \frac{POP_{co}}{POP_c} \right) \left( \frac{M_o}{POP_o} \right) \quad (2)$$

In this variable,  $o$  represents a Protestant denomination (e.g. Southern Baptist).  $POP_o$  indicates the number of church members from denomination  $o$ , and  $POP_{co}$  indicates the church members from denomination  $o$  in county  $c$ .  $M_o$  indicates the number of missionaries in denomination  $o$ .  $Exposure_c$  captures the denominational exposure to missionaries  $\left( \frac{M_o}{POP_o} \right)$  interacted with the share of denomination in each county  $\left( \frac{POP_{co}}{POP_c} \right)$ .

**Control variables.**  $\mathbf{X}_c$  include other variables to rule out confounding factors. The control variables are classified into core controls, baseline YMCA exposure (which will be explained in the next subsection), religious controls, and socioeconomic controls. Core controls include the share of all 47 denominations involved in the Student Volunteer Movement, latitude, longitude, and state fixed effects. Political controls include Congressmen's ideological

conservatism (DW-NOMINATE first dimension) and Republican Party fixed effects. These controls help isolate constituents' preferences from those of the politicians (Mian *et al.*, 2010). Religious controls include the shares of missionaries, evangelical Protestants, main-line Protestants, Catholics, Orthodox Jews, and Mormons. Socioeconomic controls include log population, urbanization rate, occupational income score, agricultural land suitability, market access, literacy rate, and the shares of college students, manufacturing workers, Northern and Western European immigrants, Eastern and Southern European immigrants, non-European immigrants, and African Americans.

## 4.2 Instrumental variable strategy

The baseline specification faces a challenge: the allocation of missionaries across denominations ( $M_o$ ) is not random. The number of missionaries from each denomination reflects not only the effect of exposure to foreign missions but also various other denominational characteristics. This self-selection into foreign missions could correlate with support for foreign aid through channels other than direct missionary exposure.

The direction of potential bias in OLS estimates of missionary exposure's effect on foreign aid support is unclear a priori. Upward bias might occur if more educated and internationally oriented denominations were more likely to host missionaries, as these denominations may have already been more supportive of foreign aid. Denominations with higher levels of education, wealth, or exposure to international trade might have both attracted missionaries and shown greater support for foreign aid, potentially leading to an overestimation of the missionaries' influence.

Conversely, a downward bias could emerge if missionary volunteers, often coming from theologically and politically conservative backgrounds, were inclined to oppose foreign aid to racially different countries. Many missionary volunteers were frequently associated with conservative theological and political views (Ramsey, 1988, p.110), which may have led to a focus on domestic concerns and a reluctance to support foreign interventions, including aid. Such denominations could have aligned with isolationist tendencies, exhibiting skepticism towards foreign aid, often seen as an extension of internationalism or global engagement. This selection bias could lead to an underestimation of the true effect of missionary exposure.

To isolate the effect of foreign missions from other denominational characteristics, I construct a push factor based on the travel locations of initial promoters of the Student Volunteer Movement. Previous research indicates that 'travel secretaries' who promoted the opportunities to serve as missionaries played a crucial role in the successful recruitment

of missionaries (Beahm, 1941; Ramsey, 1988; Parker, 1994; Gold, 2002). Travel secretaries visited college campuses to recruit students for foreign missions, and early volunteers became role models for future missionaries. Missionaries maintained close contact with colleges and seminaries, encouraging other church members to join the mission (Ramsey, 1988, p.p.55-66). During the 1880s and 1890s, before significant interracial contact occurred, the primary motivation for foreign missions was the eschatological belief that the world was nearing its end. Rapid religious conversion took precedence over establishing long-term institutions like schools or hospitals (Hollinger, 2018, p.9). The variation in exposure to travel secretaries is exogenous to interests in global development.

The proposed instrumental variable captures the indirect exposure to travel secretaries at the denomination level outside the denominations that were directly targeted by the travel secretaries. It is defined at the denomination level as follows:

$$Travel_o = \sum_{c \neq j(o)} POP_{co} \mathbf{1}[Travel_c] \quad (3)$$

where  $Travel_c$  equals one for counties visited by the travel secretaries.  $j(o)$  represents the county where denomination  $o$  has an institutional affiliation with the colleges visited by the travel secretaries. This means that directly targeted denomination is excluded from the instrumental variable. For example, if the travel secretaries visited Princeton University, which is affiliated with the Northern Presbyterians, I assume that all other denominations in that county are exposed to the travel secretaries, except for the Northern Presbyterians. This variable is constructed with 47 denominations that participated in the Student Volunteer Movement to ensure the results are not driven by denominations unrelated to foreign missions.

Travel secretaries primarily visited small, denominational colleges (Ramsey, 1988, p.12). The choice of targeted colleges would reflect the denominations expected to be most receptive to travel secretaries. The proposed instrument excludes the directly targeted denominations and captures the indirect effects on other local denominations. The travel secretaries' visits to counties were exogenous to non-targeted denominations, and the instrumental variable captures their exposure outside the directly targeted denominations.

When the outcome variables are defined at the congressional district level, the denomination-level exposure to travel secretaries is mapped onto these geographical units using the following shift-share structure:

$$TravelExposure_c = \sum_o \left( \frac{POP_{co}}{POP_c} \right) \left( \frac{Travel_o}{POP_o} \right) \quad (4)$$

which is the interaction between the share of denomination members in each county and the share of missionaries in each denomination. This specification shows that when a denomination in one area is exposed to travel secretaries, the impact spreads beyond that region to affect the same denomination nationwide through its church network.

The identifying assumption of this instrumental variable approach is that denominations indirectly influenced by travel secretaries would support foreign aid only because of their exposure to foreign missions. A potential concern with this instrument is that certain denominations may have been inherently more likely to be exposed to travel secretaries than others. For instance, if travel secretaries visited counties with specific denominational compositions or socioeconomic characteristics associated with interest in foreign missions, the instrument could be endogenous even after excluding directly targeted denominations.

To address this issue, I adopt two different approaches. First, I construct a comparable measure using counties that had YMCA chapters but were not visited by travel secretaries and control for this measure of YMCA presence unrelated to travel secretaries. The travel routes were determined in coordination with local YMCA chapters (Beahm, 1941; Ramsey, 1988). The Student Volunteer Movement began in 1886-1887, following an event at YMCA conference for a Bible study in Mount Hermon, Massachusetts. A foreign mission recruiter, initially seeking only a few volunteers, gave a speech that unexpectedly inspired 100 attendees to volunteer, sparking a large-scale missionary movement (Kelleher, 1974, p.20). Figure 1 shows the counties visited by travel secretaries and those with local YMCA chapters between 1881 and 1890. Under the assumption that travel secretaries did not account for the entire denominational composition surrounding the colleges they visited to promote foreign missions, controlling for this factor could provide further reassurance that the results are not driven by proximity to denominational colleges.

Figure 2 examines counties visited by the travel secretaries. For ease of interpretation, all variables are standardized to have a mean of zero and a standard deviation of one. As explained in the next subsection, this corresponds to a 40 percent change in the total number of missionaries across all denominations. It compares three groups of counties: those visited by travel secretaries ( $Travel_c$ ), those with YMCA chapters but without visits, and the remaining counties. Diamonds represent the regression coefficients of county covariates on indicators for counties visited by travel secretaries, while hollow triangles represent coefficients of covariates on indicators for counties with YMCA chapters but without visits.



The figure shows that counties visited by travel secretaries had more college students, were more urbanized, and had more denominations involved in the Student Volunteer Movement. To account for potential bias from certain denominations being inherently closer to the travel secretaries' routes, the empirical specification controls for baseline exposure to YMCA chapters in counties that had YMCA chapters but were not visited by travel secretaries. The figure indicates that counties with YMCA chapters are generally comparable to counties visited by travel secretaries, except for differences in the share of manufacturing workers and college students. This is consistent with the travel secretaries' focus on recruiting college students for missionary work. Although there is some imbalance due to differences in the number of college students, my instrumental variable excludes denominations affiliated with universities visited by the travel secretaries, making the counties visited by travel secretaries and those with YMCA chapters broadly comparable.

Second, after presenting the main results, I also employ an alternative approach based on temporal variations in exposure to YMCA chapters across denominations. I demonstrate that YMCA exposure just before the start of the Student Volunteer Movement influenced missionary recruitment and support for foreign aid, whereas additional exposure to the YMCA immediately after the initial travels had no effect on travel exposure or congressional support for foreign aid. This finding supports the hypothesis that exposure to travel secretaries, rather than mere exposure to YMCA chapters, was the key channel.

Table 1 presents the first-stage regression results at the denominational level, with the number of missionaries per denomination as the dependent variable. The main independent variable is the exposure to travel secretaries,  $Travel_o$ . Column (1) demonstrates that a 1,000-person increase in the exposed population correlates with 9 additional missionaries per denomination. Columns (2) and (3) incorporate additional controls, including the number of members in each denomination, an indicator for evangelical denominations, and per capita church property. Column (4) includes the control for the baseline YMCA exposure, which is constructed from counties with YMCA chapters but without visits from travel secretaries.

### 4.3 Validity of the instrument

To assess the plausibility of the identifying assumption, I check whether interest in foreign countries was higher in the exposed counties. There is a lack of comprehensive data capturing support for foreign aid prior to 1886. Private donations to foreign countries began in the early 1900s, largely driven by missionaries campaigning for famine relief (Curtis, 2018).

While no perfect measure for foreign aid exists during pre-treatment period, Wikipedia biographies offer a potential proxy for gauging interest in foreign countries. I match each individual's name to the OpenAlex bibliometric database for 1860-1945, focusing on publication titles that include foreign region names (using keywords as described in Appendix B.3). For all Wikipedia authors who can be matched with publications on foreign subjects, I use birthplace data by counting the number of authors born in each U.S. county who wrote about foreign countries (Laouenan *et al.*, 2022). Then, I estimate the following equation:

$$authors_{ct} = \sum_{\tau \neq 1880} \gamma_{\tau} (TravelExposure_c \times D_{\tau}) + \sum_k \sum_{\tau \neq 1880} \theta_{k\tau} (X_{kc} \times D_{\tau}) + \delta_c + \delta_{st} + \varepsilon_{ct} \quad (5)$$

The outcome variable is the number of Wikipedia authors born in each county  $c$  and birth cohort  $t$  (1850–1940), whose names are matched to publications on non-European countries published between 1860 and 1960. This increase reflects not only missionaries but also others influenced by early exposure to missionaries. The treatment variable,  $TravelExposure_c$ , is interacted with decade fixed effects. I also control for baseline YMCA exposure and the share of SVM denominations in 1890, both interacted with decade fixed effects, as well as county and state-year fixed effects.

Figure 3 shows that while no pre-existing trend is observed in the number of authors born across decades, there is a notable increase for individuals born after 1900 in exposed counties. As mentioned in the previous section, historian John Fairbank relied on missionaries for his research on China (Fairbank, 1983), and the observed increase in the number of authors may be related to this exposure. The results presents evidence against preexisting interest in non-European regions in areas more exposed to the travel secretaries.

#### 4.4 Results: Legislative support for foreign aid

This section documents the impact of exposure to missionaries on U.S. foreign aid by analyzing five landmark pieces of legislation: the Marshall Plan (1948), the Point Four Program (1949), the Mutual Security Act (1951), U.S. participation in the International Development Association (1960)<sup>2</sup>, and the Foreign Assistance Act of 1961, which established the U.S. Agency for International Development (USAID).

In this section, the unit of analysis is the congressional district. I examine whether

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<sup>2</sup>The International Development Association offers additional funding for developing countries within the World Bank.

congressmen representing constituents exposed to missionaries are more likely to support major foreign aid bills in Congress. I stack five different congressional periods and then control for congress fixed effects (which is similar to time fixed effects) to account for differences in each foreign aid bill.

Table 2 document the results. As before, the exposure to missionaries is standardized to have a standard deviation of one. The outcome variables are indicator variables equal to one hundred if a Congressman voted “Yea” on the passage of foreign aid. I find that representatives of constituents influenced by missionaries were more likely to vote “Yea” on major foreign aid bills. Column (1) presents an OLS estimate, controlling for YMCA exposure, the share of Protestants, congressmen’s ideology, Republican party fixed effects, and state fixed effects. It suggests that a one standard deviation increase in exposure to missionaries is associated with a zero to six percentage point increase in the probability of voting for foreign aid, though this result is not statistically significant. In contrast, the coefficients in Column (3) from the IV estimate show that a one standard deviation increase in exposure explains a 50 percentage point increase in the probability of voting for foreign aid. I run separate regressions for each foreign aid bill, as shown in Figure A.5.

**Magnitude of the effect size.** Column (7) of Table 2 shows that a one standard deviation increase in exposure to missionaries led to an 29 percentage point increase in the probability of a pro-foreign aid congressional vote. The independent variable is normalized to have a standard deviation of one, and its interpretation is explained below.

The shift-share variable has a mean of  $0.13 \times 10^{-3}$  and a standard deviation of  $0.72 \times 10^{-4}$ . The total number of missionaries with identifiable denominations is 9,755, with an average of 208 missionaries per denomination ( $=9,755/47$ ). The average number of members in each denomination is 240,256, and the average share of 47 denomination members across congressional districts is 0.21.

This suggests that reducing the number of missionaries by 82 in each of the 47 denominations would decrease the shift-share variable by one standard deviation ( $=\frac{82}{240,256} \times 0.21$ ). In other words, a 40 percent decrease ( $=82/208$ ) in the total number of missionaries is associated with a one standard deviation decrease in the shift-share variable, which in turn is associated with an 29 percentage point decrease in the probability of a pro-foreign aid congressional vote. This indicates that a 20 percent decrease in missionaries would have lowered support for foreign aid 14.5 percent points.

To facilitate comparison with other studies in the literature, I calculate the magnitude in terms of a persuasion rate, defined as the share of people who change their vote in

response to missionary exposure out of those who could have done so. The persuasion rate is expressed as

$$f = \frac{dv}{de} \frac{1}{1 - v_0} \quad (6)$$

where  $v$  represents the level of foreign aid support,  $e$  measures missionary exposure, and  $v_0$  indicates what percentage of legislators would have backed foreign aid if missionaries had never been present.

To determine  $v_0$ , I used the following method: First, I set each congressional district’s missionary exposure to the lowest level found in our data. Then, I calculated the predicted outcomes and took their average. This calculation revealed  $v_0 = 0.11$ , suggesting that without any missionary influence, only about 11 percent of congresspeople would have supported foreign aid.

Column (7) in Table 2 indicates  $\frac{dv}{de} = 0.29$ , corresponding to a 33 percent persuasion rate. This magnitude surpasses the 12 percent persuasion rate of Fox News on Republican voting (DellaVigna & Kaplan, 2007), and closely parallels the 28 percent persuasion rate that Father Coughlin’s radio broadcasts had on Roosevelt’s vote share (Wang, 2021).

#### 4.4.1 OLS-IV gap

The gap between OLS and IV estimators suggests that, under the identifying assumption of this paper, the OLS estimator is downward biased. This suggests that initially conservative denominations, which were more likely to send missionaries due to their evangelical zeal, may have been counterfactually less supportive of foreign aid without exposure to missionaries. The OLS estimate captures this initial negative correlation between missionary-sending propensity and support for foreign aid, leading to an underestimation of the effect of exposure. This interpretation is consistent with historical accounts that missionaries in the Student Volunteer Movement were theologically and socially conservative (Ramsey, 1988, p.110), and that contact with different groups led missionaries to reevaluate racial hierarchies in the United States and become more critical of them (Gallagher, 1946; Gold, 2002; Hollinger, 2018).

I provide empirical evidence related to this point. If the downward bias in the OLS estimate is driven by religious conservatism, we should expect the OLS estimator to be more biased for more conservative denominations. To test this, I split both the treatment and instrumental variables into two categories based on whether a denomination is classified

as evangelical or mainline Protestant (Steensland *et al.*, 2000). I classify 20 denominations as evangelical and group the remaining 27 as mainline Protestants.<sup>3</sup> Evangelical Protestants are considered to be more politically conservative.

I then run separate regressions for exposure to evangelical missionaries and exposure to mainline missionaries. The results in Table 3 show that the OLS estimator for evangelical missionaries is negative, while the OLS estimator for mainline missionaries is positive. When estimating the IV regressions, the OLS-IV gap is smaller for mainline Protestants, suggesting that the IV addresses the bias in the OLS estimator related to religiosity, which may have otherwise underestimated the effects of foreign missions. The IV approach, by leveraging the exogenous variation in missionary activity due to YMCA exposure, isolates the effect of missionary experiences that are independent of the initial predispositions of these denominations.

These results align with Ramsey (1988, p.86), who discusses the tension between Anglo-Saxon ethnocentrism and the egalitarian belief that all humans deserve respect during the early period of foreign missions. Similarly, Allport (1954, pp. 444-456) highlights the dual nature of religion: it can either reinforce ethnocentric prejudice and exclusivity or serve as a universalistic force being inclusive towards outgroup members. The findings of this paper suggest that interreligious contact may help foster the universalistic aspects of religion.

#### 4.4.2 Robustness

First, I construct a restricted sample after excluding counties with YMCA chapters or travel secretaries' visit. This allows us to evaluate whether the results are driven by that had a direct connection with travel secretaries or whether the indirect spillover through denominations mattered. Although the sample size got reduced, I find that the results are qualitatively and quantitatively the same (Table A.8).

The results are robust to alternative methods of calculating standard errors. Recent literature on shift-share designs highlights potential pitfalls in their use. In my research setting, the shock is assumed to be exogenous, while the shift could potentially be endogenous. Borusyak *et al.* (2022) demonstrate that under this assumption, identification can be achieved by transforming geographical data into shock-level data. Following their methods, I re-estimate the same equation at the shock level, and the results remain robust. I also

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<sup>3</sup>I first identify the evangelical denominations and group the remaining ones into mainline denominations. For example, while the Universalist Church does not strictly fall into either the evangelical or mainline categories, I classify it with the mainline denominations, as it is generally considered to have a relatively liberal political inclination.

calculate standard errors following Adao *et al.* (2019) and test for spatial correlation using 100km, 300km, and 500km distance cutoffs, following Conley (1999). These standard errors are presented in Table A.9.

There may be concerns that the shares in the shift-share variable are endogenous and that this endogeneity is driving the results. Adao *et al.* (2019) examines this issue by interacting randomly generated shifts with shares and testing whether the simulated shift-share variable over-rejects the null hypothesis. Following this method, I run the same regression with randomly generated shifts, where the share of denomination is interacted with these shifts. I find that the null hypothesis is rejected less than 5% of the time at the 95% significance level. Adjusting for standard errors as suggested by Adao *et al.* (2019) again confirms the robustness of the results.

Figure A.8 estimates the regression equation after excluding one denomination to assess whether the results are driven by a single denomination. Similarly, Figure A.9 estimates the equation by excluding one state from the sample.

#### 4.5 Alternative approach: Temporal fluctuation in exposure to YMCA

A potential threat to identification arises if travel secretaries selected the colleges they visited by considering the characteristics of surrounding denominations, even after excluding the directly targeted denominations and controlling for the baseline YMCA exposure.

This section constructs an alternative treatment variable by leveraging the temporal fluctuations in YMCA exposure to further clarify that the estimated effect arises specifically from the exposure to travel secretaries. Counties acquired YMCA chapters at different times—some before 1881, others in 1884, and still others in 1889. Since the Student Volunteer Movement began in 1886, it was not foreseeable in 1884 that a major missionary movement would soon start. Thus, YMCA exposure just before 1886 would affect travel secretaries' visits, while YMCA chapters established in 1889 should not influence missionary recruitment or support for foreign aid. By comparing the estimated coefficients of YMCA exposure before and after the start of the Student Volunteer Movement, I show that the effect is specific to YMCA exposure just before the movement's inception.

To capture this variation in exposure, I construct a measure of YMCA presence at different time points:

$$YMCA_{o\tau} = \sum_c POP_{co} \mathbf{1}[YMCA_{c\tau}] \quad (7)$$

where  $YMCA_{c\tau}$  equals one for counties that had YMCA chapters in year  $\tau$ . I construct three different measures of YMCA exposure at the denomination level: exposure up to 1881, new exposure in 1884, and new exposure in 1889. The denominational exposure is then mapped into congressional district again through the shift-share variable.

Table 4 presents the results. Columns (1)-(4) show the first-stage regression results at the denomination level, while Columns (5)-(8) display the second-stage regression results at the congressional district level. Columns (1) and (5) regress the outcome variables on the aggregate YMCA exposure between 1881 and 1889. The results indicate that YMCA exposure predicts both travel exposure and a higher likelihood of pro-foreign aid votes in Congress.

The subsequent columns split this aggregate measure of YMCA exposure into three variables based on the timing of new YMCA chapter openings. The findings reveal that YMCA exposure before 1884 (prior to the start of the Student Volunteer Movement in 1886) predicts both travel exposure and foreign aid votes.

Since YMCA exposure after the travel secretaries' initial promotion does not predict additional travel exposure, it should not increase the probability of pro-foreign aid congressional votes, as shown in columns (6)-(8). These results suggest that the YMCA's effect on foreign aid votes is tied to the timing of travel secretaries. By exploiting variation in exposure to travel secretaries, this analysis provides further evidence that the effect stems from the Student Volunteer Movement, not YMCA characteristics independent from the Student Volunteer Movement.

#### 4.6 Evidence for preference shift

The extent to which foreign aid is driven by the foreign policy preferences of constituents can be debated. To shed light on the mechanisms, I present additional evidence on preferences for international development at the denomination, congressional district, and county levels.

**Denominational support in congressional testimony.** I start by presenting evidence of preferences for international development at the denomination level. I turn to congressional hearings from 1957, a period marked by intense debate over the continuation of foreign aid programs. During these hearings, Congress sought input from various stakeholders, including religious organizations, to present their perspectives. Over a three-day period, testimonies from these religious groups, predominantly in favor of foreign aid, were documented (US Congress House Committee on Foreign Affairs, 1957; Turek, 2024).

From congressional hearing records, I identify denominations likely to support foreign



aid programs. Harper Sibley, Chairman of the Department of Church World Service of the National Council of Churches, testified at the hearings, emphasizing that denominations affiliated with the Church World Service had been actively involved in private aid distribution to developing nations. He reported that over 35 million individuals participated in these efforts, which resulted in \$110 million in aid disbursements between 1946 and 1956. These denominations were also mentioned as supporters of expanding U.S. foreign aid.

To measure a denomination’s involvement in the Church World Service, I use the list of denominations mentioned in the congressional hearings (US Congress House Committee on Foreign Affairs, 1957, p.196). By 1957, some denominations from 1890 had consolidated into larger groups. I aggregate these denominations into a single observation, reducing the sample size from 47 to 40. The independent variables include the number of missionaries, YMCA exposure in 1884 and 1890, the number of denomination members, church property per capita, and a dummy variable for evangelical denominations. The results are presented in Table 5.

Columns (1)-(3) show the regression of the outcome variable on the number of missionaries, controlling for other factors. The analysis reveals that sending 100 more missionaries is associated with a four-percentage-point increase in the probability that a denomination was involved in the Church World Service. I find that the 2SLS coefficient is approximately 15, suggesting that the OLS estimate may be underestimated. The estimated coefficients were not precise at conventional levels of statistical significance. However, the Anderson-Rubin weak IV 95 percent confidence interval was statistically significant (Andrews *et al.*, 2019), and the reduced form regression, where the outcome is directly regressed on the instrument, was significant at conventional levels.

**Congressional speech.** During the mid 1950s, advocates of foreign aid started employing the rhetoric of moral universalism to justify their stance and appeal to religious voters (Turek, 2024). Based on this observation, I present additional evidence to examine whether exposure to missionaries had any effects on moral universalism in congressional speech.

Moral universalism is characterized by the fair treatment of both ingroup and outgroup members, whereas moral particularism involves giving preferential treatment to ingroup members over outgroup members. Evidence suggests that moral universalism is linked to support for foreign aid and internationalist foreign policies (Kertzer *et al.*, 2014; Enke *et al.*, 2023), and expressions of moral universalism in congressional speech are associated with constituents’ tendencies to direct charitable donations to more distant regions (Enke *et al.*, 2024). Using moral universalism in congressional speech as an outcome, I estimate the following equation:



$$universalism_{ct} = \sum_{\tau \neq 1900} \gamma_{\tau} (TravelExposure_c \times D_{\tau}) + \sum_k \sum_{\tau \neq 1900} \theta_{k\tau} (X_{kc} \times D_{\tau}) + \delta_c + \delta_t + \varepsilon_{ct} \quad (8)$$

where  $universalism_{ct}$  is the moral universalism in congressional district  $c$  in congress  $t$ , standardized to have a mean zero and a standard deviation of one. The treatment variable interacts the exposure to travel secretaries with congress fixed effects. Controls include the share of missionary denominations and baseline YMCA exposure interacted with congress fixed effects, congress fixed effects ( $\delta_t$ ), and congressional district fixed effects ( $\delta_c$ ). All time-invariant variables are measured during the 85th Congress, when the rhetoric of moral universalism in defense of foreign aid heightened (Turek, 2024).

The results are shown in Figure 5. I find that a one standard deviation increase in exposure to travel secretaries led to a 0.3 standard deviation increase in moral universalism in congressional speeches after the 85th Congress, consistent with historical evidence of a surge in moral arguments about foreign aid during this period.

## 5. Missionaries as catalysts for regional knowledge

### 5.1 Research design

To estimate the impact of foreign missions on the understanding of foreign culture, I estimate the following equation:

$$pub_i = \theta Mission_i + \mathbf{X}_i' \rho + \varepsilon_i \quad (9)$$

where  $pub_i$  is an indicator variable for publication output or foreign intelligence during WWII.  $X_i$  is a vector of individual characteristics.

$Mission_i$  is a dummy equal to one for missionaries and zero for individuals who volunteered to be missionaries but did not become one eventually. The estimated coefficient  $\theta$  captures the effects of being missionaries under the identifying assumption that these volunteers who did not eventually become missionaries represent a counterfactual outcome, reflecting what would have happened to missionaries had they not embarked on their missions. The process of becoming a missionary began with individuals signing a declaration card, followed by filling out application blanks. The control group in this study comprises those who completed the application, indicating their serious consideration of missionary

work. This control group, consisting of individuals who were genuinely interested in missionary work and went so far as to submit applications, represents a highly comparable group and provides a compelling counterfactual outcome. An example in the control group is Kenneth Lee Pike (1912-2000), who initially applied to become a missionary but was rejected, yet later became a renowned linguist and anthropologist studying indigenous languages in Mexico (Pike, 2015).

Several factors contributed to volunteers not ultimately becoming missionaries despite initial applications. First, the Student Volunteer Movement faced significant organizational challenges in reaching out to applicants (Ramsey, 1988, p.67). One notable issue was the frequent loss of contact with applicants, often because students changed their residences after graduation or relocation, making it difficult to maintain communication (Student Volunteer Movement, 1891, p.36). Second, budget constraints played a role. The Movement relied heavily on volunteers and a small donor base, operating with limited financial resources (Parker, 1994, p.111). Despite an abundance of volunteers, some had to wait years for missionary opportunities. Third, as time passed, alternative career paths emerged. Some students who initially considered missionary work pursued other vocations, such as ministries in city churches (Ramsey, 1988, p.57). Fourth, various idiosyncratic factors intervened, including parental opposition, responsibilities towards aging parents, health issues, and college tuition concerns (Parker, 1994, p.180). In some cases, even highly committed individuals who submitted multiple applications did not become missionaries for reasons that remain unclear (Parker, 1994, p.180). Previous research has also considered this group of applicants comparable to those who became missionaries (Ramsey, 1988, p.65).

The impact of foreign missions may be overestimated if mission boards selected applicants who were already knowledgeable about foreign countries before becoming missionaries or who would have become experts on foreign countries regardless of their mission experience. However, religiosity, rather than interest in world affairs or foreign cultures, was considered the most crucial criterion for missionary selection. “The boards do not commission mere physicians or school-teachers, but missionaries...[which should be] above everything else a spiritual guide.” (Brown, 1907, p.77) These individuals had limited knowledge about their destinations and were primarily motivated by religious devotion when applying to become missionaries (Ramsey, 1988, p.148). Missionaries were often described as having a “blank slate” mindset due to their general lack of understanding of non-European countries before beginning their work (Bays & Widmer, 2009, p.17). Those who showed interest in foreign countries without strong religious commitment were screened out by the missions board (Student Volunteer Movement, 1891, p.54). Since religiosity was the main selection

criterion, it is plausible to assume that both missionaries and runners-up came from similar backgrounds and likely had a comparable level of interest in foreign countries.

To further address concerns that selection into missionary work may not be as random as assumed, Appendix A.16 leverages political instability in Turkey as an exogenous source of variation, which disrupted exposure to foreign cultures. Specifically, I compare missionaries in Turkey with those in other countries before and after 1915, showing that missionaries in Turkey, who often had to return to the U.S. for exogenous reasons, were less likely to publish works about Turkey or other foreign countries.

## 5.2 Balance checks

To evaluate the plausibility of the identifying assumption, I compare missionaries with volunteers who did not ultimately become missionaries. I match archival records to 1900 or 1910 individual-level census data using names, birth years, and addresses, employing the iterative census linkage method (Abramitzky *et al.*, 2021). The analysis compares volunteers to the general population, and missionaries to non-missionary volunteers, controlling for birth year, census year (1900 and 1910), gender, and county fixed effects. I restrict the sample to individuals younger than twenty. As shown in Appendix A.1, the probability of being matched in the census is higher for volunteers who did not eventually become missionaries. This is because missionaries who sailed abroad were not enumerated in the census. However, the difference in matching probability disappears when the sample is restricted to individuals born after 1890, as the census enumeration occurred before they became missionaries, during the period when they were still living with their parents.

Panel (a) of Figure 6 shows the differences between missionaries and non-missionaries under various specifications. I take the average of the 1900 and 1910 covariates when an individual is matched to both censuses. Hollow circles represents the raw difference after only controlling for birth year and female fixed effects. The results indicate that missionaries and volunteers differ in terms of whether their parents had religious occupations, foreign-born status, parents' foreign-born status, city population, and farm household status. As has been shown in Table A.10, these differences decrease when county fixed effects are controlled for, but differences still remain for parents having religious occupations, parents being immigrants, and farm household status. In the filled diamonds, I present differences after reweighting the control group so that the mean and variance of parents having religious occupations, parents' foreign-born status, foreign-born status, city population, and farm household status are made equal, following Hainmueller (2012). The results indicate that the

variables not directly targeted by entropy balancing remain well-balanced after reweighting.

### 5.3 Results

Panel (b) of 6 presents the estimated effect on knowledge output with and without controlling for county or denomination fixed effects, as well as inverse probability-weighted estimates based on entropy balancing. Table 6 presents regression results from Equation 9, with unadjusted estimates (without birth year and gender controls) and those with controls for birth year, gender, county, and denomination fixed effects. The results indicate that the estimated coefficients remain consistent across all specifications.

The probability of writing about foreign subjects in the control group is 0.7 percent, while the estimated effects of missionary experience range from 2.4 to 3 percentage points. This indicates that being a missionary increases the probability of becoming an author who writes about foreign subjects by three to four times. The analysis reveals a 0.6 percentage point increase in the probability of missionaries writing graduate dissertations about foreign countries, with no effect on other topics. This suggests that missionary experience specifically enhanced knowledge production about foreign countries, rather than increasing overall graduate school attendance. These findings suggest that foreign missions contributed to increased knowledge production about foreign countries without significantly influencing the overall likelihood of becoming authors or attending graduate school.

### 5.4 Evaluating the role of missionaries against other experts

While previous analyses document that missionary experience increased expertise on foreign countries, potentially valued by the government in times of crisis, it does not fully address the missionaries' importance in a broader landscape of knowledge about non-European countries. The landscape of foreign expertise could be more diverse, including academics and journalists without missionary background who facilitated cross-cultural understanding. To investigate whether the role of missionaries in understanding foreign countries in a broader landscape, I compare missionaries with other prominent individuals from Wikipedia biographies. The outcome variable is an indicator whether an individual served as a regional expert during WWII, and the independent variable is whether an individual is a missionary or a child of missionaries.

Table 7 displays the results. I find that having a missionary background (being a missionary or their child) is associated with about a 7 percent point higher probability of joining intelligence missions relative to other prominent individuals. The coefficients become even

larger when I narrow down my sample to individuals whose names can be matched to publications about foreign countries in OpenAlex (Column (3)) or those whose names were mentioned in the Smithsonian roster (Column (4)). Even among selected group of regional experts, missionaries were more likely to work for the government during WWII.

Appendix Table A.14 documents additional results. I find that the coefficient size was larger for missionary children and those who did their work in China or Japan. This suggests that longer exposure to foreign cultures from an earlier age enabled them to develop a deeper understanding. Interestingly, missionaries in the Philippines were less likely to become intelligence officers. This is likely because the US already had substantial knowledge about the Philippines due to colonization. This also suggests that for much of the non-European world that was not a US colony, missionaries played a crucial role.

A potential concern with this analysis is that the sample is restricted to individuals who appear on Wikipedia. Selection into having a biography may be non-random. The estimated coefficients could be biased upwards if less talented individuals are more likely to be included in Wikipedia if they are missionaries (negative selection). I test this possibility in Appendix A.18, using the number of views for each biography as the outcome, and find evidence against negative selection.

## 6. Longer-run effects: Peace Corps volunteers

The declining importance of religion in the United States, coupled with the proliferation of alternative channels for international engagement, may diminish the role of missionaries as conduits for foreign cultural understanding. However, missionaries could still maintain a lasting influence by shaping long-term preferences toward international engagement and cultural understanding.

To broaden the scope of analysis, I turn my attention to the Peace Corps. This extension is for three reasons. First, the number of Peace Corps volunteers can serve as a measure of preference for international development efforts in a non-religious context over the long term. Second, there are notable parallels between missionaries and Peace Corps volunteers, as both groups predominantly consist of white middle-class individuals sent abroad to facilitate economic development (Bays & Widmer, 2009, p. 18). Third, foreign aid is not just about monetary transfers; survey evidence indicates that the American public prefers channeling aid through private organizations or the Peace Corps rather than direct contributions to recipient countries (Lumsdaine, 1993, p. 149).

The results are presented in Table 8. While the OLS estimates lose statistical significance

upon the inclusion of additional controls, the IV estimates and reduced-form estimates remain robust. To address the skewed distribution, I applied the inverse hyperbolic sine transformation. The results are robust to using a log transformation after excluding counties without Peace Corps volunteers. The results show that a one standard deviation increase in the exposure to missionaries could account for 82 percent (0.6 log points) larger numbers of Peace Corps volunteers between 1961-2000.

## 7. Conclusion

Using hand-collected archival data, this study documents that missionaries played a crucial role in fostering understanding of non-European countries and shaping preferences for development assistance during the period when the concept of development assistance was being institutionalized.

The findings of this paper align with recent evidence highlighting the significance moral dimensions in policy preferences (Enke *et al.*, 2023). Myrdal (1970) and Lumsdaine (1993) argues that, in persuading voters to support foreign aid, the notion of a moral obligation to assist those in need has proven more effective than appeals to national interest or incentive-based arguments that foreign aid benefits donor countries.

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Table 1: YMCA exposure and foreign missions in Protestant denominations (first stage)

|                                     | (1)                        | (2)               | (3)               | (4)              |
|-------------------------------------|----------------------------|-------------------|-------------------|------------------|
|                                     | DV: Number of missionaries |                   |                   |                  |
|                                     | Mean DV: 207.55            |                   |                   |                  |
| Travel exposure (thousand)          | 9.51***<br>(1.20)          | 7.97***<br>(2.35) | 7.63***<br>(2.25) | 6.55**<br>(2.72) |
| YMCA without travel (thousand)      |                            |                   |                   | 0.00<br>(0.00)   |
| Members (thousand)                  |                            | 0.18<br>(0.20)    | 0.21<br>(0.20)    |                  |
| D(Evangelical)                      |                            |                   | -10.10<br>(54.45) |                  |
| Church property<br>per capita (std) |                            |                   | 20.13<br>(60.85)  |                  |
| Observations                        | 47                         | 47                | 47                | 47               |
| R-squared                           | 0.76                       | 0.76              | 0.77              | 0.78             |
| p[Travel exposure=YMCA]             |                            |                   |                   | 0.02             |

Note: Unit of analysis is a Protestant denomination that had at least one volunteers in the Student Volunteer Movement. Regressions of a number of missionaries from each denomination on the marginal exposure to YMCA in 1884. Number of missionaries from each denomination is taken from the Student Volunteer Movement records. Evangelical indicates whether a denomination is evangelical (Steensland *et al.*, 2000). Church property and the number of church attendants from each denomination is taken from Census of Religious Bodies, 1890. Robust standard errors in parentheses. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Table 2: Landmark legislation on foreign aid

| DV: D[Pro foreign aid vote]×100 | (1)             | (2)             | (3)                | (4)                | (5)                | (6)                 | (7)                |
|---------------------------------|-----------------|-----------------|--------------------|--------------------|--------------------|---------------------|--------------------|
|                                 | Mean DV: 68.58  |                 |                    |                    |                    |                     |                    |
| Exposure to missionaries (std)  | -0.43<br>(2.59) | 6.73*<br>(4.01) | 50.36**<br>(25.46) | 40.62**<br>(19.10) | 46.18**<br>(18.09) | 39.76***<br>(14.39) | 29.34**<br>(12.68) |
| Estimation method               | OLS             | OLS             | IV                 | IV                 | IV                 | IV                  | IV                 |
| Observations                    | 1,951           | 1,951           | 1,951              | 1,951              | 1,951              | 1,951               | 1,951              |
| R-squared                       | 0.21            | 0.36            | -0.02              | 0.07               | 0.21               | 0.29                | 0.35               |
| Mean outcome                    | 68.58           | 68.58           | 68.58              | 68.58              | 68.58              | 68.58               | 68.58              |
| Core controls                   | Y               | Y               | Y                  | Y                  | Y                  | Y                   | Y                  |
| Baseline YMCA                   |                 | Y               |                    | Y                  | Y                  | Y                   | Y                  |
| Political controls              |                 | Y               |                    |                    | Y                  | Y                   | Y                  |
| Religious controls              |                 | Y               |                    |                    |                    | Y                   | Y                  |
| Socioeconomic controls          |                 |                 |                    |                    |                    |                     | Y                  |
| AR weak IV [p-value]            |                 |                 | 0.00               | 0.00               | 0.00               | 0.00                | 0.00               |
| KP F-stat                       |                 |                 | 8.32               | 14.52              | 15.27              | 39.85               | 28.01              |

Note: Note: Unit of analysis is a congressional district over five Congresses (80th, 81st, 82nd, 86th, and 87th). Regressions of a pro-foreign aid vote on the exposure to missionaries instrumented by the exposure to travel secretaries. Core controls include the share of denominations involved in the Student Volunteer Movement, latitude, longitude, state fixed effects, and congress fixed effects. Baseline YMCA exposure is denominational exposure to YMCA chapters in counties with YMCA chapters and without travel secretaries' visit. Political controls include DW-NOMINATE score (first dimension) and Republican Party fixed effects. Religious controls include the shares of evangelical Protestants, mainline Protestants, Catholics, Orthodox Jews, and Mormons. Socioeconomic controls include log population, urbanization rate, occupational income score, agricultural land suitability, market access, literacy rate, and the shares of college students, manufacturing workers, Northern and Western European immigrants, Eastern and Southern European immigrants, non-European immigrants, and African Americans. Robust standard errors clustered by state in parentheses. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Table 3: Heterogeneous effects by denomination

| DV: D(Pro foreign aid vote)×100 | (1)              | (2)               | (3)               | (4)               | (5)              | (6)                |
|---------------------------------|------------------|-------------------|-------------------|-------------------|------------------|--------------------|
| Estimation method:              | OLS              | IV                | OLS               | IV                | OLS              | IV                 |
| Mainline missionaries (std)     | 7.19**<br>(3.03) | 12.27**<br>(5.50) |                   |                   | 6.81**<br>(3.01) | 30.82**<br>(12.42) |
| Evangelical missionaries (std)  |                  |                   | -2.76**<br>(1.31) | 22.79*<br>(13.70) | -2.50*<br>(1.30) | 25.49*<br>(13.07)  |
| Observations                    | 1,951            | 1,951             | 1,951             | 1,951             | 1,951            | 1,951              |
| R-squared                       | 0.23             | 0.23              | 0.23              | 0.07              | 0.23             | 0.02               |
| Core controls                   | Y                | Y                 | Y                 | Y                 | Y                | Y                  |
| Baseline YMCA                   | Y                | Y                 | Y                 | Y                 | Y                | Y                  |
| Religious controls              | Y                | Y                 | Y                 | Y                 | Y                | Y                  |
| AR weak IV [p-value]            |                  | 0.01              |                   | 0.07              |                  |                    |
| KP F-stat                       |                  | 49.44             |                   | 8.55              |                  | 5.82               |

Note: Unit of analysis is a congressional district over five Congresses (80th, 81st, 82nd, 86th, and 87th). Regressions of a pro-foreign aid vote on the exposure to missionaries instrumented by the exposure to travel secretaries. Treatment and instrumental variables are divided into two categories respectively, based on whether or not a denomination belongs to evangelical denominations. Core controls include the share of denominations involved in the Student Volunteer Movement, latitude, longitude, state fixed effects, and congress fixed effects. Baseline YMCA exposure is denominational exposure to YMCA chapters in counties with YMCA chapters and without travel secretaries' visit. Religious controls include the shares of evangelical Protestants, mainline Protestants, Catholics, Orthodox Jews, and Mormons. Robust standard errors clustered by state in parentheses. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1



Table 4: Alternative identification strategy: YMCA exposure as an instrument

|                     | (1)                 | (2)               | (3)               | (4)               | (5)                         | (6)               | (7)               | (8)                |
|---------------------|---------------------|-------------------|-------------------|-------------------|-----------------------------|-------------------|-------------------|--------------------|
|                     | DV: Travel exposure |                   |                   |                   | D(Pro foreign aid vote)×100 |                   |                   |                    |
| YMCA exposure       | 0.38***<br>(0.02)   |                   |                   |                   | 18.15***<br>(5.87)          |                   |                   |                    |
| YMCA 1881           |                     | 0.58*<br>(0.30)   | 1.04***<br>(0.24) |                   |                             | 19.29*<br>(10.64) | 18.87**<br>(7.80) |                    |
| YMCA 1884           |                     | 0.83***<br>(0.22) |                   | 1.33***<br>(0.19) |                             | 3.68<br>(7.06)    |                   | 16.63***<br>(4.92) |
| YMCA 1889           |                     | -0.22<br>(0.22)   | -0.09<br>(0.25)   | 0.01<br>(0.18)    |                             | -4.51<br>(9.48)   | -8.77<br>(6.79)   | -7.16<br>(4.90)    |
| Observations        | 47                  | 47                | 47                | 47                | 1,951                       | 1,951             | 1,951             | 1,951              |
| R-squared           | 0.92                | 0.96              | 0.93              | 0.94              | 0.21                        | 0.22              | 0.39              | 0.39               |
| p[Equal coef.]      |                     | 0.00              | 0.02              | 0.00              |                             | 0.47              | 0.05              | 0.01               |
| Core controls       |                     |                   |                   |                   | Y                           | Y                 | Y                 | Y                  |
| Additional controls |                     |                   |                   |                   |                             |                   | Y                 | Y                  |

Note: Unit of analysis is denomination in Columns (1)-(4) and congressional district in Columns (5)-(8). Additional controls in Columns (5)-(8) refer to all controls in Table 2. YMCA exposure is constructed based on the denominational composition of counties with local YMCA chapters between 1881 and 1889. Columns (2)-(4) and (6)-(8) divide YMCA exposure into three groups: exposure from counties with YMCA chapters established before 1881 (YMCA 1881), counties with chapters established between 1881 and 1884 (YMCA 1884), and counties with chapters established between 1884 and 1889 (YMCA 1889). \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Table 5: Denominational support for foreign aid

|                                              | (1)                             | (2)      | (3)        | (4)        | (5)     | (6)    |
|----------------------------------------------|---------------------------------|----------|------------|------------|---------|--------|
|                                              | DV: D(Church World Service)×100 |          |            |            |         |        |
|                                              | Mean DV: 50                     |          |            |            |         |        |
| Estimation method                            | OLS                             | OLS      | IV         | IV         | RF      | RF     |
| Missionaries (hundred)                       | 4.50**                          | 4.02*    | 15.20*     | 15.09      |         |        |
|                                              | (1.74)                          | (2.05)   | (8.54)     | (10.41)    |         |        |
| AR weak IV 95 percent<br>confidence interval |                                 |          | [6.8,48.7] | [4.8,55.9] |         |        |
| Travel exposure (thousand)                   |                                 |          |            |            | 0.76*** | 0.58** |
|                                              |                                 |          |            |            | (0.25)  | (0.28) |
| YMCA without travel (thousand)               |                                 |          |            |            |         | -0.16  |
|                                              |                                 |          |            |            |         | (0.18) |
| Members (thousand)                           | -0.01                           | -0.01    | -0.09      | -0.09      | -0.04** |        |
|                                              | (0.02)                          | (0.02)   | (0.06)     | (0.07)     | (0.02)  |        |
| D(Evangelical)                               |                                 | -36.68** |            | -37.88**   | -33.72* |        |
|                                              |                                 | (17.26)  |            | (17.31)    | (17.67) |        |
| Church property<br>per capita (std)          |                                 | -3.06    |            | -11.89     | -4.64   |        |
|                                              |                                 | (9.61)   |            | (10.99)    | (9.06)  |        |
| Observations                                 | 40                              | 40       | 40         | 40         | 40      | 40     |
| R-squared                                    | 0.12                            | 0.24     | -0.20      | -0.08      | 0.27    | 0.14   |
| p[Travel exposure=YMCA]                      |                                 |          |            |            |         | 0.11   |
| KP F-stat                                    |                                 |          | 3.16       | 2.64       |         |        |

Note: Unit of analysis is a Protestant denomination that had at least one volunteer in the Student Volunteer Movement. Regressions of an indicator whether a denomination participated in the Church World Service on the number of missionaries from each denomination. Denominations that were consolidated by 1957 are aggregated into one observation. Number of missionaries from each denomination is taken from the Student Volunteer Movement records. Evangelical indicates whether a denomination is evangelical denomination (Steensland *et al.*, 2000). Church property and the number of church attendants from each denomination is taken from Census of Religious Bodies, 1890. Robust standard errors clustered by state in parentheses. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Table 6: Effects of foreign missions on knowledge production

| Outcome:              | Grad. dissertation<br>foreign subjects |                   | Grad. dissertation<br>nonforeign subjects |                   |
|-----------------------|----------------------------------------|-------------------|-------------------------------------------|-------------------|
| (dummy×100)           | (1)                                    | (2)               | (3)                                       | (4)               |
| Mission               | 0.64***<br>(0.08)                      | 0.95***<br>(0.12) | -0.32*<br>(0.16)                          | 0.31<br>(0.22)    |
| Control outcome mean: | 0.13                                   | 0.13              | 2.46                                      | 2.46              |
| Observations          | 37,106                                 | 29,380            | 37,106                                    | 29,380            |
| R-squared             | 0.00                                   | 0.07              | 0.00                                      | 0.08              |
| controls              |                                        | Yes               |                                           | Yes               |
| Outcome:              | Author foreign subjects                |                   | Author nonforeign subjects                |                   |
| (dummy×100)           | (5)                                    | (6)               | (7)                                       | (8)               |
| Mission               | 2.37***<br>(0.16)                      | 3.04***<br>(0.22) | -0.73***<br>(0.26)                        | -0.06<br>(0.34)   |
| Control outcome mean: | 0.7                                    | 0.7               | 6.35                                      | 6.35              |
| Observations          | 37,106                                 | 29,380            | 37,106                                    | 29,380            |
| R-squared             | 0.01                                   | 0.09              | 0.00                                      | 0.08              |
| controls              |                                        | Yes               |                                           | Yes               |
| Outcome:              | Smithsonian Institution roster         |                   | Office of Strategic Services              |                   |
| (dummy×100)           | (9)                                    | (10)              | (11)                                      | (12)              |
| Mission               | 1.00***<br>(0.09)                      | 1.21***<br>(0.13) | 0.11***<br>(0.04)                         | 0.14***<br>(0.05) |
| Control outcome mean: | 0.07                                   | 0.07              | 0.05                                      | 0.05              |
| Observations          | 37,106                                 | 29,380            | 37,106                                    | 29,380            |
| R-squared             | 0.01                                   | 0.07              | 0.00                                      | 0.08              |
| controls              |                                        | Yes               |                                           | Yes               |

Note: Unit of analysis is an individual. Regressions of an indicator of non-European regional expertise on an indicator of being missionaries conditional on being volunteers to the Student Volunteer Movement. The outcome variables in Columns (1) to (4) indicate whether individual names can be matched to the ProQuest Dissertations Theses database, while Columns (5) to (8) indicate whether individual names can be matched to publications about foreign subjects using OpenAlex. The outcomes in Columns (9) and (10) reflect whether an individual's name can be matched to the Smithsonian roster (Bennett, 1947). The outcomes in Columns (11) and (12) capture whether an individual's name can be matched to the Office of Strategic Services roster or publications about foreign subjects according to OpenAlex. Odd-numbered columns report raw differences, while even-numbered columns document differences after controlling for birth year, gender, county, and denomination fixed effects. Robust standard errors, clustered by county, are provided in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Table 7: Regional expertise during WWII: Evidence from Wikipedia

|                          | (1)                                          | (2)               | (3)                | (4)                 |
|--------------------------|----------------------------------------------|-------------------|--------------------|---------------------|
| DV: dummy $\times 100$   | Intelligence or language experts during WWII |                   |                    |                     |
| missionaries or children | 7.33***<br>(1.37)                            | 7.03***<br>(1.39) | 15.63***<br>(4.78) | 63.15***<br>(13.71) |
| Control outcome mean     | 3.00                                         | 3.00              | 4.31               | 7.80                |
| Missionaries             | 489                                          | 489               | 70                 | 7                   |
| Observations             | 74,966                                       | 74,966            | 4,474              | 148                 |
| birth cohort FE          |                                              | Y                 | Y                  | Y                   |
| female FE                |                                              | Y                 | Y                  | Y                   |
| occupation FE            |                                              | Y                 | Y                  | Y                   |
| pub. foreign             |                                              |                   | Y                  |                     |
| Smithsonian roster       |                                              |                   |                    | Y                   |
| R-squared                | 0.00                                         | 0.02              | 0.01               | 0.23                |

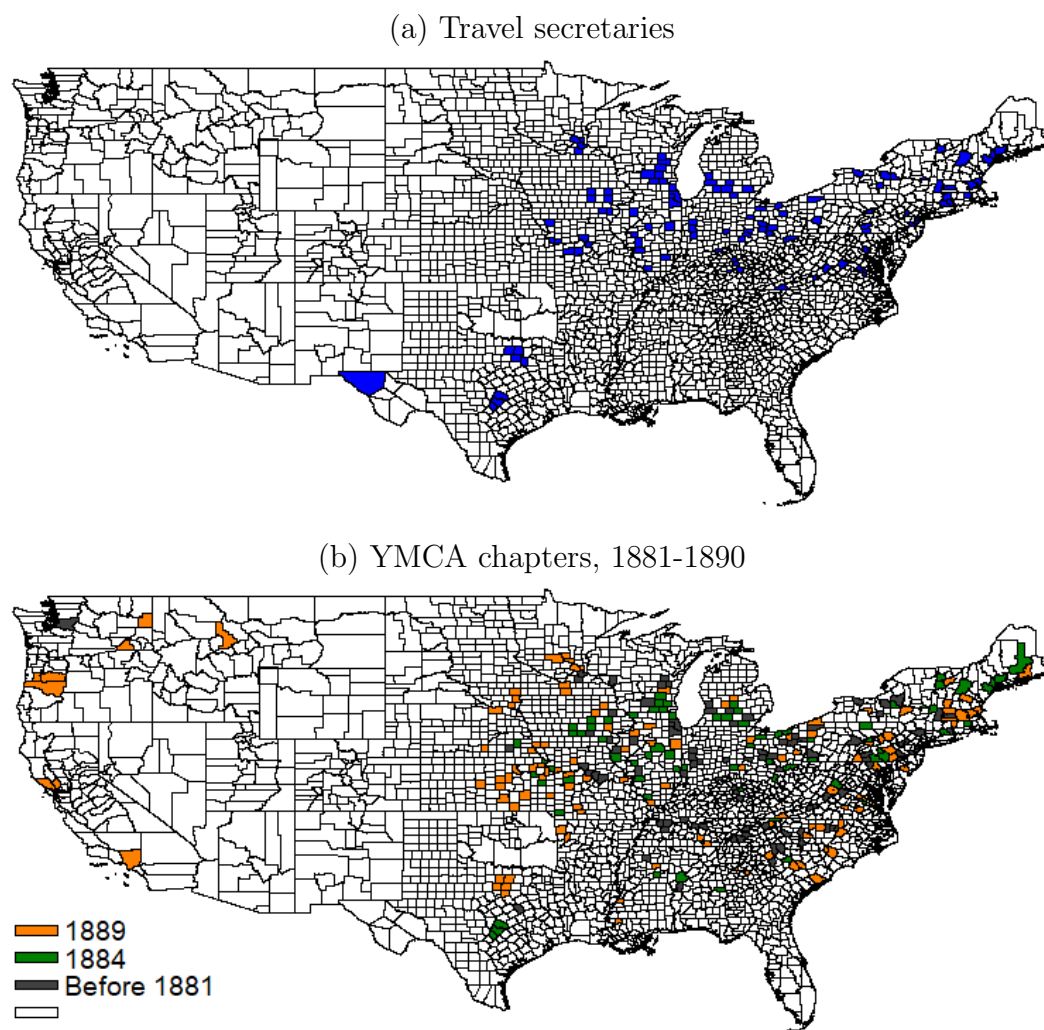
Note: Unit of analysis is an individual. Regressions of an indicator of whether an individual served as an intelligence or language expert during WWII on an indicator of whether an individual has a missionary background. The variable “missionary or children” is an indicator equal to one if an individual is a missionary or the child of missionaries. The sample is restricted to individuals with a Wikipedia biography who were active in the United States, born after 1880, before 1925, and who died after 1945. The sample in Column (3) is restricted to individuals whose names can be matched to publications about foreign subjects. The sample in Column (4) is restricted to individuals whose names can be matched to the Smithsonian Institution roster. Occupation is taken from Laouenan *et al.* (2022). Robust standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Table 8: Peace Corps volunteers

|                                | (1)              | (2)               | (3)              | (4)             | (5)               | (6)               |
|--------------------------------|------------------|-------------------|------------------|-----------------|-------------------|-------------------|
| DV: Peace Corps volunteers     |                  |                   |                  |                 |                   |                   |
| Estimation method              | OLS              | OLS               | IV               | IV              | IV                | IV                |
| Exposure to missionaries (std) | 0.19**<br>(0.08) | 0.23***<br>(0.08) | 0.65**<br>(0.25) | 0.53*<br>(0.29) | 0.58***<br>(0.18) | 0.60***<br>(0.17) |
| DV transform:                  | asinh            | asinh             | asinh            | asinh           | asinh             | log               |
| Observations                   | 2,807            | 2,807             | 2,808            | 2,808           | 2,785             | 2,469             |
| R-squared                      | 0.56             | 0.58              | 0.55             | 0.57            | 0.63              | 0.61              |
| Baseline controls              | Y                | Y                 | Y                | Y               | Y                 | Y                 |
| Religious controls             |                  | Y                 |                  | Y               | Y                 | Y                 |
| Socioeconomic controls         |                  |                   |                  |                 | Y                 | Y                 |
| KP F-stat                      |                  |                   | 20.23            | 19.37           | 18.43             | 33.70             |
| AR weak IV [p-value]           |                  |                   | 0.00             | 0.05            | 0.00              | 0.00              |

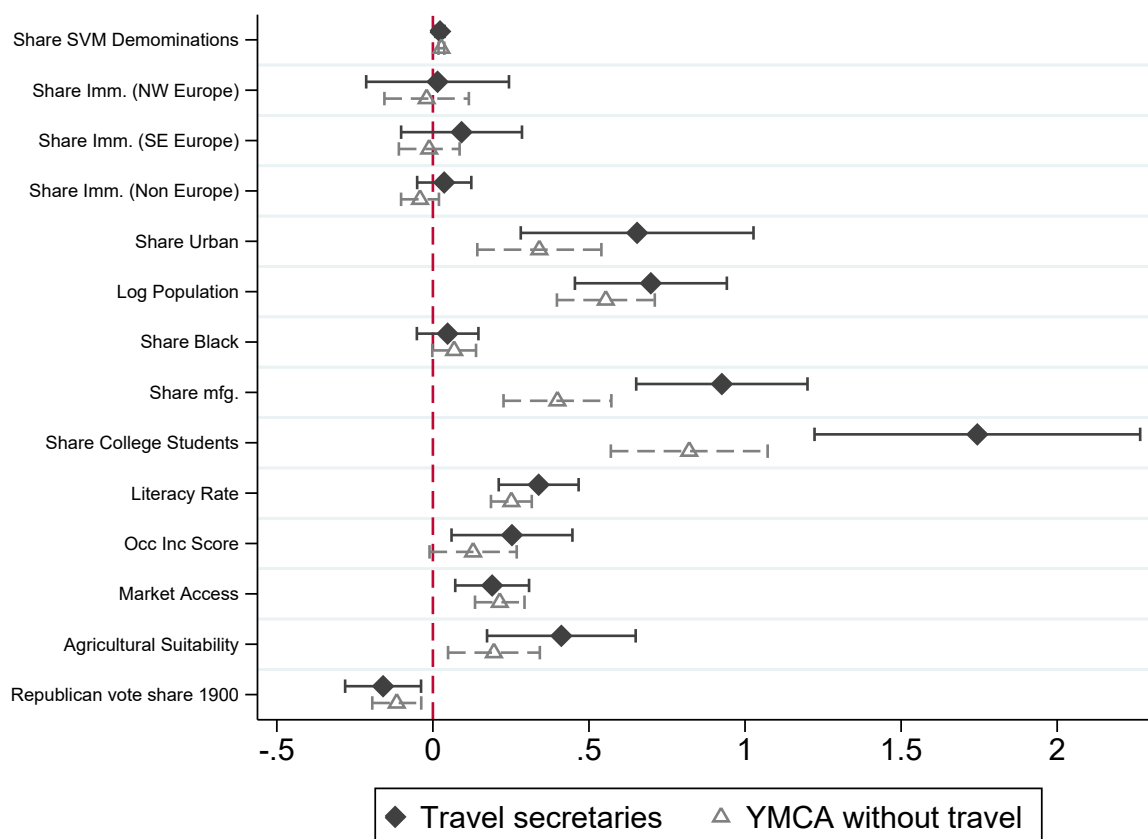
Note: Unit of analysis is a county. Regressions of the number of Peace Corps volunteers on the exposure to missionaries instrumented by the exposure to travel secretaries during 1886-87. Baseline controls include log population, latitude, longitude, the share of denominations involved in the Student Volunteer Movement, baseline YMCA exposure, and state fixed effects. Religious controls include the shares of evangelical Protestants, mainline Protestants, Catholics, Orthodox Jews, and Mormons. Socioeconomic controls include urbanization rate, occupational income score, agricultural land suitability, market access, literacy rate, and the shares of college students, manufacturing workers, Northern and Western European immigrants, Eastern and Southern European immigrants, non-European immigrants, and African Americans. Robust standard errors clustered by state in parentheses. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Figure 1: YMCA chapters and travel secretaries



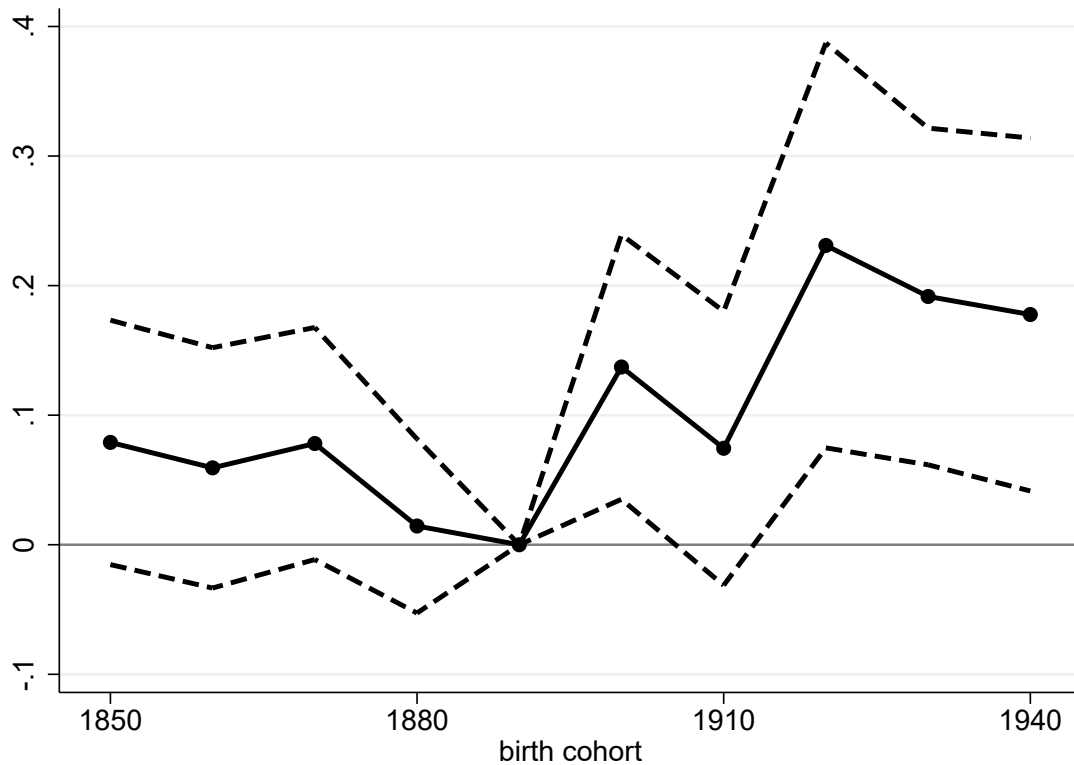
Note: Figure (a) shows the locations visited by travel secretaries. Figure (b) illustrates the spatial expansion of local YMCA chapters between 1881 and 1890. Data on travel secretaries' itinerary is constructed from *Missionary Review of the World* magazine, February 1887 to June 1887 (Beahm, 1941, p. 80). Data on local YMCA chapters is taken from *Intercollegian* magazine published between 1881 and 1890.

Figure 2: Balance in travel routes



Note: The unit of analysis is the county. All variables are standardized to have a mean of zero and a standard deviation of one. Coefficient plot from regressions of covariates on indicator variables for counties visited by the travel secretaries during 1886-87 (diamonds) and counties with YMCA chapters that did not receive a visit (hollow triangles). The results control for state fixed effects, and error bars represent 95 percent confidence intervals. Standard errors are clustered at the state level.

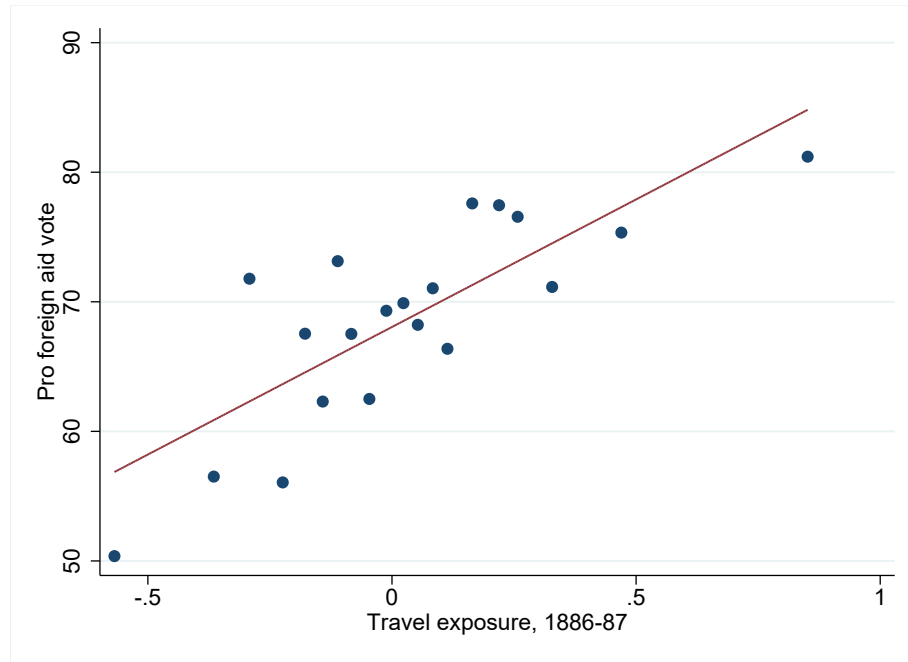
Figure 3: Effects on the number of Wikipedia authors writing about non-European countries



Note: Unit of analysis is county-birth cohort (decade). The outcome variable is the number of Wikipedia authors, matched to publication records between 1860 and 1960, born in each county for each birth cohort. The independent variable is the interaction between travel exposure and decade fixed effects. Controls include the share of Student Volunteer Movement denominations, baseline YMCA exposure (both interacted with decade fixed effects), as well as state-by-year and county fixed effects. Standard errors are clustered at the county level, and error bars represent 95 percent confidence intervals.

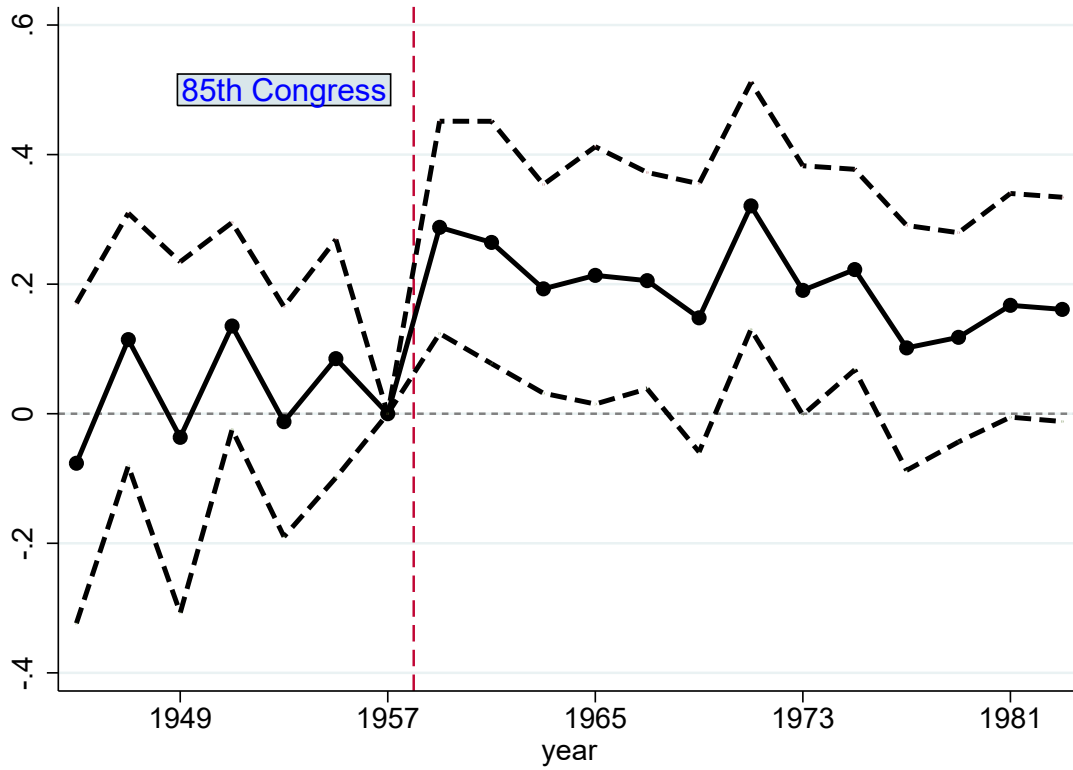


Figure 4: Effects on foreign aid legislation



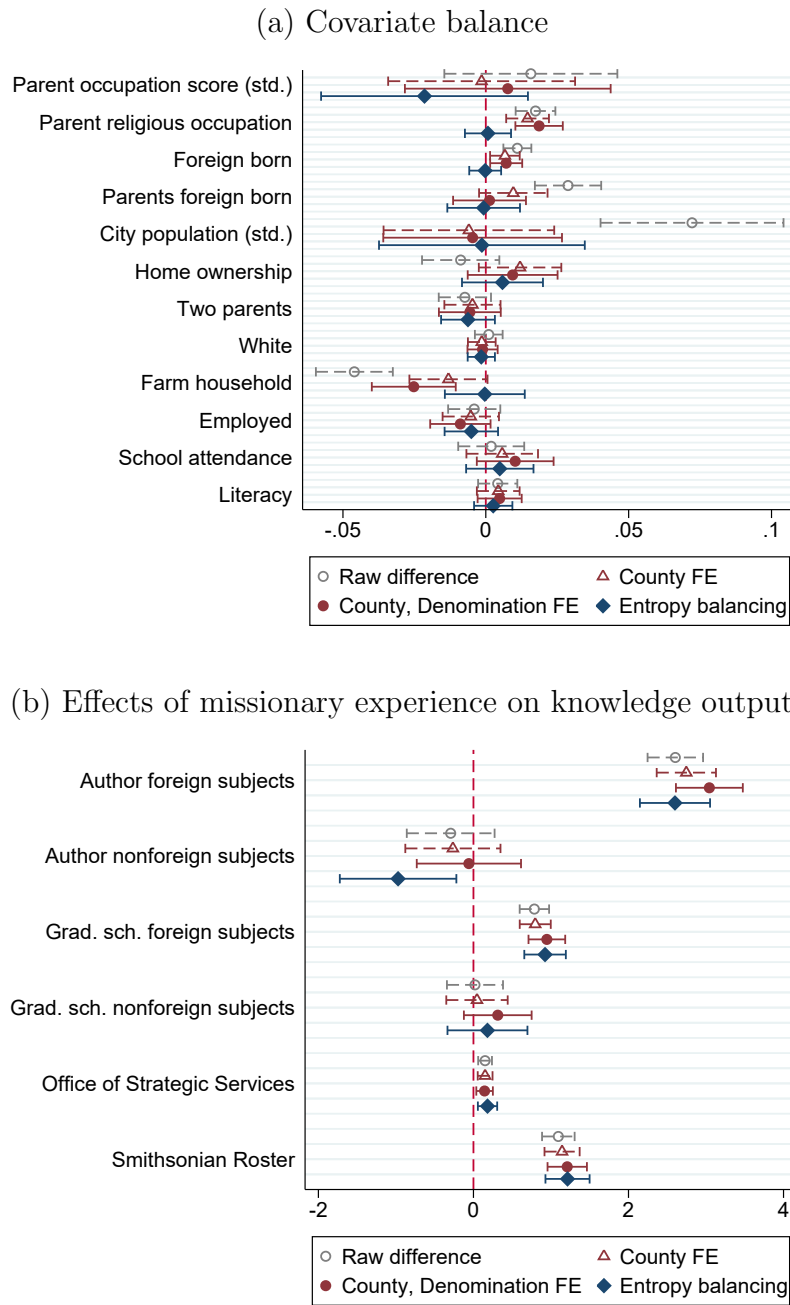
Note: Unit of analysis is a congressional district by congressional terms. Coefficients from reduced-form regression results of a Congressman vote “Yea” for the foreign aid bill on the marginal YMCA exposure in 1884, after controlling for the baseline YMCA exposure, share of SVM denominations, Republican party fixed effects, DW-NOMINATE score (first dimension), latitude, longitude, and state fixed effects.

Figure 5: Moral universalism in Congressional speech



Note: Unit of analysis is a congressional district by congressional term. Coefficients from the regressions of moral universalism in congressional speeches on the marginal exposure to YMCA in 1884 (Equation 8). Outcome variable is a standardized score of moral universalism in congressional speech (Enke, 2020). The marginal exposure to YMCA in 1884 is interacted with congress fixed effects. Control variables include the YMCA control, the share of missionary denominations interacted with congressional period fixed effects, congressional district fixed effects and congress fixed effects. Error bars indicate 95 percent confidence intervals. Error terms are clustered by state.

Figure 6: Covariate balance and effects of missionary experience: Individual-level analysis



Note: Raw difference indicates the difference between missionaries and volunteers, controlling for birth year and female fixed effects. The hollow triangle additionally controls for county fixed effects, while the filled circle adds controls for denomination fixed effects. The filled diamond represents differences after reweighting the control group using entropy balancing, where indicators such as parents' religious occupation, foreign born status, parents' foreign born status, farm household status, as well as continuous variables like city population, birth year, and female indicators are weighted to have equal means and variances, following Hainmueller (2012). Indicators in Panel (a) are scaled to one, while those in Panel (b) are scaled to one hundred. The control group means for each variable are 0.7, 0.13, 6.35, 2.46, 0.05, and 0.07, respectively. Error bars indicate 95 percent confidence intervals, based on robust standard errors.

## Appendix

### A. Additional results

#### A.1 Census match

I employ an iterative record linking procedure to match archival records to census data (Abramitzky *et al.*, 2021). An individual and a census record are considered matched if they share the same NYSIIS-standardized first and last name, middle initial, address, and have estimated birth years within two years of each other. If only one census observation matches only one archival sample, I consider those records linked and remove them from the pool of possible matches. If the match is not unique, the records are not linked.

I match individuals under forty years old across each census, beginning with the most stringent criteria. I then repeat the procedure, gradually relaxing the matching variables by using less detailed names or geographic identifiers. The matching process follows these steps:

- Match based on first and last names, middle initial, birth year, and county of residence.
- Match based on first and last names, birth year, and county of residence.
- Match based on first and last names, middle initial, birth year, and state of residence (rather than county).
- Match based on first and last names, birth year, and state of residence (rather than county).
- Match based on first and last names, middle initial, and birth year.
- Match based on first and last names and birth year.

The data includes 33,535 individuals with name and birth year information. Of these, 33,478 individuals have birth years after 1858, comprising the sample matched to the 1900 census (40 years of age in 1900 plus two years of allowable gap for census matching). Among them, 28,256 individuals are matched to either the 1900 or 1910 census records, resulting in a census match rate of 84 percent.

The automated linking procedure may result in some false positives. As suggested by Bailey *et al.* (2020), a more unique match can enhance matching accuracy. To improve

the reliability of the matches, I further restrict the sample to individuals who are unique within  $\pm 2$  years of age (Abramitzky *et al.*, 2021). This refinement yields 22,402 matched individuals, achieving a match rate of 67 percent.

In the baseline matching procedure, I consider an individual matched if there is only one unique combination of name, birth year, and location in both the census record and the archival sample. For example, a match would be made if there is only one John Smith born in 1897 who lives in Illinois in 1900 in both datasets.

For a more conservative matching approach, I implement additional stringency. In this case, I drop matches if there are similar individuals within a close age range in the same location. For instance, I would drop the match for John Smith born in 1897 in Illinois if there is another John Smith in Illinois born in 1895.

This match rate is at the higher end of the distribution compared to existing studies that match external data to individual census records (Bleemer & Quincy, 2021; Moreira & Pérez, 2022; Aneja & Xu, 2022) or perform census-to-census or census-to-patent matching (Sarada *et al.*, 2019; Abramitzky *et al.*, 2020), which report 20 to 40 percent match rates. This higher rate is presumably due to the Student Volunteer Movement records containing detailed data on county of residence and birth years. Matching procedures that involve more accurate birth years or county of residence allow for more unique matches between external data and census records, resulting in about 57 to 69 percent match rates in such cases (Michelman *et al.*, 2022; Ang & Chinoy, 2023).

For the 1900 census, the match rate is 71 percent, as 32,658 individuals have birth years between 1858 and 1902, and 23,059 individuals were matched. For the 1910 census, the match rate is 72 percent, as 32,484 individuals have birth years between 1868 and 1912, and 23,461 individuals are matched.

To examine whether the census match rate differs by treatment status, I regress an outcome variable indicating whether an individual is matched to either the 1900 or 1910 census on the treatment indicator, controlling for baseline characteristics such as birth year, gender, and county fixed effects. The results are presented in Table A.1.

Panel A displays the results from the baseline matching procedure. I find a notable difference: missionaries were 3.45 percentage points less likely to be matched to census records. This discrepancy is likely attributable to missionaries residing abroad at the time of the census, thus not being enumerated in the U.S. census.

However, when I narrow the sample to individuals born after 1890 (who were younger than twenty at the time of the census survey), this difference disappears. This observation

provides an additional rationale for restricting attention to individuals younger than 20 in the balance test presented in Table [A.10](#).

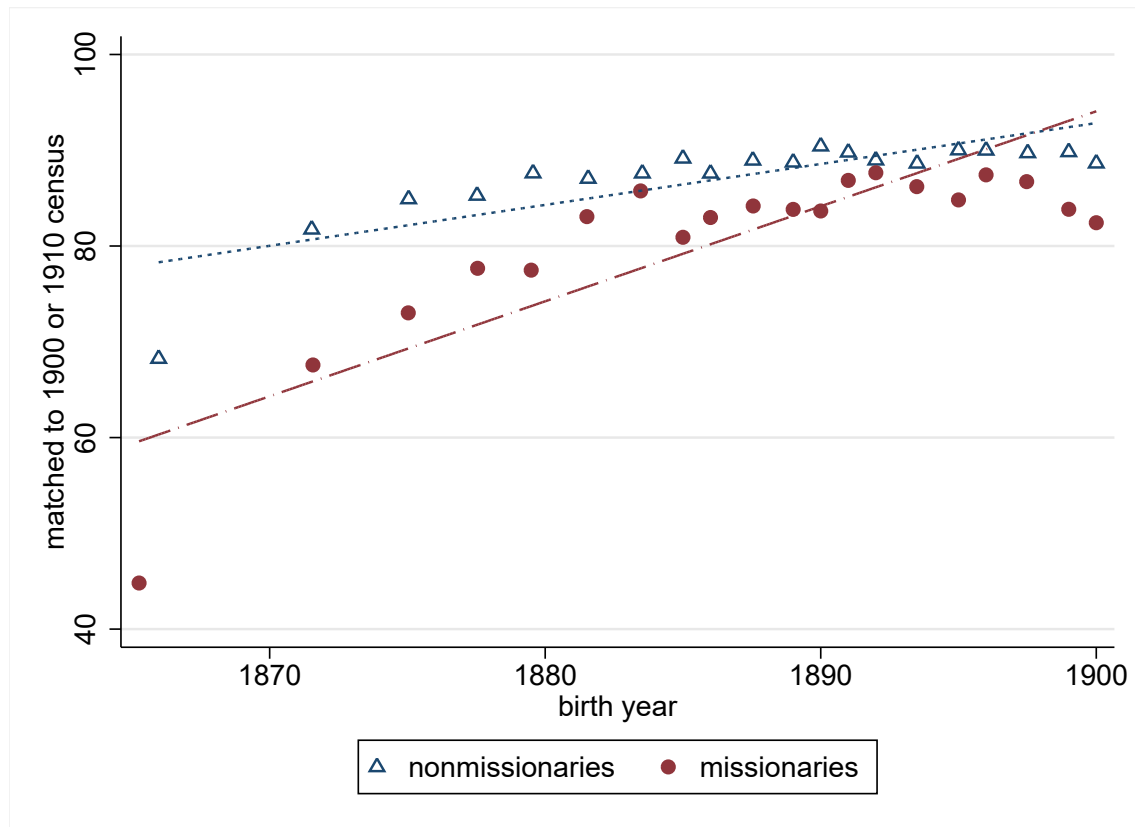
Table A.1: Matching to the census

|                                | (1)                            | (2)                | (3)                |
|--------------------------------|--------------------------------|--------------------|--------------------|
| DV:                            | Matched to 1900 or 1910 census |                    |                    |
| Panel A: Baseline matching     |                                |                    |                    |
| Mission                        | -3.45***<br>(0.50)             | -0.24<br>(0.81)    | -5.20***<br>(0.64) |
| Observations                   | 29,351                         | 11,859             | 16,854             |
| R-squared                      | 0.10                           | 0.14               | 0.16               |
| Control outcome mean           | 87.09                          | 88.55              | 85.77              |
| Controls                       | Yes                            | Yes                | Yes                |
| Sample restriction             | no                             | birthyear<br>>1890 | birthyear<br>≤1890 |
| Panel B: Conservative matching |                                |                    |                    |
| Mission                        | -3.65***<br>(0.65)             | 0.12<br>(1.01)     | -6.11***<br>(0.84) |
| Observations                   | 29,351                         | 11,859             | 16,854             |
| R-squared                      | 0.09                           | 0.14               | 0.13               |
| Control outcome mean           | 69.65                          | 73.03              | 66.90              |
| Controls                       | Yes                            | Yes                | Yes                |
| Sample restriction             | no                             | birthyear<br>>1890 | birthyear<br>≤1890 |

Note: Unit of analysis is an individual in the Student Volunteer Movement archival records. Regressions of an indicator equal to one hundred if an individual is matched to either 1900 or 1910 census data on an indicator for missionaries. Outcome variable in Panel A is defined by the baseline matching procedure described in Section A.1. Outcome variable in Panel B is defined by a more conservative matching restricting the matched sample to those unique within  $\pm 2$  years of age. Control variables include birth year, female, denomination, and county fixed effects. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

## A.2 Census match rate by birth year

Figure A.1: Census match rate by birth year



Note: Binscatter plot where the vertical axis indicates whether an individual is matched to 1900 or 1910 census and the horizontal axis is the birth year.



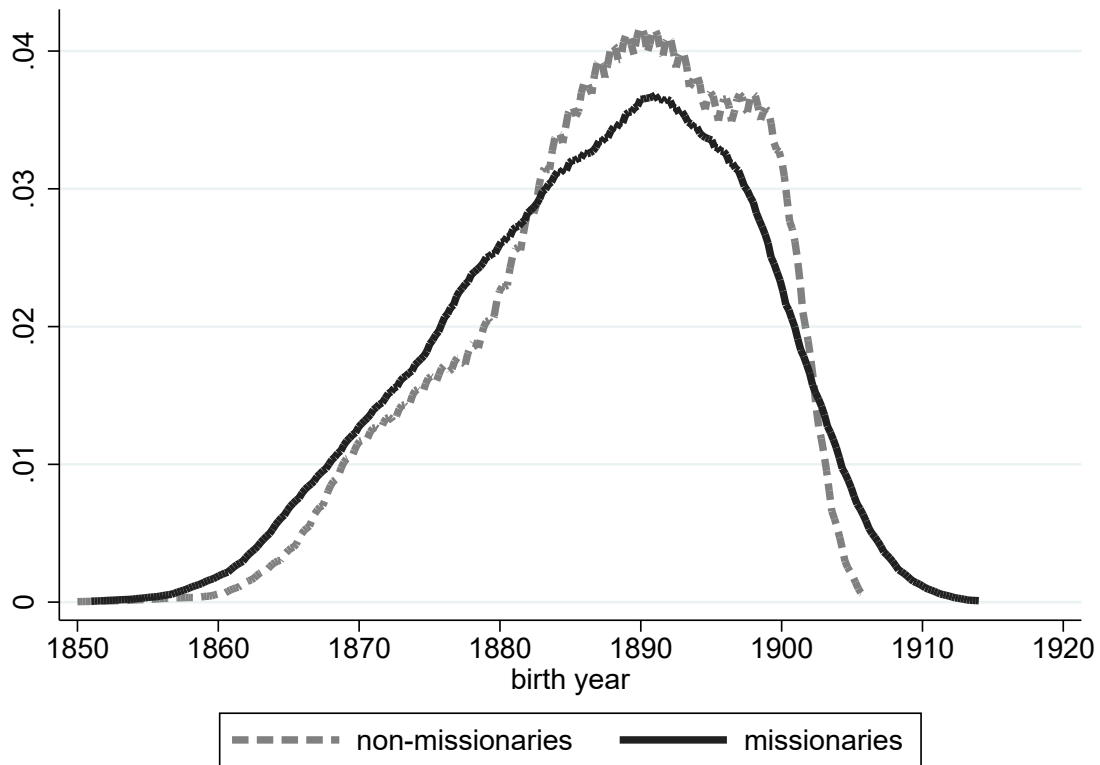
Table A.2: Balance test with a more conservative census match

| Variables                  | (1)<br>diff (s.e.) | (2)<br>control mean | (3)<br>N | (4)<br>age |
|----------------------------|--------------------|---------------------|----------|------------|
| Parental occ. score (\$00) | 0.15 (0.23)        | 20.96               | 21,419   | [0,20]     |
| Parent religious occ.      | 2.38*** (0.45)     | 6.13                | 21,419   | [0,20]     |
| Foreign born               | 0.51** (0.24)      | 1.64                | 21,419   | [0,20]     |
| Foreign born parents       | 1.11* (0.63)       | 17.01               | 21,419   | [0,20]     |
| City pop. (000s)           | -1.22 (2.12)       | 106.61              | 21,419   | [0,20]     |
| Home ownership             | 1.27 (0.82)        | 63.21               | 21,419   | [0,20]     |
| Two parents                | 0.47 (0.46)        | 90.41               | 21,419   | [0,20]     |
| White                      | -0.03 (0.19)       | 98.26               | 21,419   | [0,20]     |
| Farm household             | -1.38* (0.78)      | 39.35               | 21,419   | [0,20]     |
| Employed                   | -0.01 (0.60)       | 11.49               | 11,138   | [10,20]    |
| School attendance          | -0.20 (0.78)       | 84.89               | 11,138   | [10,20]    |
| Literacy                   | -0.33 (0.21)       | 98.74               | 11,138   | [10,20]    |

Note: This is the same results in Table A.10 except that this table restricts the sample to the matches between the archival records and the census that are unique within  $\pm 2$  years of age to reduce the false positives in the census matching process. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

### A.3 Demographics of missionaries

Figure A.2: Kernel density plot of birth year



Note: Kernel density plots of birth years in the archival records by missionaries and non-missionaries. Ten individuals born before 1850 have been excluded for presentational purpose. N=33,401. Samples include everyone in the records who volunteered to be missionaries.

Table A.3: Number of volunteers by state

| State                | Volunteers | State                | Volunteers per million |
|----------------------|------------|----------------------|------------------------|
| ILLINOIS             | 2724       | OKLAHOMA             | 5854.384               |
| OHIO                 | 2624       | OREGON               | 1574.417               |
| PENNSYLVANIA         | 2609       | WASHINGTON           | 1451.1                 |
| NEW YORK             | 2220       | CALIFORNIA           | 1270.559               |
| IOWA                 | 2126       | IDAHO                | 1232.447               |
| KANSAS               | 1536       | SOUTH DAKOTA         | 1164.814               |
| CALIFORNIA           | 1535       | IOWA                 | 1111.985               |
| MICHIGAN             | 1341       | COLORADO             | 1091.708               |
| INDIANA              | 1338       | KANSAS               | 1076.312               |
| MASSACHUSETTS        | 1165       | NORTH DAKOTA         | 1034.375               |
| MISSOURI             | 1152       | NEBRASKA             | 931.1462               |
| NEBRASKA             | 986        | OHIO                 | 714.5355               |
| TEXAS                | 979        | ILLINOIS             | 711.9054               |
| MINNESOTA            | 924        | MINNESOTA            | 709.7723               |
| NORTH CAROLINA       | 820        | MICHIGAN             | 640.4351               |
| VIRGINIA             | 739        | INDIANA              | 610.289                |
| WISCONSIN            | 667        | SOUTH CAROLINA       | 555.9663               |
| NEW JERSEY           | 666        | MONTANA              | 552.365                |
| SOUTH CAROLINA       | 640        | VERMONT              | 526.4393               |
| GEORGIA              | 594        | MASSACHUSETTS        | 520.3348               |
| KENTUCKY             | 532        | NORTH CAROLINA       | 506.8151               |
| TENNESSEE            | 508        | PENNSYLVANIA         | 496.195                |
| WASHINGTON           | 507        | WYOMING              | 477.7201               |
| OREGON               | 494        | NEW JERSEY           | 460.921                |
| COLORADO             | 450        | CONNECTICUT          | 448.9064               |
| SOUTH DAKOTA         | 383        | VIRGINIA             | 446.2614               |
| OKLAHOMA             | 362        | TEXAS                | 437.9288               |
| CONNECTICUT          | 335        | ARIZONA              | 436.0953               |
| MISSISSIPPI          | 335        | MISSOURI             | 429.9817               |
| ALABAMA              | 327        | FLORIDA              | 406.2112               |
| MARYLAND             | 275        | WISCONSIN            | 395.4045               |
| WEST VIRGINIA        | 274        | NEW YORK             | 370.1324               |
| ARKANSAS             | 248        | WEST VIRGINIA        | 359.2058               |
| MAINE                | 206        | DISTRICT OF COLUMBIA | 325.5321               |
| NORTH DAKOTA         | 189        | GEORGIA              | 323.2912               |
| VERMONT              | 175        | MAINE                | 311.6085               |
| FLORIDA              | 159        | NEW HAMPSHIRE        | 310.7322               |
| NEW HAMPSHIRE        | 117        | RHODE ISLAND         | 303.9021               |
| RHODE ISLAND         | 105        | TENNESSEE            | 287.4087               |
| IDAHO                | 104        | KENTUCKY             | 286.2316               |
| LOUISIANA            | 82         | MARYLAND             | 263.8168               |
| DISTRICT OF COLUMBIA | 75         | MISSISSIPPI          | 259.7705               |
| MONTANA              | 73         | NEW MEXICO           | 247.4071               |
| NEW MEXICO           | 38         | ARKANSAS             | 219.8233               |
| DELAWARE             | 36         | NEVADA               | 218.5267               |
| WYOMING              | 29         | ALASKA               | 218.3951               |
| ARIZONA              | 26         | ALABAMA              | 216.1245               |
| UTAH                 | 24         | DELAWARE             | 213.6587               |
| NEVADA               | 10         | UTAH                 | 115.4373               |
| HAWAII               | 9          | HAWAII               | 100.0111               |
| ALASKA               | 7          | LOUISIANA            | 73.30677               |

Note: Number of volunteers (missionaries and volunteers who did not become missionaries) by state. Data is sourced from the Student Volunteer Movement records. The number of missionaries per million residents is calculated using the 1890 state population. State address is available for 33,879 out of 37,106 volunteers.

Table A.4: Number of missionaries by state

| State                | Missionaries | State                | Missionaries per million |
|----------------------|--------------|----------------------|--------------------------|
| PENNSYLVANIA         | 976          | OKLAHOMA             | 1261.442                 |
| ILLINOIS             | 940          | CALIFORNIA           | 451.1104                 |
| OHIO                 | 930          | WASHINGTON           | 403.5605                 |
| NEW YORK             | 822          | IDAHO                | 379.2143                 |
| IOWA                 | 656          | KANSAS               | 346.1575                 |
| CALIFORNIA           | 545          | OREGON               | 344.2045                 |
| KANSAS               | 494          | IOWA                 | 343.1149                 |
| MASSACHUSETTS        | 453          | SOUTH DAKOTA         | 334.5417                 |
| MICHIGAN             | 433          | NORTH DAKOTA         | 317.4273                 |
| INDIANA              | 349          | NEBRASKA             | 283.3102                 |
| MISSOURI             | 340          | COLORADO             | 276.5661                 |
| MINNESOTA            | 339          | MINNESOTA            | 260.4035                 |
| NEBRASKA             | 300          | OHIO                 | 253.2462                 |
| TEXAS                | 268          | ILLINOIS             | 245.6649                 |
| NEW JERSEY           | 265          | MICHIGAN             | 206.7922                 |
| VIRGINIA             | 256          | MASSACHUSETTS        | 202.3276                 |
| WISCONSIN            | 224          | VERMONT              | 195.5346                 |
| NORTH CAROLINA       | 205          | CONNECTICUT          | 186.2627                 |
| SOUTH CAROLINA       | 177          | PENNSYLVANIA         | 185.6214                 |
| KENTUCKY             | 168          | NEW JERSEY           | 183.3995                 |
| GEORGIA              | 166          | INDIANA              | 159.186                  |
| TENNESSEE            | 152          | ALASKA               | 155.9965                 |
| WASHINGTON           | 141          | VIRGINIA             | 154.5912                 |
| CONNECTICUT          | 139          | SOUTH CAROLINA       | 153.7594                 |
| COLORADO             | 114          | NEW YORK             | 137.049                  |
| SOUTH DAKOTA         | 110          | WISCONSIN            | 132.7895                 |
| OREGON               | 108          | DISTRICT OF COLUMBIA | 130.2129                 |
| ALABAMA              | 93           | MISSOURI             | 126.9043                 |
| MARYLAND             | 93           | NORTH CAROLINA       | 126.7038                 |
| MAINE                | 80           | MONTANA              | 121.0663                 |
| OKLAHOMA             | 78           | MAINE                | 121.013                  |
| WEST VIRGINIA        | 76           | TEXAS                | 119.8825                 |
| MISSISSIPPI          | 65           | WYOMING              | 115.3118                 |
| VERMONT              | 65           | WEST VIRGINIA        | 99.63372                 |
| NORTH DAKOTA         | 58           | NEW HAMPSHIRE        | 95.60991                 |
| ARKANSAS             | 57           | RHODE ISLAND         | 95.51209                 |
| NEW HAMPSHIRE        | 36           | KENTUCKY             | 90.38891                 |
| FLORIDA              | 34           | GEORGIA              | 90.34736                 |
| RHODE ISLAND         | 33           | MARYLAND             | 89.21805                 |
| IDAHO                | 32           | FLORIDA              | 86.86277                 |
| DISTRICT OF COLUMBIA | 30           | TENNESSEE            | 85.9963                  |
| LOUISIANA            | 17           | ARIZONA              | 83.86448                 |
| MONTANA              | 16           | ALABAMA              | 61.46659                 |
| NEW MEXICO           | 9            | NEW MEXICO           | 58.59642                 |
| WYOMING              | 7            | ARKANSAS             | 50.5239                  |
| UTAH                 | 6            | MISSISSIPPI          | 50.40323                 |
| ALASKA               | 5            | HAWAII               | 44.44938                 |
| ARIZONA              | 5            | DELAWARE             | 29.67482                 |
| DELAWARE             | 5            | UTAH                 | 28.85933                 |
| HAWAII               | 4            | LOUISIANA            | 15.19775                 |
| NEVADA               | 0            | NEVADA               | 0                        |

Note: Number of missionaries by state. Data is sourced from the Student Volunteer Movement records. The number of missionaries per million residents is calculated using the 1890 state population. State address is available for 10,974 out of 12,265 missionaries.

Table A.5: Number of volunteers by denomination

| Denomination                                               | Volunteers | Denomination                                               | Volunteers per million |
|------------------------------------------------------------|------------|------------------------------------------------------------|------------------------|
| Methodist Episcopal Church                                 | 9547       | Dunkard (Old Order)                                        | 109499                 |
| Presbyterian Church in the United State of America (North) | 4927       | Christian Missionary Association                           | 26525.2                |
| Regular Baptist (North) Church                             | 4101       | Mennonite Church                                           | 15809.81               |
| Congregationalist Church                                   | 2679       | Free Methodist Church                                      | 11935.81               |
| Regular Baptist (South) Church                             | 1633       | Associate Reformed Synod of the South                      | 8940.125               |
| Presbyterian Church in the United States (Southern)        | 1291       | Christian (Christian Connection) Church                    | 8840.583               |
| Disciples of Christ Church                                 | 1142       | Reformed Presbyterian Church in the United States (Synod)  | 8416.872               |
| Christian (Christian Connection) Church                    | 802        | United Presbyterian Church                                 | 8124.828               |
| United Presbyterian Church                                 | 767        | Friends (Orthodox) Church                                  | 7844.117               |
| Methodist Episcopal Church (South)                         | 688        | Presbyterian Church in the United States (Southern)        | 7232.655               |
| Friends (Orthodox) Church                                  | 628        | Wesleyan Methodist Connection of America Church            | 6801.069               |
| United Brethren in Christ Church                           | 601        | Presbyterian Church in the United State of America (North) | 6309.783               |
| Protestant Episcopal Church                                | 587        | Congregationalist Church                                   | 5240.672               |
| Lutheran Synodical Conference Church                       | 550        | Christian Reformed Church                                  | 5212.51                |
| Dunkard (Old Order)                                        | 483        | Regular Baptist (North) Church                             | 5145.894               |
| Evangelical Association Church                             | 426        | Reformed Church in America                                 | 4539.099               |
| Reformed Church in America                                 | 422        | Methodist Episcopal Church                                 | 4283.379               |
| Free Methodist Church                                      | 270        | Seventh-day Adventist                                      | 4187.576               |
| Mennonite Church                                           | 270        | Moravian Church                                            | 3331.909               |
| Reformed Church in the United States                       | 207        | Evangelical Association Church                             | 3196.207               |
| Freewill Baptist Church                                    | 175        | United Brethren in Christ Church                           | 2964.973               |
| United Norwegian Lutheran Church of America                | 164        | Mennonite Brethren in Christ                               | 2695.418               |
| Cumberland Presbyterian Church                             | 161        | Plymouth Brethren                                          | 2554.47                |
| Methodist Protestant Church                                | 127        | German Augsburg Synod (Lutheran)                           | 2425.107               |
| Seventh-day Adventist                                      | 121        | United Synod in the South (Lutheran) Church                | 2189.177               |
| Wesleyan Methodist Connection of America Church            | 112        | Evangelical Missionary Church                              | 2103.049               |
| German Evangelical Synod of North America Church           | 95         | Freewill Baptist Church                                    | 1990.944               |
| Reformed Presbyterian Church in the United States (Synod)  | 89         | Disciples of Christ Church                                 | 1790.089               |
| United Synod in the South (Lutheran) Church                | 82         | Lutheran Synodical Conference Church                       | 1541.743               |
| Associate Reformed Synod of the South                      | 76         | United Norwegian Lutheran Church of America                | 1367.886               |
| Christian Reformed Church                                  | 65         | Church of God                                              | 1290.323               |
| Moravian Church                                            | 39         | Regular Baptist (South) Church                             | 1289.231               |
| Church of God                                              | 28         | Protestant Episcopal Church                                | 1121.946               |
| General Council (Lutheran) Church                          | 23         | Seventh-day Baptist                                        | 1093.733               |
| Christian Missionary Association                           | 20         | Reformed Church in the United States                       | 1016.115               |
| Norwegian Lutheran Church in America                       | 18         | Cumberland Presbyterian Church                             | 983.4403               |
| German Augsburg Synod (Lutheran)                           | 17         | Methodist Protestant Church                                | 903.0662               |
| Plymouth Brethren                                          | 17         | Welsh Calvinistic Methodist Church                         | 864.6439               |
| African Methodist Episcopal Church                         | 12         | Methodist Episcopal Church (South)                         | 573.9264               |
| Advent Christian Church                                    | 11         | German Evangelical Synod of North America Church           | 506.8505               |
| Welsh Calvinistic Methodist Church                         | 11         | Amish Mennonite Church                                     | 495.0005               |
| Seventh-day Baptist                                        | 10         | Advent Christian Church                                    | 426.0923               |
| Amish Mennonite Church                                     | 5          | Norwegian Lutheran Church in America                       | 324.6051               |
| Mennonite Brethren in Christ                               | 3          | General Council (Lutheran) Church                          | 70.9338                |
| Evangelical Missionary Church                              | 2          | Universalist Church                                        | 40.76142               |
| Universalist Church                                        | 2          | African Methodist Episcopal Church                         | 26.62188               |
| Joint Lutheran Synod of Ohio and Other States Church       | 1          | Joint Lutheran Synod of Ohio and Other States Church       | 14.41857               |

Note: Number of missionaries by denomination. Data is sourced from the Student Volunteer Movement records. Samples are restricted to denominations with at least one volunteer in the Student Volunteer Movement records. The number of volunteers per million for each denomination is calculated using the 1890 Census of Religious Bodies. Denominational affiliation is available for 33,507 out of 37,106 volunteers.

Table A.6: Number of missionaries by denomination

| Denomination                                               | Missionaries | Denomination                                               | Missionaries per million |
|------------------------------------------------------------|--------------|------------------------------------------------------------|--------------------------|
| Methodist Episcopal Church                                 | 2520         | Dunkard (Old Order)                                        | 29698.48                 |
| Presbyterian Church in the United State of America (North) | 1774         | Christian Missionary Association                           | 7957.56                  |
| Regular Baptist (North) Church                             | 1088         | Mennonite Church                                           | 4860.054                 |
| Congregationalist Church                                   | 922          | Reformed Presbyterian Church in the United States (Synod)  | 3782.864                 |
| Presbyterian Church in the United States (Southern)        | 378          | Free Methodist Church                                      | 3050.263                 |
| Regular Baptist (South) Church                             | 360          | Associate Reformed Synod of the South                      | 2940.83                  |
| Disciples of Christ Church                                 | 296          | United Presbyterian Church                                 | 2828.33                  |
| United Presbyterian Church                                 | 267          | Christian (Christian Connection) Church                    | 2381.005                 |
| Christian (Christian Connection) Church                    | 216          | Presbyterian Church in the United State of America (North) | 2271.88                  |
| Methodist Episcopal Church (South)                         | 193          | Friends (Orthodox) Church                                  | 2148.389                 |
| Protestant Episcopal Church                                | 179          | Presbyterian Church in the United States (Southern)        | 2117.695                 |
| Friends (Orthodox) Church                                  | 172          | Congregationalist Church                                   | 1803.621                 |
| United Brethren in Christ Church                           | 172          | Wesleyan Methodist Connection of America Church            | 1760.991                 |
| Reformed Church in America                                 | 161          | Reformed Church in America                                 | 1731.741                 |
| Lutheran Synodical Conference Church                       | 150          | Christian Reformed Church                                  | 1523.657                 |
| Dunkard (Old Order)                                        | 131          | Regular Baptist (North) Church                             | 1365.212                 |
| Evangelical Association Church                             | 112          | Seventh-day Adventist                                      | 1280.498                 |
| Mennonite Church                                           | 83           | Plymouth Brethren                                          | 1202.104                 |
| Free Methodist Church                                      | 69           | Methodist Episcopal Church                                 | 1130.629                 |
| Reformed Church in the United States                       | 68           | Moravian Church                                            | 1110.636                 |
| United Norwegian Lutheran Church of America                | 60           | United Brethren in Christ Church                           | 848.5446                 |
| Freewill Baptist Church                                    | 43           | Evangelical Association Church                             | 840.3172                 |
| Reformed Presbyterian Church in the United States (Synod)  | 40           | Church of God                                              | 691.2442                 |
| Cumberland Presbyterian Church                             | 38           | United Synod in the South (Lutheran) Church                | 640.7347                 |
| Seventh-day Adventist                                      | 37           | German Augsburg Synod (Lutheran)                           | 570.6134                 |
| Wesleyan Methodist Connection of America Church            | 29           | United Norwegian Lutheran Church of America                | 500.4462                 |
| Methodist Protestant Church                                | 27           | Freewill Baptist Church                                    | 489.2034                 |
| German Evangelical Synod of North America Church           | 27           | Disciples of Christ Church                                 | 463.9811                 |
| Associate Reformed Synod of the South                      | 25           | Lutheran Synodical Conference Church                       | 420.4755                 |
| United Synod in the South (Lutheran) Church                | 24           | Protestant Episcopal Church                                | 342.1267                 |
| Christian Reformed Church                                  | 19           | Reformed Church in the United States                       | 333.7964                 |
| Church of God                                              | 15           | Amish Mennonite Church                                     | 297.0003                 |
| Moravian Church                                            | 13           | Regular Baptist (South) Church                             | 284.2149                 |
| General Council (Lutheran) Church                          | 9            | Welsh Calvinistic Methodist Church                         | 235.812                  |
| Plymouth Brethren                                          | 8            | Cumberland Presbyterian Church                             | 232.1164                 |
| Christian Missionary Association                           | 6            | Seventh-day Baptist                                        | 218.7466                 |
| Advent Christian Church                                    | 5            | Advent Christian Church                                    | 193.6783                 |
| German Augsburg Synod (Lutheran)                           | 4            | Methodist Protestant Church                                | 191.9904                 |
| Norwegian Lutheran Church in America                       | 4            | Methodist Episcopal Church (South)                         | 160.9997                 |
| Amish Mennonite Church                                     | 3            | German Evangelical Synod of North America Church           | 144.0522                 |
| Welsh Calvinistic Methodist Church                         | 3            | Norwegian Lutheran Church in America                       | 72.13446                 |
| Seventh-day Baptist                                        | 2            | General Council (Lutheran) Church                          | 27.7567                  |
| African Methodist Episcopal Church                         | 2            | Joint Lutheran Synod of Ohio and Other States Church       | 14.41857                 |
| Joint Lutheran Synod of Ohio and Other States Church       | 1            | African Methodist Episcopal Church                         | 4.43698                  |
| Evangelical Missionary Church                              | 0            | Evangelical Missionary Church                              | 0                        |
| Mennonite Brethren in Christ                               | 0            | Mennonite Brethren in Christ                               | 0                        |
| Universalist Church                                        | 0            | Universalist Church                                        | 0                        |

Note: Number of missionaries by denomination. Data is sourced from the Student Volunteer Movement records. Samples are restricted to denominations with at least one volunteer in the Student Volunteer Movement records. The number of missionaries per million for each denomination is calculated using the 1890 Census of Religious Bodies. Denominational affiliation is available for 9,755 out of 12,265 missionaries.

Table A.7: Top 50 destination list

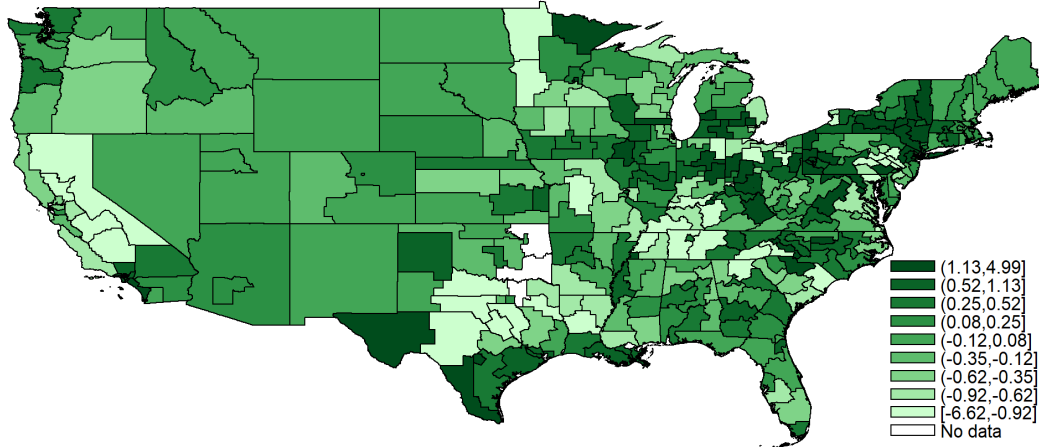
| destination   | count | destination   | count |
|---------------|-------|---------------|-------|
| China         | 2497  | Ceylon        | 19    |
| India         | 1452  | United States | 16    |
| Africa        | 785   | Singapore     | 16    |
| Japan         | 635   | Micronesia    | 15    |
| South America | 492   | Palestine     | 15    |
| Korea         | 322   | Jamaica       | 13    |
| Mexico        | 230   | Europe        | 12    |
| Philippines   | 196   | Madagascar    | 11    |
| Turkey        | 186   | Java          | 10    |
| Burma         | 165   | Peru          | 10    |
| Egypt         | 107   | Panama        | 9     |
| Persia        | 99    | Gabon         | 8     |
| Syria         | 84    | Italy         | 7     |
| Brazil        | 77    | Uruguay       | 7     |
| Cuba          | 74    | Bolivia       | 7     |
| West Indies   | 74    | Russia        | 7     |
| Siam          | 71    | France        | 6     |
| Alaska        | 70    | Sierra Leone  | 6     |
| Malaysia      | 55    | Tibet         | 5     |
| Hawaii        | 53    | Cyprus        | 5     |
| Puerto Rico   | 52    | Belgium       | 5     |
| Chile         | 37    | Colombia      | 5     |
| Saudi Arabia  | 26    | Venezuela     | 5     |
| Laos          | 23    | Bulgaria      | 5     |
| Argentina     | 23    | Congo         | 4     |

Note: Number of missionaries by destination: Destination data is available for 8,195 out of 12,265 missionaries. This list highlights the top 50 most frequent destinations, based on records from the Student Volunteer Movement.

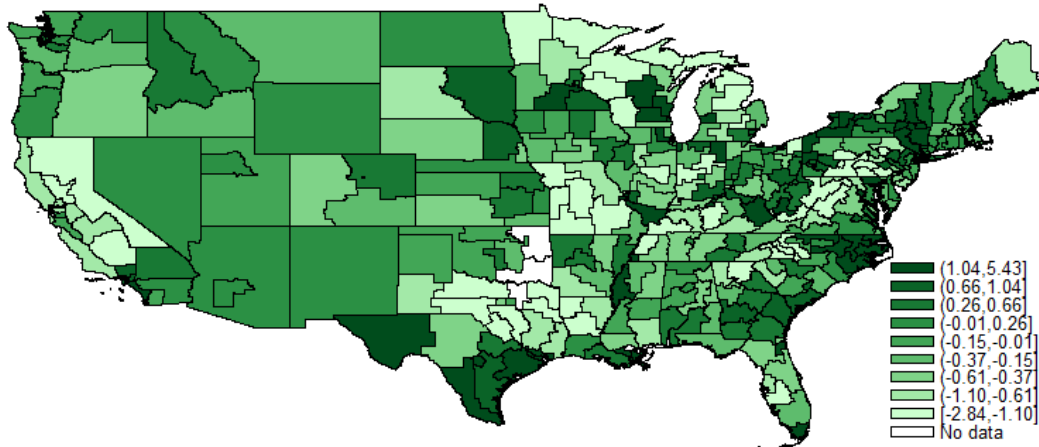
## A.4 Treatment and instrumental variables

Figure A.3: Treatment and instrumental variables

(a) Exposure to missionaries



(b) Exposure to travel secretaries, 1886-87

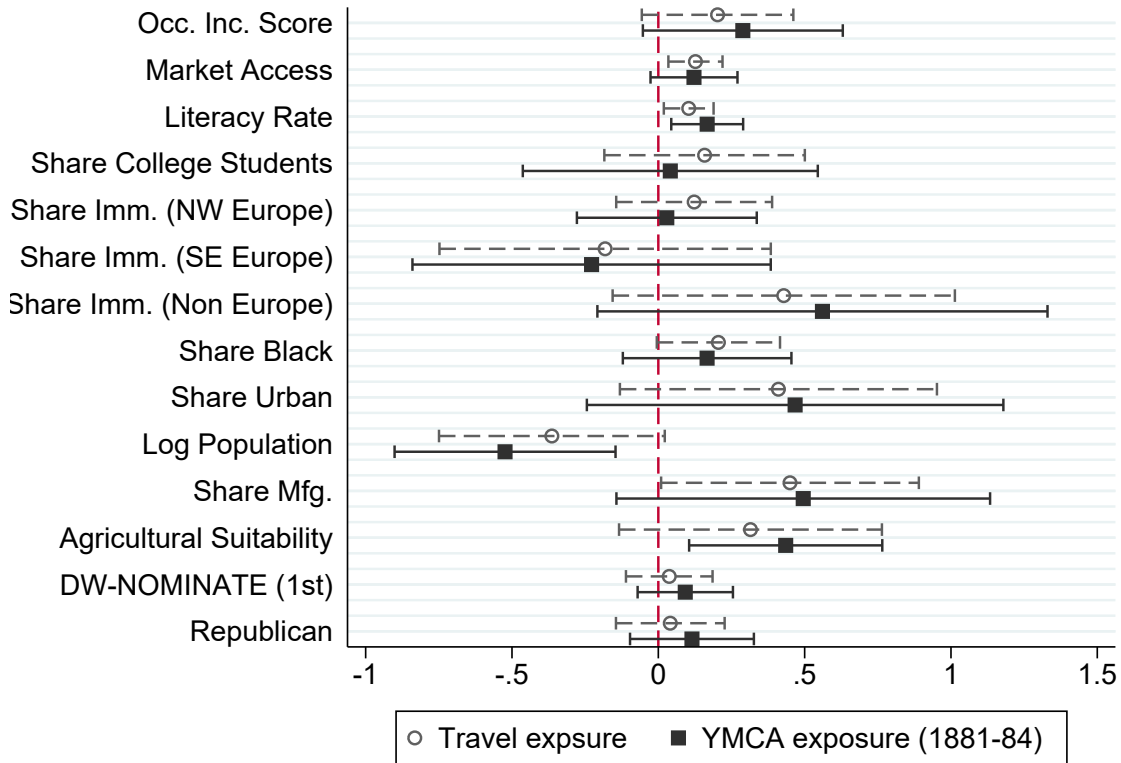


Note: Exposure to missionaries and exposure to travel secretaries are standardized after controlling for state fixed effects. Polygons indicate congressional districts in the 87th Congress.



## A.5 Balance in exposure to the instrumental variables

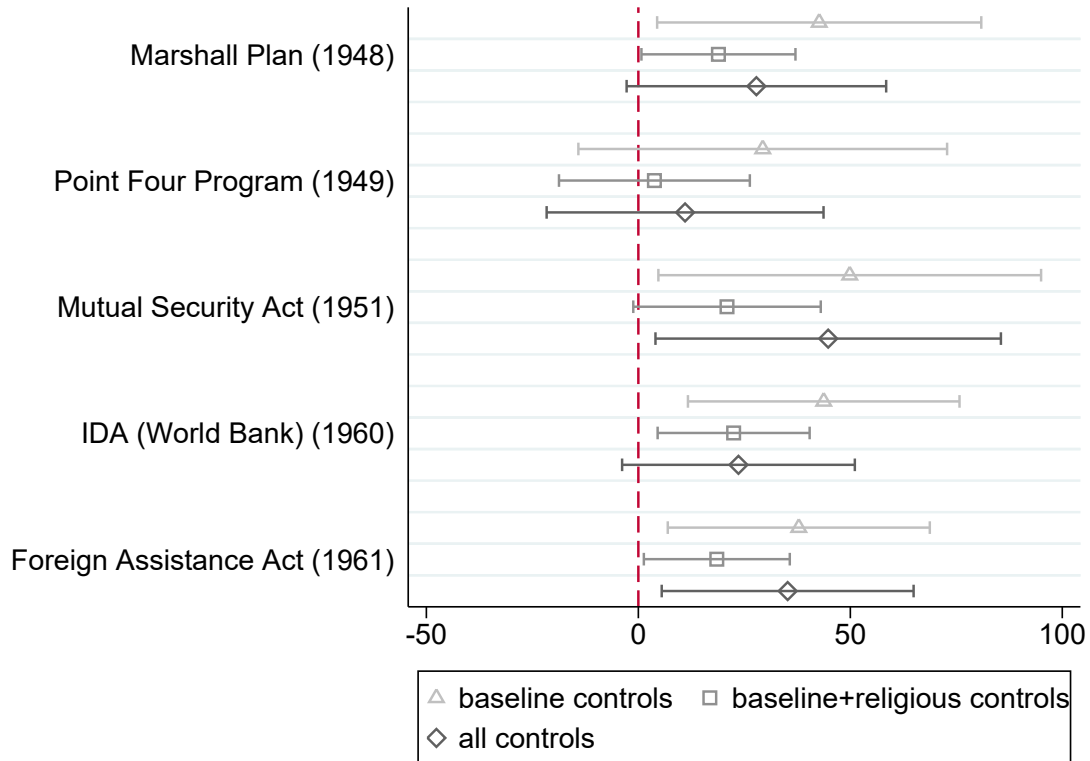
Figure A.4: Balance in exposures



Note: Unit of analysis is a congressional district by congressional terms. The circles represent coefficients from regressions of congressional district covariates on travel exposure, controlling for baseline YMCA exposure. Square markers indicate coefficients from regressions of congressional district covariates on YMCA exposure in 1881-84, controlling for YMCA exposure in 1889 (as explained in Section 4.5). All regressions include state fixed effects, time fixed effects, and the share of missionary denominations. Error bars indicate 95 percent confidence intervals, and standard errors are clustered at the state level.

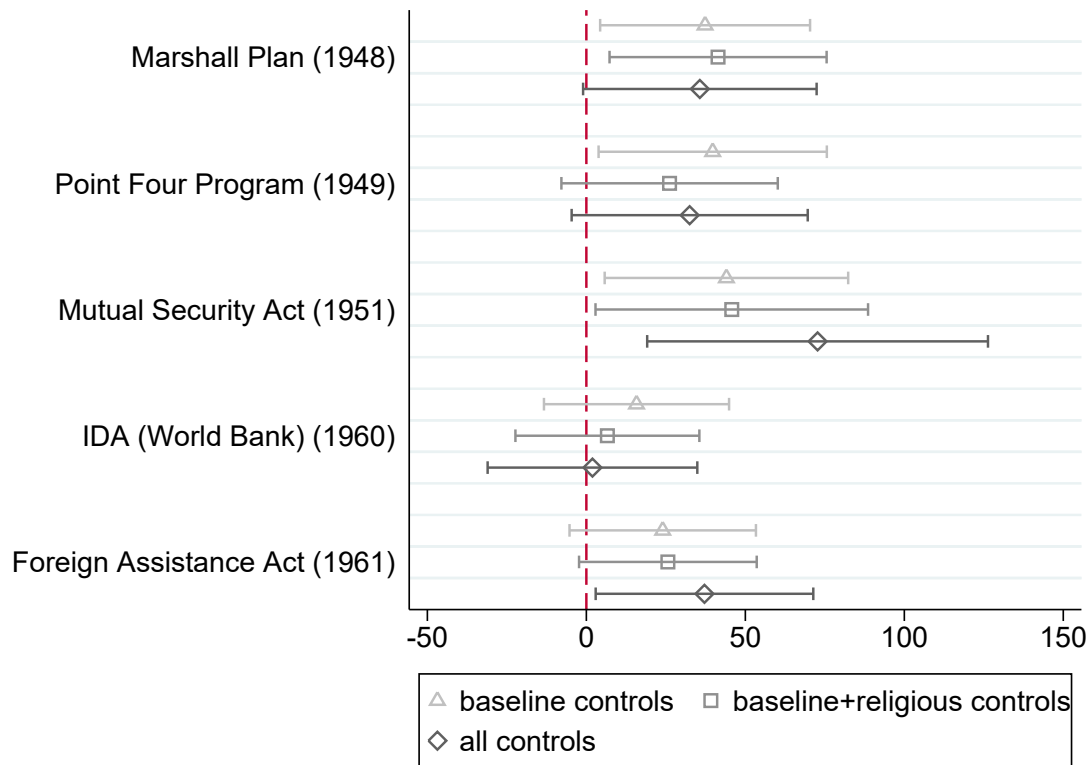
## A.6 Effects on congressional votes: Bill-specific estimates

Figure A.5: Effects on foreign aid legislation: Travel exposure as an IV



Note: Unit of analysis is a congressional district. Coefficients from IV regression results of a Congressman vote “Yea” for the foreign aid bill on the exposure to missionaries instrumented by the travel exposure. Core controls include the YMCA controls, share of SVM denominations, Republican party fixed effects, DW-NOMINATE score (first dimension), latitude, longitude, and state fixed effects. Socioeconomic controls include log population, urbanization rate, occupational income score, agricultural land suitability, market access, literacy rate, and the shares of college students, manufacturing workers, Northern and Western European immigrants, Eastern and Southern European immigrants, non-European immigrants, African Americans. Religious controls include the shares of missionaries, evangelical Protestants, mainline Protestants, Catholics, Orthodox Jews, and Mormons. Error bars indicate 95 percent confidence intervals. Standard errors are robust to heteroscedasticity.

Figure A.6: Effects on foreign aid legislation: YMCA exposure as an IV



Note: Unit of analysis is a congressional district. Coefficients from IV regression results of a Congressman vote “Yea” for the foreign aid bill on the exposure to missionaries instrumented by YMCA exposure in 1881 and 1884. The instrumental variable is explained in Section 4.5. Core controls include the YMCA exposure in 1889, share of SVM denominations, Republican party fixed effects, DW-NOMINATE score (first dimension), latitude, longitude, and state fixed effects. Socioeconomic controls include log population, urbanization rate, occupational income score, agricultural land suitability, market access, literacy rate, and the shares of college students, manufacturing workers, Northern and Western European immigrants, Eastern and Southern European immigrants, non-European immigrants, African Americans. Religious controls include the shares of missionaries, evangelical Protestants, mainline Protestants, Catholics, Orthodox Jews, and Mormons. Error bars indicate 95 percent confidence intervals. Standard errors are robust to heteroscedasticity.

## A.7 Congressional vote: Excluding districts with YMCA chapters

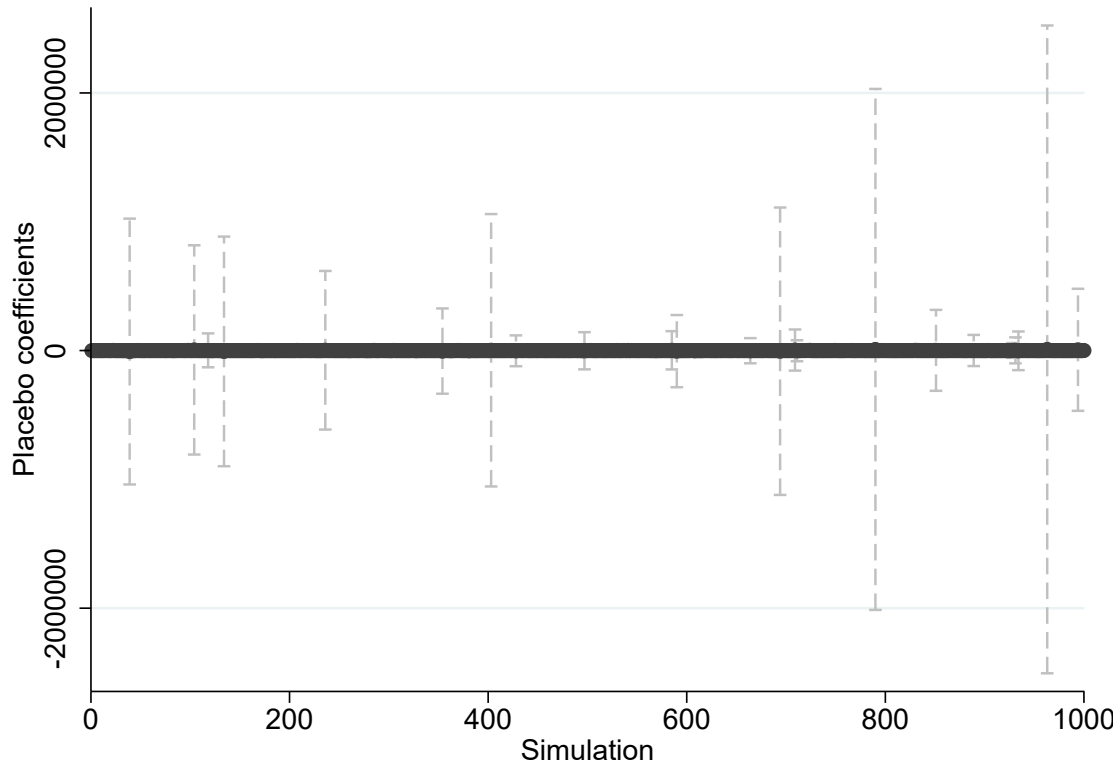
Table A.8: Landmark legislation on foreign aid: Excluding districts with YMCA chapters

| DV: D[Pro foreign aid vote]×100 | (1)            | (2)              | (3)                | (4)               | (5)                | (6)                | (7)                |
|---------------------------------|----------------|------------------|--------------------|-------------------|--------------------|--------------------|--------------------|
| Mean DV: 66.33                  |                |                  |                    |                   |                    |                    |                    |
| Exposure to missionaries (std)  | 2.84<br>(2.98) | 8.06**<br>(3.36) | 38.98**<br>(19.38) | 34.55*<br>(20.32) | 36.23**<br>(17.40) | 26.59**<br>(11.82) | 22.39**<br>(10.99) |
| Observations                    | 1,678          | 1,678            | 1,678              | 1,678             | 1,678              | 1,678              | 1,678              |
| R-squared                       | 0.21           | 0.36             | 0.13               | 0.15              | 0.29               | 0.34               | 0.36               |
| Core controls                   | Y              | Y                | Y                  | Y                 | Y                  | Y                  | Y                  |
| Baseline YMCA                   |                | Y                |                    | Y                 | Y                  | Y                  | Y                  |
| Political controls              |                | Y                |                    |                   | Y                  | Y                  | Y                  |
| Religious controls              |                | Y                |                    |                   |                    | Y                  | Y                  |
| Socioeconomic controls          |                |                  |                    |                   |                    |                    | Y                  |
| AR weak IV [p-value]            |                |                  | 0.01               | 0.01              | 0.00               | 0.00               | 0.01               |
| KP F-stat                       |                |                  | 16.66              | 10.56             | 10.99              | 26.74              | 24.51              |

Note: Unit of analysis is a congressional district over five Congresses (80th, 81st, 82nd, 86th, and 87th). Regressions of a pro-foreign aid vote on the exposure to missionaries instrumented by the exposure to travel secretaries. Core controls include the share of denominations involved in the Student Volunteer Movement, latitude, longitude, state fixed effects, and congress fixed effects. Baseline YMCA exposure is denominational exposure to YMCA chapters in counties with YMCA chapters and without travel secretaries' visit. Political controls include DW-NOMINATE score (first dimension) and Republican Party fixed effects. Religious controls include the shares of evangelical Protestants, mainline Protestants, Catholics, Orthodox Jews, and Mormons. Socioeconomic controls include log population, urbanization rate, occupational income score, agricultural land suitability, market access, literacy rate, and the shares of college students, manufacturing workers, Northern and Western European immigrants, Eastern and Southern European immigrants, non-European immigrants, and African Americans. Robust standard errors clustered by state in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

## A.8 Randomly generated shift

Figure A.7: Randomly generated shift



Note: Unit of analysis is a congressional district by five congressional terms (80th, 81st, 82nd, 86th, and 87th). The sample excludes congressional districts that include local YMCA chapters between 1881 and 1890 and districts visited by the travel secretaries of the Student Volunteer Movement. Instrumental variables regressions of the pro foreign aid vote on the share of exposure to missionaries. The independent variable is instrumented using a shift-share instrument from randomly generated shift. Specifically, The instrument is constructed by interacting denominational share of county population with a random shift (with mean zero and variance one). The figure shows the coefficients and 95 percent confidence intervals from instrumental variables regressions where the instrument was generated with 1,000 random shifts. Regressions include the set of all YMCA, baseline, socioeconomic, and religious controls in Table 2 and cluster standard errors by state.

## A.9 Robustness to alternative standard errors

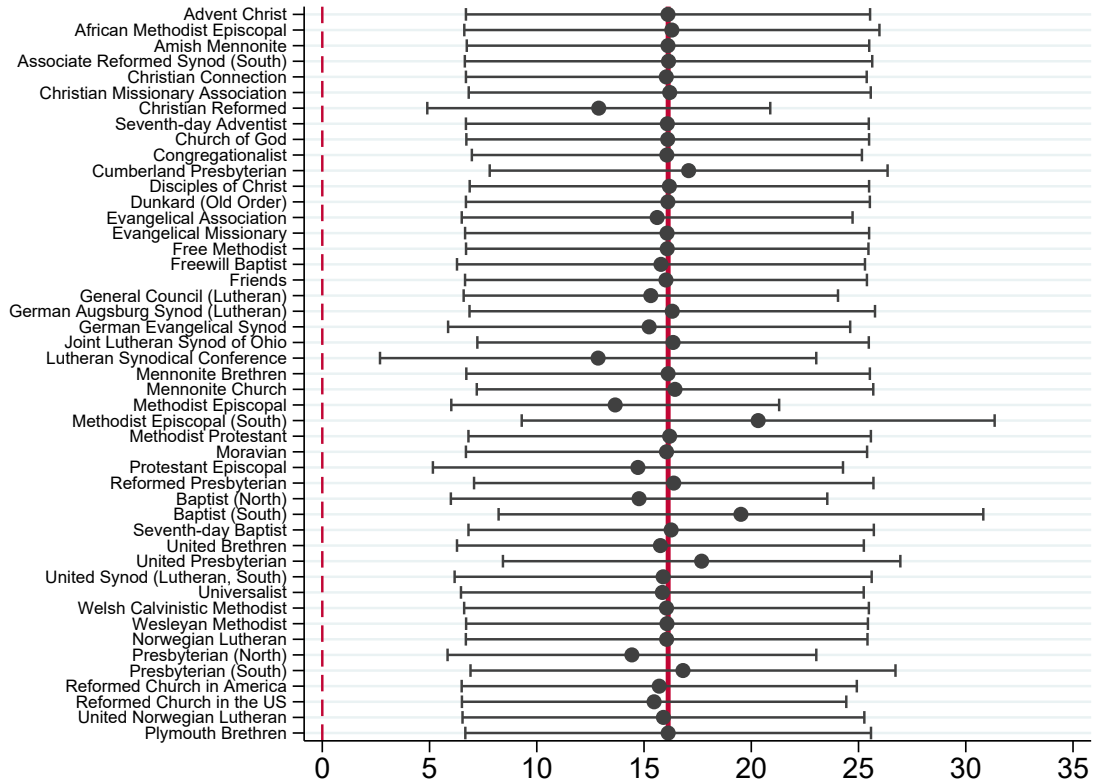
Table A.9: Robustness to alternative standard errors

| DV: D(Pro foreign aid vote)×100            | (1)    | (2)    | (3)    | (4)    | (5)    |
|--------------------------------------------|--------|--------|--------|--------|--------|
| Travel exposure (std.)                     | 19.14  | 19.33  | 21.71  | 20.37  | 15.91  |
| Standard errors:                           |        |        |        |        |        |
| Clustered by state                         | (5.22) | (5.72) | (4.75) | (5.35) | (5.14) |
| Adao <i>et al.</i> (2019)                  | (3.92) | (3.94) | (3.19) | (3.13) | (3.27) |
| Borusyak <i>et al.</i> (2022) <sup>+</sup> | (5.59) | (5.86) | (3.73) | (3.69) | (4.32) |
| Conley (1999) (100km)                      | (4.01) | (4.20) | (4.42) | (4.17) | (4.48) |
| Conley (1999) (300km)                      | (3.63) | (3.74) | (4.25) | (4.80) | (4.16) |
| Conley (1999) (500km)                      | (4.27) | (4.38) | (4.99) | (4.98) | (4.98) |
| Observations                               | 1,951  | 1,951  | 1,951  | 1,951  | 1,951  |
| R-squared                                  | 0.22   | 0.22   | 0.38   | 0.38   | 0.39   |
| Core controls                              | Y      | Y      | Y      | Y      | Y      |
| Baseline YMCA                              |        | Y      | Y      | Y      | Y      |
| Political controls                         |        |        | Y      | Y      | Y      |
| Religious controls                         |        |        |        | Y      | Y      |
| Socioeconomic controls                     |        |        |        |        | Y      |

+Number of observations for Borusyak *et al.* (2022)'s standard error is 235 since the unit of analysis is denomination by congressional terms ( $235 = 47 \times 5$ ). Standard errors clustered by denomination. Note: \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

## A.10 Leave one denomination out

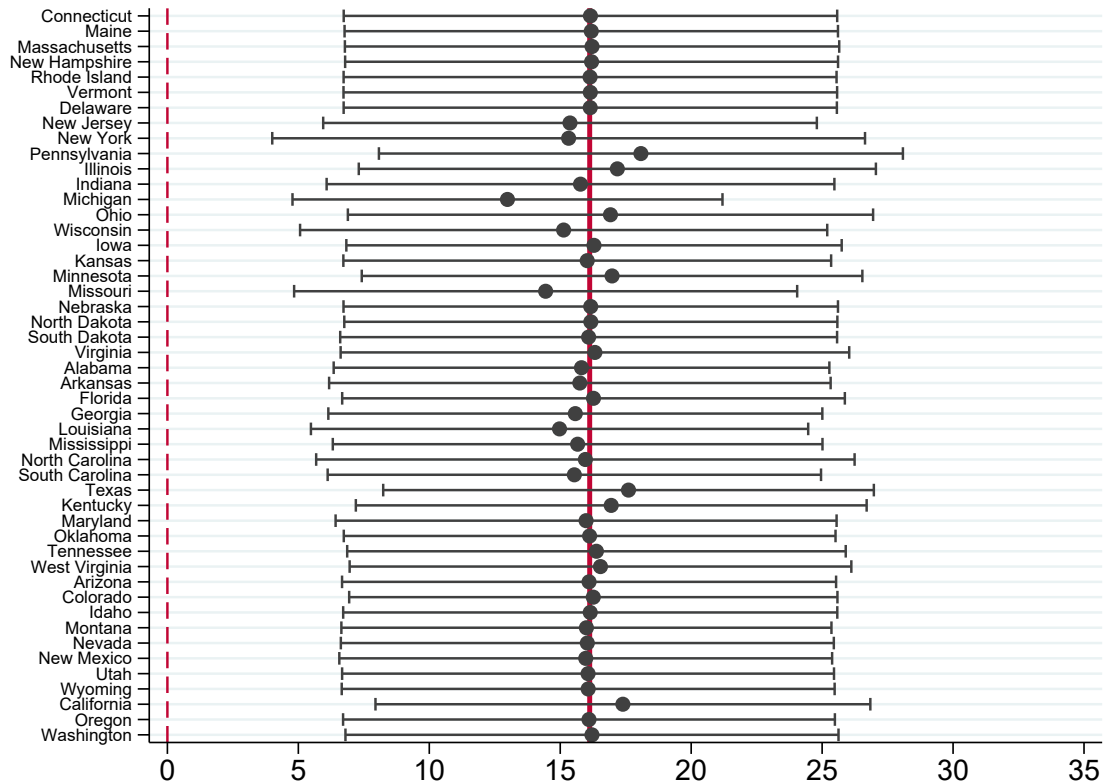
Figure A.8: Leave one denomination out estimator



Note: Unit of analysis is a congressional district by congressional terms. Reduced form regressions of the foreign aid vote on the marginal exposure to YMCA in 1884. Regressions include the set of all YMCA, baseline, socioeconomic, and religious controls in Table 2 and cluster standard errors by state. Regressions with all 47 denominations are plotted in the solid red line. Regressions after leaving out one denomination is drawn with 95 percent confidence intervals.

## A.11 Leave one state out

Figure A.9: Leave one state out estimator

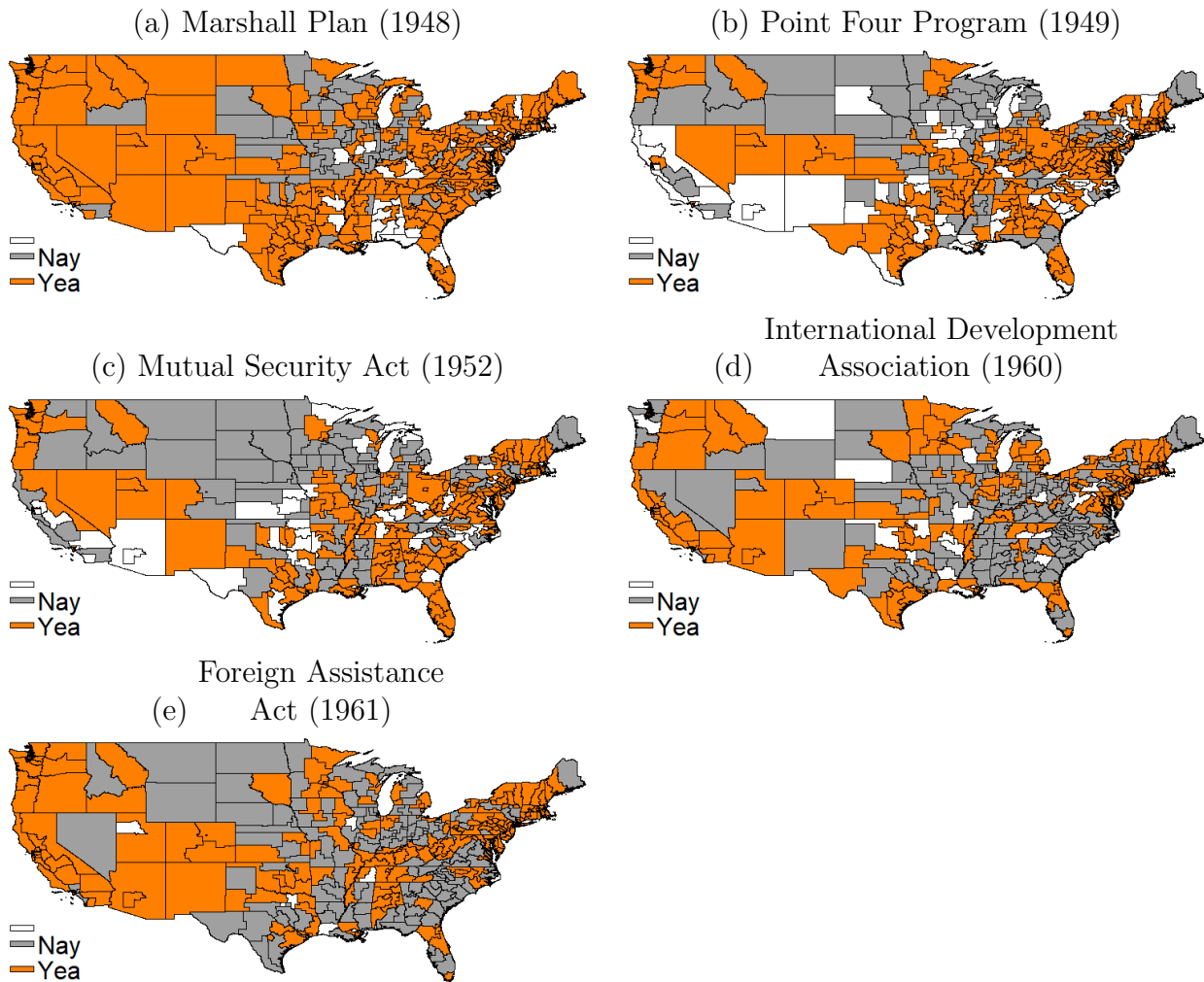


Note: Unit of analysis is a congressional district by congressional terms. Reduced form regressions of the foreign aid vote on the marginal exposure to YMCA in 1884. Regressions include the set of all YMCA, baseline, socioeconomic, and religious controls in Table 2 and cluster standard errors by state. Regressions with all 48 states are plotted in the solid red line. Regressions after leaving out one state is drawn with 95 percent confidence intervals.



## A.12 Congressional voting on foreign aid

Figure A.10: Congressional voting on foreign aid



Note: Congressional voting on key foreign aid legislation. Congressional districts that voted 'Yea' on the passage of foreign aid are highlighted in orange, while those that voted 'Nay' are shown in gray. Districts where no vote was recorded are left blank.

### A.13 Publication

Table A.10 presents both the selection into volunteering and the selection into becoming missionaries, conditional on being volunteers. Column (1) compares volunteers to the general population, revealing several differences. First, people who applied to become missionaries are more likely to have parents with higher incomes and in religious occupations<sup>4</sup>. Second, applicants are less likely to be foreign-born or have foreign-born parents. Third, volunteers are more likely to live in owned houses, be white, attend school, demonstrate higher literacy rates, more likely to have both parents present, and less likely to live in farm households.

Column (4) shows that many of the previously observed differences diminish when the sample is restricted to volunteers, comparing missionaries to non-missionaries within this group. However, some significant distinctions remain. Missionaries are more likely to have parents in religious occupations, even among volunteers. Notably, missionaries are also more likely to be foreign-born compared to volunteers who did not embark on missions. Yet, as column (5) indicates, the vast majority were white, suggesting they were primarily European immigrants with limited exposure to non-European regions. It is reassuring that missionaries and non-missionary volunteers do not differ significantly in terms of parental income, race, or school attendance.

In columns (1) to (8) of A.11, I categorize the ProQuest dissertation data into four groups: social sciences, philosophy & religion, STEM, and language. A dissertation is classified as social sciences if the `category` variable in the ProQuest database is assigned as "social sciences". Similarly, "philosophy & religion" is assigned when the category is "philosophy", and STEM when it is either "Applied sciences", "Biological sciences", "Earth sciences", "Health and environmental sciences", or "Pure sciences". The "language" category is assigned when the category is "language".

In the OpenAlex bibliometric data, each publication output is assigned 'concepts' based on an automated classifier trained on Microsoft Academic Graph's corpus.<sup>5</sup> Using this variable, I classified foreign-related publications as social science if any of the concepts include the following terms: agriculture, demography, development, economics, history, political science, or sociology. A foreign-subject publication is categorized as medicine and nursing if the concepts include bacteria, immunology, medicine, nursing, pathology, or virology. Publications are classified as religion if the concepts include religion, Christianity, or theology. These keywords were chosen based on my manual inspection of frequent concepts in

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<sup>4</sup>Religious occupations are defined as household heads with occ1950 codes for clergymen (009) or religious workers (078).

<sup>5</sup><https://docs.openalex.org/api-entities/concepts>

foreign-related publication outputs published between 1880-1945.

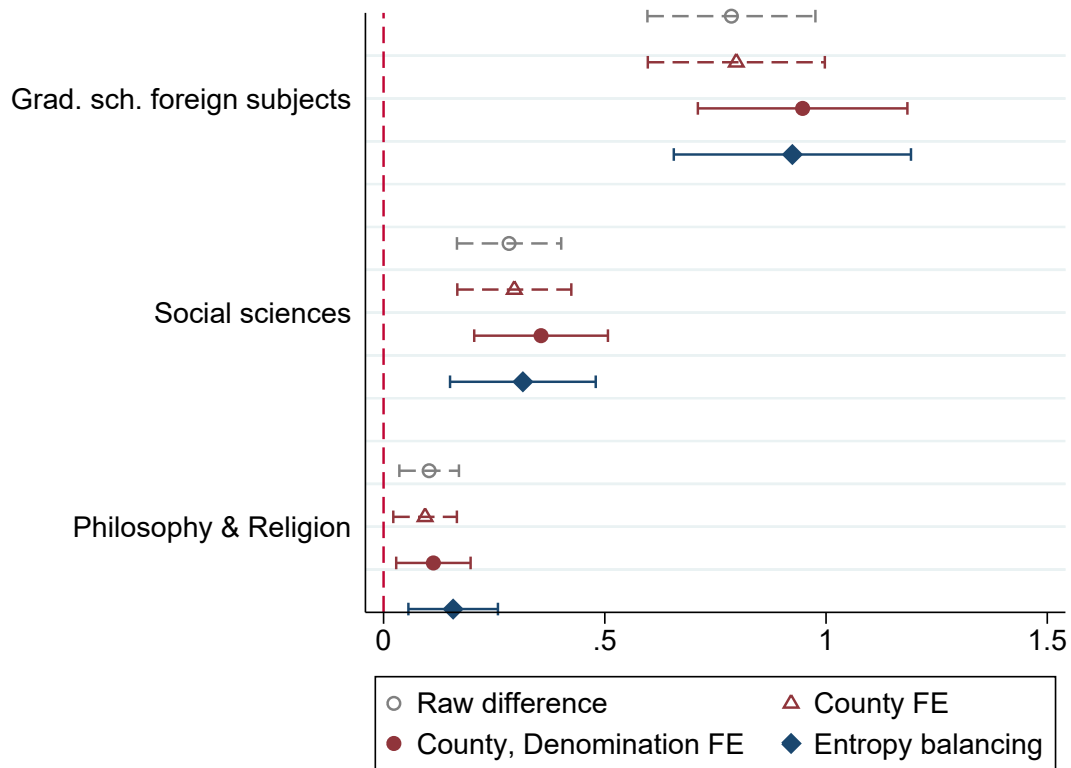
The results in Figure A.11 and Table A.11 indicate that the effect on graduate dissertations and publications about foreign subjects was primarily driven by output related to social sciences or topics relevant to economic development. In contrast, the probability of pursuing graduate studies in STEM-related fields or language was statistically indistinguishable from zero.

Table A.10: Balance checks: volunteers and missionaries

|                            | volunteers – general population |         |            | missionaries – volunteers |         |        | age     |
|----------------------------|---------------------------------|---------|------------|---------------------------|---------|--------|---------|
|                            | diff (s.e.)                     | control | N          | diff (s.e.)               | control | N      |         |
|                            |                                 | mean    |            |                           | mean    |        |         |
|                            | (1)                             | (2)     | (3)        | (4)                       | (5)     | (6)    | (7)     |
| Parental occ. score (\$00) | 2.10 (0.10)***                  | 18.85   | 65,816,681 | 0.01 (0.19)               | 20.69   | 27,532 | [0,20)  |
| Parent religious occ.      | 5.54 (0.18)***                  | 0.42    | 65,816,681 | 1.86 (0.36)***            | 5.48    | 27,532 | [0,20)  |
| Foreign born               | -1.02 (0.16)***                 | 2.87    | 65,816,681 | 0.38 (0.20)*              | 1.76    | 27,532 | [0,20)  |
| Foreign born parents       | -7.76 (0.44)***                 | 25.73   | 65,816,681 | 1.13 (0.56)**             | 18.26   | 27,532 | [0,20)  |
| City pop. (000s)           | -3.21 (5.49)                    | 282.05  | 65,816,681 | -0.84 (2.11)              | 131.22  | 27,532 | [0,20)  |
| Home ownership             | 8.39 (0.37)***                  | 46.70   | 65,816,681 | 1.13 (0.72)               | 60.45   | 27,532 | [0,20)  |
| Two parents                | 1.54 (0.21)***                  | 89.57   | 65,816,681 | 0.10 (0.41)               | 90.27   | 27,532 | [0,20)  |
| White                      | 4.95 (0.30)***                  | 87.45   | 65,816,681 | -0.23 (0.21)              | 97.11   | 27,532 | [0,20)  |
| Farm household             | -1.53 (0.36)***                 | 39.60   | 65,816,681 | -0.98 (0.66)              | 38.98   | 27,532 | [0,20)  |
| Employed                   | -8.00 (0.29)***                 | 21.41   | 25,522,632 | 0.36 (0.55)               | 12.51   | 14,084 | (10,20) |
| School attendance          | 14.65 (0.33)***                 | 64.82   | 25,522,632 | -0.42 (0.76)              | 82.62   | 14,084 | (10,20) |
| Literacy                   | 2.33 (0.16)***                  | 92.74   | 25,522,632 | -0.16 (0.20)              | 98.29   | 14,084 | (10,20) |

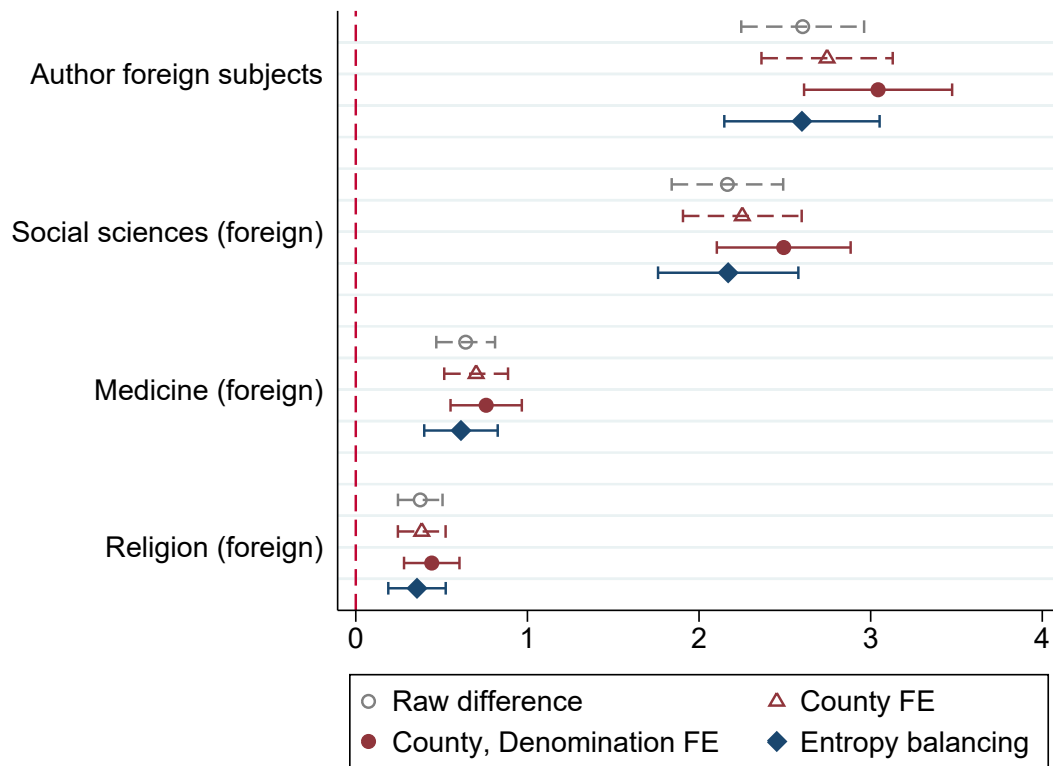
Note: Unit of analysis is an individual. Column (1) reports the differences between the general population and volunteers to the Student Volunteer Movement (including those who did not eventually become missionaries). Column (4) measures the differences between missionaries and non-missionaries, conditional on being volunteers. Both columns provide differences after controlling for census year (1900 and 1910), birth year, gender, and county fixed effects. Columns (2) and (5) indicate the mean of each variable for non-volunteers and non-missionaries. Columns (3) and (6) document the number of observations. The samples are drawn from archival records matched to the 1900 and 1910 censuses and are limited to individuals under 20 years old from family households. All variables, except for parental occupational score and city population, are indicator variables multiplied by 100. The parental occupational score reflects the median total income (in hundreds of 1950 dollars) of all individuals' parents with that occupation in 1950. City population is measured in thousands. Robust standard errors clustered by county in parentheses. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Figure A.11: Graduate dissertation by field



Note: The figure presents results from Equation 9 across different graduate school subfields. “Social Sciences” refers to graduate dissertation output on foreign subjects within the social sciences. “Philosophy & Religion” captures dissertation output on foreign subjects in philosophy or religion. Error bars indicate 95 percent confidence intervals.

Figure A.12: Publication by field



Note: The figure presents results from Equation 9 across different concepts. Each concepts are explained in Section A.13. Error bars indicate 95 percent confidence intervals.

Table A.11: Effects on publication by subfield

| Outcome:           | Grad. dissertation<br>social sciences      |         | Grad. dissertation<br>philosophy & religion   |         |
|--------------------|--------------------------------------------|---------|-----------------------------------------------|---------|
| (dummy×100)        | (1)                                        | (2)     | (3)                                           | (4)     |
| Mission            | 0.18*                                      | 0.44*** | 0.09***                                       | 0.11*** |
|                    | (0.10)                                     | (0.14)  | (0.03)                                        | (0.04)  |
| Observations       | 33,733                                     | 29,380  | 33,733                                        | 29,380  |
| R-squared          | 0.00                                       | 0.06    | 0.00                                          | 0.05    |
| Control group mean | 0.57                                       | 0.57    | 0.01                                          | 0.01    |
| Controls           | Y                                          |         | Y                                             |         |
| Outcome:           | Grad. dissertation<br>STEM                 |         | Grad. dissertation<br>language                |         |
| (dummy×100)        | (5)                                        | (6)     | (7)                                           | (8)     |
| Mission            | -0.26***                                   | -0.16   | -0.07                                         | -0.03   |
|                    | (0.10)                                     | (0.12)  | (0.06)                                        | (0.09)  |
| Observations       | 33,733                                     | 29,380  | 33,733                                        | 29,380  |
| R-squared          | 0.00                                       | 0.07    | 0.00                                          | 0.08    |
| Control group mean | 0.81                                       | 0.81    | 0.30                                          | 0.30    |
| Controls           | Y                                          |         | Y                                             |         |
| Outcome:           | Author foreign subjects<br>social sciences |         | Author foreign subjects<br>medicine & nursing |         |
| (dummy×100)        | (9)                                        | (10)    | (11)                                          | (12)    |
| Mission            | 2.01***                                    | 2.49*** | 0.64***                                       | 0.76*** |
|                    | (0.16)                                     | (0.20)  | (0.09)                                        | (0.12)  |
| Observations       | 33,733                                     | 29,380  | 33,733                                        | 29,380  |
| R-squared          | 0.01                                       | 0.09    | 0.00                                          | 0.06    |
| Control group mean | 0.54                                       | 0.54    | 0.08                                          | 0.08    |
| Controls           | Y                                          |         | Y                                             |         |

Note: Unit of analysis is an individual. Regressions of an indicator of publication output on an indicator of being missionaries conditional on being volunteers to the Student Volunteer Movement. Odd-numbered columns report raw differences, while even-numbered columns document differences after controlling for birth year, gender, county, and denomination fixed effects. Definition of outcome variables are available in Appendix A.13. Robust standard errors, clustered by county, are provided in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

## A.14 Heterogeneous effects by denomination

Table A.12: Effects on regional expertise during WWII

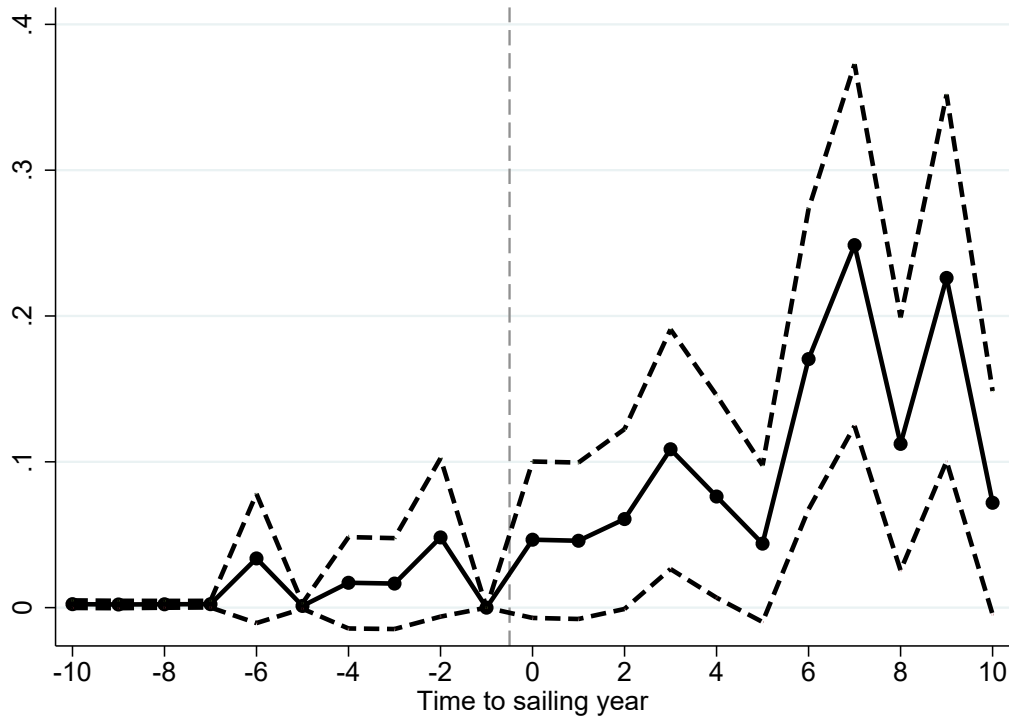
| DV:(dummy×100)              | Author foreign subjects |                   | Grad. dissertation<br>foreign subjects |                   | Grad. dissertation<br>social sciences |                   |
|-----------------------------|-------------------------|-------------------|----------------------------------------|-------------------|---------------------------------------|-------------------|
|                             | (1)                     | (2)               | (3)                                    | (4)               | (5)                                   | (6)               |
| Mission                     | 2.30***<br>(0.56)       | 2.49***<br>(0.36) | 1.56***<br>(0.40)                      | 1.35***<br>(0.25) | 0.67<br>(0.46)                        | 0.84***<br>(0.27) |
| Mision<br>× Mainline        | 0.83<br>(0.59)          |                   | -0.69*<br>(0.41)                       |                   | -0.27<br>(0.47)                       |                   |
| Mission<br>× Hocking report |                         | 0.75*<br>(0.44)   |                                        | -0.56**<br>(0.26) |                                       | -0.56*<br>(0.29)  |
| Observations                | 29,380                  | 29,380            | 29,380                                 | 29,380            | 29,380                                | 29,380            |
| R-squared                   | 0.09                    | 0.09              | 0.07                                   | 0.07              | 0.06                                  | 0.06              |
| Control group mean          | 0.65                    | 0.65              | 0.65                                   | 0.65              | 0.65                                  | 0.65              |
| DV:(dummy×100)              | Smithsonian roster      |                   | Office of Strategic Services           |                   | WWII Army Enlistment                  |                   |
|                             | (7)                     | (8)               | (9)                                    | (10)              | (11)                                  | (12)              |
| Mission                     | 0.57**<br>(0.27)        | 0.75***<br>(0.19) | 0.01<br>(0.05)                         | 0.02<br>(0.06)    | -0.14<br>(0.09)                       | 0.07<br>(0.15)    |
| Mission<br>× Mainline       | 0.72**<br>(0.28)        |                   | 0.15**<br>(0.06)                       |                   | 0.18<br>(0.12)                        |                   |
| Mission<br>× Hocking report |                         | 0.63***<br>(0.23) |                                        | 0.17**<br>(0.08)  |                                       | -0.06<br>(0.16)   |
| Observations                | 29,380                  | 29,380            | 29,380                                 | 29,380            | 29,380                                | 29,380            |
| R-squared                   | 0.07                    | 0.07              | 0.08                                   | 0.08              | 0.07                                  | 0.07              |
| Control group mean          | 0.07                    | 0.07              | 0.05                                   | 0.05              | 0.24                                  | 0.24              |

Note: Unit of analysis is an individual. Regressions of an indicator for serving as an area expert during WWII on an indicator for being missionaries. All outcome variables are indicator variables. Columns (1) to (4) represent whether individual names can be matched to the Smithsonian Institution roster of world specialists, while Columns (5) to (8) indicate whether individual names can be matched to the Office of Strategic Services and the publication output about foreign countries in OpenAlex. Mainline is an indicator variable for the mainline protestant denominations (Steensland *et al.*, 2000) and Hocking report is an indicator variable for denominations that were involved in Hocking (1932)'s report. Robust standard errors clustered by county in parentheses. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1



### A.15 Leveraging variation in sailing years

Figure A.13: Effect on publication about foreign subjects



Note: Differences-in-difference estimator where the outcome is the first publication about foreign countries of an individual relative to the years in which they were sent as missionaries. Control groups include volunteers who did not become missionaries and not-yet-treated missionaries. Estimators based on local projections approach (Busch & Girardi, 2023; Dube *et al.*, 2023). Error bars indicate 95 percent confidence interval where standard errors are clustered by individuals.

### A.16 Estimating the effect of exposure to foreign countries: Serving in Turkey

To further address concerns that the differences between missionaries and volunteers who did not eventually sail may not be as good as random, I compare missionaries in Turkey with other missionaries before and after 1915. Turkey was undergoing significant political instability during this period, including the Armenian Genocide and the Turkish War of Independence. These political changes were exogenous to the missionaries' choices, yet they disrupted their opportunities to engage with and learn about foreign cultures.

For example, missionary Henry H. White initially went to Turkey in 1915 but had to return because his school was closed during the Armenian Genocide.<sup>6</sup> A 1921 local newspaper article also reports that Mustafa Kemal opposed many foreign influences and another missionary in my sample, Theodore D. Riggs, was expelled from Turkey (Morning Union, 1921). In the case of Henry H. White, he later served as a missionary in China. However, many other missionaries who went to Turkey after 1915 were likely less exposed to foreign countries.

If missionaries were purely attracted to their mission out of curiosity about foreign cultures, those whose work in Turkey was disrupted would have likely pursued other occupations and continued to write about foreign countries. To test whether the reduced exposure to foreign culture in Turkey due to political instability affected publication output, I regress three different types of publication output on the interaction between serving in Turkey and sailing after 1915.

The sample is restricted to missionaries who sailed after 1900 (the first year of missionary sailing to Turkey in my data), whose sailing year is available, and whose destination country is recorded. The control group consists of missionaries who went to other countries, not volunteers who did not sail. Out of 6,263 such missionaries, 138 missionaries served in Turkey between 1901-1923. Using these missionaries, I estimate the following equation

$$pub_{idt} = \beta_t(Turkey_d \times After1915_t) + \delta_d + \delta_{c(d),t} + \varepsilon_{idt} \quad (10)$$

where  $pub_{idt}$  represents the publication output of a missionary  $i$  who was sent to destination  $d$  in year  $t$ . The estimated coefficient  $\beta_t$  captures the effect of being assigned to Turkey after 1915 on publication output.  $\delta_d$  controls for destination fixed effects, while  $\delta_{c(d),t}$  accounts for continent-by-sailing-year fixed effects. Error terms are clustered at the destination region level.

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<sup>6</sup>I learned this from the personal memoir I received from his grandson and an economist Lawrence H. White.

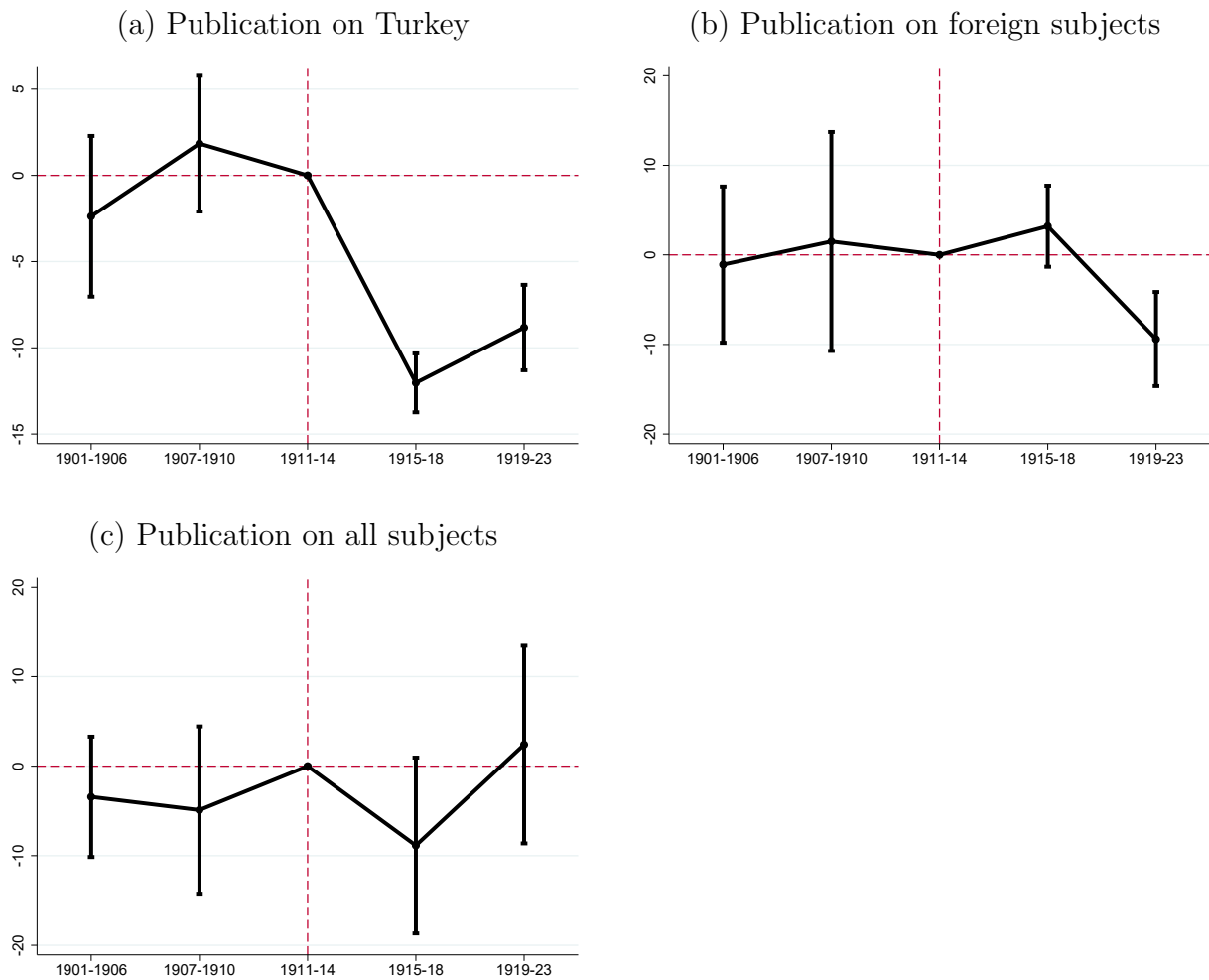
In Table A.13 and Figure A.14, I find that the instability in Turkey reduced publication output not only about Turkey but also about foreign subjects. If the selection into missionary work, compared to the runner-up volunteers, was purely driven by interest in foreign countries rather than religiosity, we would expect the coefficients in columns (3) and (4) to be zero, as these individuals would have eventually learned about foreign cultures through other channels. However, that was not the case. Lastly, I checked for authorship on non-foreign subjects and found no significant differences. Out of 138 missionaries in Turkey, only one person's name could be matched to graduate dissertation data on foreign subjects, so I do not include this outcome in my analysis.

Table A.13: Missionaries in Turkey after 1915

|                                 | Pub. Turkey         |                    | Pub. foreign subjects |                    | Pub. all subjects |                |
|---------------------------------|---------------------|--------------------|-----------------------|--------------------|-------------------|----------------|
| DV: (dummy $\times$ 100)        | (1)                 | (2)                | (3)                   | (4)                | (5)               | (6)            |
| Turkey $\times$ Sail after 1915 | -10.09***<br>(0.04) | -9.58***<br>(0.97) | -10.11***<br>(0.46)   | -7.08***<br>(2.40) | 3.63***<br>(0.67) | 1.78<br>(2.93) |
| Observations                    | 6,263               | 5,459              | 6,263                 | 5,459              | 6,263             | 5,459          |
| R-squared                       | 0.10                | 0.11               | 0.02                  | 0.05               | 0.02              | 0.05           |
| Control group mean              | 0.05                | 0.05               | 4.00                  | 4.00               | 6.62              | 6.62           |
| Destination FE                  | Y                   | Y                  | Y                     | Y                  | Y                 | Y              |
| After 1915 FE                   | Y                   | Y                  | Y                     | Y                  | Y                 | Y              |
| Individual controls             |                     | Y                  |                       | Y                  |                   | Y              |
| Continent-by-After 1915 FE      |                     | Y                  |                       | Y                  |                   | Y              |

Note: The unit of analysis is a missionary, for whom the sail year and destination country are available. Regressions are performed with an indicator of publication output as the dependent variable, focusing on the interaction between serving in Turkey and a dummy variable equal to one for missionaries who sailed in 1915 or later. The outcome variables in Columns (1) and (2) indicate whether individual names can be matched to publications with titles that include the words "Turkey," "Turkish," or "Ottoman," using data from OpenAlex. The outcome variables in Columns (3) and (4) capture whether individuals can be matched to publications about foreign subjects, also using OpenAlex. In contrast, the outcome variables in Columns (5) and (6) reflect whether individuals can be matched to authors who did not write about foreign subjects. Odd-numbered columns show raw differences, controlling for destination fixed effects and a time dummy set to one if the sailing year is 1915 or later. Even-numbered columns add further controls for birth year fixed effects, gender fixed effects, and a destination continent-by-time dummy for the year 1915 or later. Robust standard errors, clustered by 49 destination regions, are reported in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Figure A.14: Missionaries in Turkey after 1915



Note: The panel names represent the outcome variables. The estimated coefficients, along with 95 percent confidence intervals clustered by destination region, are displayed. The coefficients reflect the interaction between serving as a missionary in Turkey and the time indicator for the sailing year. Controls include destination, birth year, gender, denomination fixed effects and continent by sailing year bin fixed effects.

### A.17 Differential effects of missionary experience

Table A.14: Differential effects of missinoary experience: Evidence from Wikipedia

|                                                  | (1)      | (2)      |
|--------------------------------------------------|----------|----------|
| DV: Intelligence or language experts during WWII |          |          |
| missionaries                                     | 2.97**   |          |
|                                                  | (1.36)   |          |
| missionary children                              | 13.90*** |          |
|                                                  | (2.96)   |          |
| Japan                                            |          | 21.58*** |
|                                                  |          | (6.92)   |
| China                                            |          | 9.64***  |
|                                                  |          | (3.55)   |
| US or UK                                         |          | -0.69    |
|                                                  |          | (1.46)   |
| Philippines                                      |          | -9.84*** |
|                                                  |          | (3.73)   |
| Observations                                     | 74,966   | 74,966   |
| R-squared                                        | 0.02     | 0.02     |
| Control outcome mean                             | 3.00     | 3.00     |
| birth cohort FE                                  | Yes      | Yes      |
| female FE                                        | Yes      | Yes      |
| occupation FE                                    | Yes      | Yes      |
| Equal coef [p-value]                             | 0.00     | 0.00     |

Note: Unit of analysis is an individual. Regressions of an indicator of whether an individual served as an intelligence or language expert during WWII on an indicator of whether an individual has a missionary background. The variable “missionaries” is equal to one if an individual was a missionary. The variable “missionary children” is equal to one if an individual was the child of a missionary. Country names in the independent variables represent the missionary experience in the respective countries. The sample is restricted to individuals with a Wikipedia biography who were active in the United States, born after 1880, before 1925, and who died after 1945. Robust standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

## A.18 Selection into Wikipedia

This section examines the possibility of differential selection. One potential concern with the results in Table 7 is that missionaries might be more likely to appear in Wikipedia biographies. Consider two experts who worked for the government during WWII: one a missionary, the other not. It is possible that descendants of missionaries are more likely to add their ancestors to Wikipedia than those of non-missionaries. Even if there's no difference in the probability of working for the government, the regression results from Wikipedia may yield a significant effect due to this selection bias.

To address this concern, I replace the outcome variable with the number of views each biography received from 2015 to 2018, using data from Laouenan *et al.* (2022). This outcome variable serves as a proxy for the individuals' general fame or ability, which is arguably outside the control of those who initially wrote the Wikipedia biographies. If the results were driven by differential selection into Wikipedia biographies, we would expect to find a negative association between being a missionary and the number of views. However, the results in Table A.15 show that this is not the case.

Table A.15: Testing for differential selection

|                          | (1)                          | (2)               | (3)              | (4)             | (5)               | (6)               |
|--------------------------|------------------------------|-------------------|------------------|-----------------|-------------------|-------------------|
| DV: (dummy×100)          | ln(Wikipedia views, 2015-18) |                   |                  |                 |                   |                   |
| missionaries or children | 0.28***<br>(0.08)            | 0.39***<br>(0.08) | 0.67**<br>(0.27) | 2.09*<br>(1.07) |                   |                   |
| missionaries             |                              |                   |                  |                 | 0.22**<br>(0.09)  |                   |
| missionary children      |                              |                   |                  |                 | 0.64***<br>(0.13) |                   |
| missionary background    |                              |                   |                  |                 |                   |                   |
| in Japan                 |                              |                   |                  |                 |                   | 0.44<br>(0.30)    |
| in China                 |                              |                   |                  |                 |                   | 0.70***<br>(0.17) |
| in India                 |                              |                   |                  |                 |                   | 0.26<br>(0.26)    |
| in Burma                 |                              |                   |                  |                 |                   | -0.37<br>(0.59)   |
| in Korea                 |                              |                   |                  |                 |                   | 0.02<br>(0.27)    |
| in US or UK              |                              |                   |                  |                 |                   | -0.15<br>(0.14)   |
| in the Philippines       |                              |                   |                  |                 |                   | 0.19<br>(0.46)    |
| Observations             | 74,951                       | 74,951            | 1,134            | 88              | 74,951            | 74,951            |
| R-squared                | 0.02                         | 0.10              | 0.01             | 0.11            | 0.10              | 0.10              |
| Control outcome mean     | 8.41                         | 8.41              | 8.23             | 8.28            | 8.41              | 8.41              |
| birth cohort FE          |                              | Yes               | Yes              | Yes             | Yes               | Yes               |
| female FE                |                              | Yes               | Yes              | Yes             | Yes               | Yes               |
| occupation FE            |                              | Yes               |                  |                 | Yes               | Yes               |
| Equal coef [p-value]     |                              |                   |                  |                 | 0.01              | 0.02              |
| sample restriction       |                              |                   | academics        | academics       |                   |                   |
| Smithsonian roster       |                              |                   |                  | Yes             |                   |                   |
| pub. foreign             |                              |                   | Yes              |                 |                   |                   |

Note: Unit of analysis is an individual. Robust standard errors in parentheses.

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1



### A.19 Preference for internationalist foreign policy

To understand the foreign policy preferences of missionaries or their children, this section examines the association between missionary experience and the propensity to express political opinions to understand missionaries' attitudes towards foreign individuals. China is a particularly interesting case for this investigation. It was one of the most common destination countries for missionaries and was undergoing significant transformations during this period. These changes included the rise of the Chinese Communist Party, the conflict between the Nationalist and Communist parties, and invasions by various foreign powers. This geopolitical instability created circumstances in which foreign individuals were compelled to take a stance on many issues. The limited information about China due to the communist takeover also highlights the importance of missionaries who were present before these events.

To explore this idea, I focus on individuals with a missionary background and their political expressions related to China. I use Wikipedia biographies, restricting my sample to individuals whose names can be linked to publications about developing countries and whose biographies mention China. I manually read through the biographies and exclude any individuals who were not physically present in China before World War II.

I construct outcome variables to determine whether an individual showed political expressions or actions in favor of China or the Chinese. Examples include "steering American foreign policy and popular sentiment in favor of the Kuomintang leader," "overtly pro-Chinese Communist sympathies," "backed admitting China to the United Nations," "outrage at Japanese atrocities in China," and "enduring concern was to restore contact between China and the United States." Mere charitable activities, such as running a hospital or teaching in China, do not count.

The analysis involves regressing these measures of political actions on whether an individual is a missionary or their child, along with other controls. I compare missionaries with foreign-related publications to non-missionaries with similar publications to see if their political actions, as evidenced in Wikipedia, differ. The results are presented in Table A.16. The results indicate that missionaries were about 20-30 percentage points more likely to publicly express opinions in favor of China. In column (3), I exclude individuals who visited China as soldiers during the war, and the results remain robust. In columns (4) and (5), I split the treatment indicators between individuals who were missionaries themselves and those who were children of missionaries. In contrast to the previous results on World War II intelligence, I found that the propensity to express political opinions in favor of China

does not differ between missionaries and their children.

Table A.16: Political actions or expressions in favor of China

| Political expressions<br>in favor of China ( $\times 100$ ) | (1)                | (2)                | (3)                | (4)                 | (5)               |
|-------------------------------------------------------------|--------------------|--------------------|--------------------|---------------------|-------------------|
| missionaries or children                                    | 31.77***<br>(9.57) | 27.14***<br>(9.73) | 24.64**<br>(10.27) |                     |                   |
| missionaries                                                |                    |                    |                    | 23.25*<br>(13.13)   | 25.08*<br>(14.01) |
| missionary children                                         |                    |                    |                    | 39.15***<br>(13.21) | 24.28*<br>(14.45) |
| Observations                                                | 161                | 159                | 118                | 161                 | 118               |
| R-squared                                                   | 0.13               | 0.26               | 0.32               | 0.14                | 0.32              |
| Control outcome mean                                        | 7.52               | 7.52               | 10.75              | 7.52                | 10.75             |
| birth cohort, female, occupation FE                         |                    | Yes                | Yes                |                     | Yes               |
| excluding military contact                                  |                    |                    | Yes                |                     | Yes               |
| Equal coef [p-value]                                        |                    |                    |                    | 0.39                | 0.97              |
| Note: *** p<0.01, ** p<0.05, * p<0.1                        |                    |                    |                    |                     |                   |

## B. Data appendix

### B.1 Student Volunteer Movement records

The Student Volunteer Movement (SVM) records are available at the Yale Divinity School Library in both table and individual application formats.<sup>7</sup> I photographed the following sources:

#### Series I: Volunteer and Inquirer Forms and Statistics—Statistics and Records

- Box 126: Two volumes recording sailed volunteers alphabetically by name (1886-1926)
- Box 131: Two volumes (1886-1905)
- Oversized boxes 23-25: Statistics by denomination (1905-1922)

These records are available in table formats (See Figure B.15). Box 126 contains a list of missionaries from 1886 to 1926, including individual names, denominations, destination countries, sailing years, and serial numbers. This source comprises 734 images. It does not include information about the universities the individuals attended or their addresses.

Box 131 includes a list of missionaries and volunteers from 1886 to 1905. These records contain names, denominations, addresses, school names, and serial numbers. For missionaries, the address column lists destination country names instead of addresses. I supplemented this information with data from application blanks (which will be explained below). This source consists of 1,458 images.

Oversized boxes 23 to 25 contain lists of missionaries and volunteers from 1905 to 1922. These records include names, denominations, addresses, school names, and serial numbers. This source contains 1,765 images. All records are handwritten. I digitized all available data into table formats.

#### Series I: Volunteer and Inquirer forms and statistics—Application Blanks

The application blanks are available in individual application format (See Figure B.16). I photographed all available application blanks, totaling 62,693 images. This data includes papers from individuals who filled out the application form multiple times. Application blanks began to appear after 1890, so the earliest missionaries and volunteers are missing from this record. The names of the missionaries who sailed before 1890 are all provided in the table format data described above.

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<sup>7</sup><https://archives.yale.edu/repositories/4/resources/250>

The application blanks cover all available individuals after 1890 and helps identify all missionaries who initially appeared in the table data as volunteers but eventually became missionaries.

Due to the historical nature of these handwritten documents, they had to be photographed rather than scanned. Consequently, OCR software such as Transkribus, Amazon Textract, or Microsoft Azure did not perform well. As a result, human research assistants were hired to manually transcribe the data.

Only application forms stamped with ‘SAILED’ were digitized, indicating that the individual eventually became a missionary. Due to budget constraints, not all application forms were digitized. The drawback of not digitizing applications without the ‘SAILED’ stamp is the potential loss of some volunteers in the control group, especially those who sailed after 1922, when participation in the movement began to decline.

### **Additional sources on the list of missionaries**

The list of missionaries who sailed between 1901 and 1923 is available in certain issues of *Intercollegian* magazine and several Student Volunteer Convention reports.

- 1901-1905: *Intercollegian* magazine<sup>8</sup>
- 1906-1909: Turner (1910)<sup>9</sup>
- 1910-1913: Turner (1914)<sup>10</sup>
- 1914-1923: Stauffer (1924)<sup>11</sup>

**Linking process** After photographing and digitizing all archival records, I hired two independent groups of assistants: one group transcribed the data, and the other reviewed it for spelling errors.

The table format data provides a comprehensive list of individuals who volunteered to be missionaries between 1886 and 1922. However, whether an individual ultimately served as a missionary remains unclear in many cases. For instance, if a volunteer filled out an application in 1920, they are recorded in the table format data; however, they may have become a missionary after the table data was compiled, such as in 1927. To accurately identify those who ultimately served as missionaries, I rely on individual application data,

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<sup>8</sup><https://www.nypl.org/research/research-catalog/bib/cb6079643>

<sup>9</sup>[https://babel.hathitrust.org/cgi/pt?id=uc1.\\$b53209&seq=537](https://babel.hathitrust.org/cgi/pt?id=uc1.$b53209&seq=537)

<sup>10</sup><https://www.nypl.org/research/research-catalog/bib/b13455128>

<sup>11</sup><https://babel.hathitrust.org/cgi/pt?id=wu.89065733792&seq=489>

which is available until 1938. This ensures that I do not mistakenly include individuals who volunteered in 1922 but became missionaries later in life.

After digitizing all available records, I combine and harmonize the data using Stata's `reclink2` function, supplemented by manual review. To link records across different archives, I use individual names, serial numbers, birth years, and addresses.

Figure B.15: Sample pages from the Student Volunteer Movement records

## (a) Records on sailed volunteers, 1886-1926

|         |    |                                        |          |    |   |
|---------|----|----------------------------------------|----------|----|---|
| ✓ 1021  | 28 | Borbo, Minnie M. (Mrs E. F. Bliss)     | China    | 98 | ✓ |
| ✓ 8774  | 32 | Borland, Robert                        | China    | 98 | ✓ |
| ✓ 3865  | 47 | Botner Ragnhild                        | China    | 98 | ✓ |
| ✓ 2443  | 1  | Bowes, John Wm                         | Kenya    | 98 | ✓ |
| ✓ 782   | 9  | Boyd, Harry W. wife (Margaret Fellner) | China    | 99 | ✓ |
| ✓ 11096 | 19 | Bowen Chas. E.                         | Malaysia | 99 | ✓ |
| ✓ 8592  | 18 | Bowen Mabel (Mrs G. G. Crozier)        | Assam    | 00 | ✓ |
|         |    | Bowles, Gilbert wife (Minnie Pickett)  | Japan    | 01 | ✓ |

## (b) Records on volunteers, 1886-1905

|     |                                 |                |             |   |
|-----|---------------------------------|----------------|-------------|---|
| 432 | Lobenstein Edwin C              | Yale Univ.     | Auburn T.S. | S |
|     | China                           | Union T.S.     |             |   |
| 368 | Long Clifford S.                | Bible Inst     |             | S |
| 236 | Luce Henry W                    | Union T.S.     |             | S |
|     | China                           | Princeton T.S. |             |   |
| 287 | Matter Althea S. (Mrs Doretine) | U of Illinois  |             | S |
|     | San Juan                        |                |             |   |

## (c) Records on volunteers, 1905-1922

|     |                     |                     |                |                    |
|-----|---------------------|---------------------|----------------|--------------------|
| 387 | Adkins Thomas H.    | Clarksville Ark     | Lincoln Coll.  | 1203.              |
| 389 | Ainslie, Charles A. | Oxnard Calif        | Pomona C.      | 1313.S.            |
| 385 | Allison, Samuel P.  | College Ave         | Woster O.      | Woster U. 1603.    |
| 387 | Bischoff, Carl H.   | S. Edwards Hall     | Princeton N.J. | Princeton U. 1403. |
| 389 | Buck, John Lossing  | La Grangeville N.Y. | Cornell U.     | 1403.              |
| 387 | Churchill, Clark R. | Bancroft Neb.       | State Farm     |                    |

Sources: Yale Divinity School Archives. Student Volunteer Movement for Foreign Missions records. (a) Box 126: Two volumes recording sailed volunteers alphabetically by name (1886-1926) (b) Box 131: Two volumes (1886-1905) (c) Oversized boxes 23-25: Statistics by denomination (1905-1922). The above records show missionaries such as Gilbert Bowles, Henry W. Luce, and John Lossing Buck.



Figure B.16: Sample pages from the Student Volunteer Movement records

## (a) Application paper

1916 DEC 28 1916  
CORRESPONDENCE NUMBER 58829  
STUDENT VOLUNTEER MOVEMENT FOR FOREIGN MISSIONS  
25 Madison Avenue, New York  
STUDENT VOLUNTEER BLANK  
Date Dec 28 1916  
SAILED  
98/27

CONFIDENTIAL

Before writing any answers please read all the questions on both sides of the blank.

1. Name in full (initials are not sufficient)? *Walter Henry Judd.*

2. (A) If a married man, the maiden name of your wife; or, if a married woman, your maiden name and your husband's name.  
*Miriam Louise Pearson # 641027*

(b) If engaged to be married, the name of person to whom engaged. *Robert Jacobi # 34031*

3. Year of birth *1898*

4. Present address *403 Y.M.C.A. Lincoln, Nebr.*

Home or permanent address (i.e., an address, if possible, which will always reach you)? *Rising City, Nebr.*

Of what branch of the Church are you a member? (As there are in the United States 17 Methodist bodies, 12 Presbyterian, 13 Baptist, etc., please give the full title.)  
*First Congregational Church, Rising City.*

7. Have any members of your family been missionaries? (give facts.)  
*No missionaries*

In what institution are you a student? (If not now a student give name of institution you last attended.)  
*University of Nebraska.*

In what year will you graduate? *Probably 1924* What degree will you receive *B.Sc. M.D.*

In what other institutions have you been a student, and courses taken  
*Rising City Public and High School*

Source: Yale Divinity School Archives. Student Volunteer Movement for Foreign Missions records. Series I: Volunteer and Inquirer forms and statistics—Application Blanks. The image above is an application submitted by Walter Judd, who served as a medical missionary and later became a Republican Congressman.

## B.2 Variable definitions and sources

### B.2.1 Individual data

- Individual-level census data (1900, 1910): Ruggles *et al.* (2010). Sample in the balance test is restricted to children in the household younger than twenty (**relate** variable in the census: Child [301], Adopted child [302], Stepchild [303], Adopted [304], Child-in-law [401], Step child-in-law [402], Grandchild [901], Step grandchild [903], Grandchild-in-law [904])
  - Parental occupation score (\$00): Occupational income score (**occscore**) of the household head (**relate**=101).
  - Parent religious occupation: Household head has one of the following two occupations: Clergymen (**occ1950**=9) or Religious workers (**occ1950**=78)
  - Foreign born: **nativity**=5
  - Parent foreign born: Father foreign, mother native (**nativity**=2), Mother foreign, father native (**nativity**=3), Both parents foreign (**nativity**=4)
  - City pop (000)s: **citypop**
  - Home ownership: Owned or being bought (**ownership**=10)
  - Two parents: Married-couple family household (**hhtype**=1)
  - White: Race: white (**racwht**=2)
  - Farm household: Farm status (**farm**=2)
  - Employed: In the labor force (**labforce**=2)
  - School attendance: In school (**school**=2)
  - Literacy: able to read and write (**lit**=4)
- Publication output
  - Book or article publication: OpenAlex
  - Graduate dissertation: Dissertations - ProQuest accessed through Proquest TDM studio<sup>12</sup>. To reduce false positives, I exclude matches where the gap between the degree year and the birth year of volunteers or missionaries is less than twenty years or more than forty years, ensuring the individual's age at degree completion is between twenty and forty.

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<sup>12</sup><https://tdmstudio.proquest.com/home>



- Wikipedia biography: Laouenan *et al.* (2022)
  - Additional details on keyword search is explained in Appendix B.4.
- Smithsonian Roster: Sourced from Smithsonian Institution Archives. Record Unit 87—Ethnogeographic Board (Washington, D. C.).<sup>13</sup> I digitize the following records:
  - Personnel lists of Africa (Installments I-IV)
  - Personnel list of Asia
  - Personnel lists of Oceania (Installments I-VI)
  - Asiatic geographers compiled by George B. Cressey
  - African list 127
  - Partial list of personnel with field experience in Burma
  - American Political Science Association—Political scientists with competence regarding foreign countries
- Office of Strategic Services: Sourced from the National Archives webpage.<sup>14</sup>

### B.2.2 Regional data

- County boundary map (1900): National Historical Geographic Information System (NHGIS)
- County & congressional district boundary adjustment: Adjusted by population weight (`m2_weight`). Ferrara *et al.* (2024)
- County population, 1890: National Historical Geographic Information System (NHGIS)—1890 Census: Population, Housing, Agriculture Manufacturing Data (1890\_cPHAM)
- County characteristics, 1900: National Historical Geographic Information System (NHGIS)—1900 Census: Population, Housing, Agriculture Manufacturing Data (1900\_cPHAM) and individual level census (IPUMS USA)
  - Urbanization rate: Population in Cities of 25,000 and Over/Total population
  - Share African Americans: (male black Americans+female black Americans)/Total population

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<sup>13</sup>[https://siarchives.si.edu/collections/siris\\_arc\\_216694](https://siarchives.si.edu/collections/siris_arc_216694)

<sup>14</sup><https://www.archives.gov/files/iwg/declassified-records/rg-226-oss/personnel-database.pdf>

- Occupational income score: Average occupational income score in Ruggles *et al.* (2010) for individuals who are older than fifteen
- Literacy rate: Share of individuals older than fifteen who can read and write (`lit=4`)
- Share of manufacturing workers: Total Number of Salaried Officials, Clerks, etc. in Manufacturing/Total population
- Northern and Western European immigrants: foreign born individuals from Australia, Austria, Belgium, Canada (English), Canada (French), Denmark, England, Finland, France, Germany, Holland, Ireland, Luxembourg, Norway, Norway and Denmark, Scotland, Sweden, Switzerland, Wales
- Eastern and Southern European immigrants: foreign born individuals from Greece, Hungary, Bohemia, Italy, Poland (not otherwise specified), Poland (Austrian), Poland (German), Poland (other), Poland (Russian), Poland (unknown), Portugal, Romania, Russia, Spain, Turkey
- Non-European immigrants: foreign born individuals from Asia (country unspecified), Atlantic Islands, Japan, China, Cuba, Central America (country unspecified), Mexico, Pacific, Islands, South America (country unspecified), West Indies, Asia (except China)
- County market access, 1900: Hornbeck & Rotemberg (2021)
- Agricultural land suitability: Ramankutty *et al.* (2002)
- Number of college students for each county in 1890: Xiong & Zhao (2023)
- Census of Religious Bodies: Association of Religion Data Archives webpage—Data Archives—Categories—US Church Membership Data—County-level data—Statistics of Churches in the United States, County File, 1890
  - Number of evangelical and mainline Protestants: Classify each denomination according to Steensland *et al.* (2000)
  - Catholics: Number of Roman Catholic Church communicants or members (`rcathmem`)
  - Orthodox Jews: Number of Jewish (Orthodox) communicants or members (`jewormem`)
  - Mormons: Number of the Church of Jesus Christ of Latter-day Saints communicants or members (`latdymem`)

- Church property and members at the denomination level: Value of church property and the number of communicants or members of each denomination is aggregated at the national level.
- Participation in Church World Service (at the denomination level): Sourced from (US Congress House Committee on Foreign Affairs, 1957, p.196)
- Local YMCA chapters: *Intercollegian* magazine. I digitized the records in the following five issues: January 1881, January 1884, November 1884, December 1889, November 1890. Records are available in microfilm at New York Public Library.<sup>15</sup>. Sample page is shown in Figure B.18.
- Itineraries of the Travel Secretaries: *Missionary Review of the World* magazine from February 1887 to June 1887.<sup>16</sup>
- Congressional voting on foreign aid bills: House of Representatives’ Congressional votes were sourced from VoteView (Poole & Rosenthal, 2000).
  - Considered to vote “Yea” if `cast_code` is 1 or 2. Considered to vote “Nay” if `cast_code` is 5 or 6. Considered absent if `cast_code` is 7 or 8.
  - Marshall Plan: 80th Congress, roll number: 109
  - Point Four Program: 81st Congress, roll number: 190
  - Mutual Security Act: 82nd Congress, roll number: 153
  - Participation in International Development Association: 86th Congress, roll number: 159
  - Foreign Assistance Act of 1961: 87th Congress, roll number: 87
- Peace Corps volunteers: Raw data on the number of volunteers by zipcode is obtained from Freedom of Information Act request. Coordinates of zipcode were sourced from US Census Bureau’s 2016 TIGER/Line® Shapefiles: ZIP Code Tabulation Areas, [simplemaps.com](https://simplemaps.com), and [unitedstateszipcodes.org](https://unitedstateszipcodes.org), and then were overlaid onto the 1900 county shape file.

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<sup>15</sup><https://www.nypl.org/research/research-catalog/bib/cb6079643>

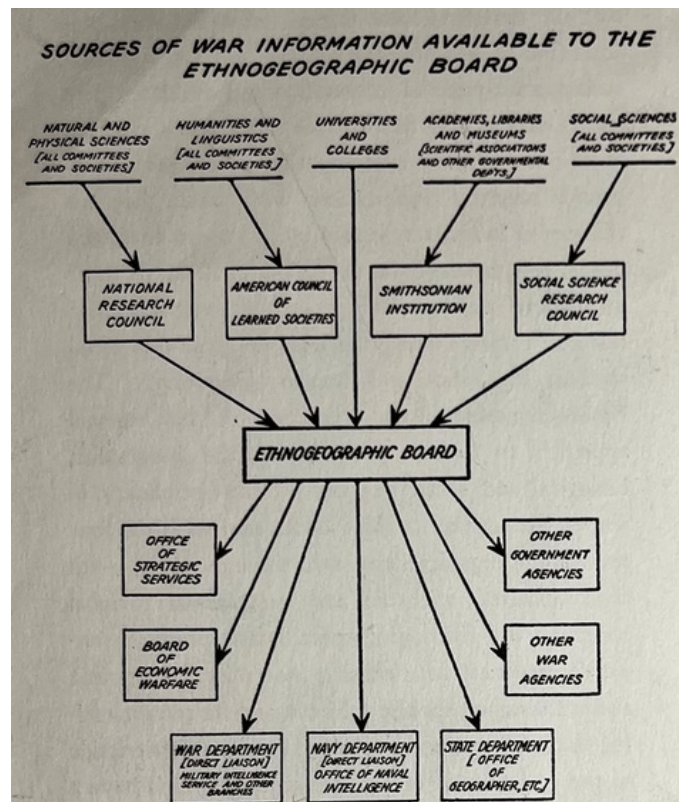
<sup>16</sup><https://onlinebooks.library.upenn.edu/webbin/serial?id=missionrevworld>

Figure B.17: Sample records from the Ethnographic Board (Smithsonian Roster)

## (a) Personnel list of Oceania—Installment IV

Boot, Dr. Earnest R. (b. 1916, Wisc.), historian. Box 824, R.F.D. No. 3, Annapolis, Md. (2474); Dept. of English and History, U.S. Naval Academy, Annapolis, Md.  
 Bowles, Dr. Gordon T. (b. 1904, Japan), anthropologist. 134 Hilldale Road, Lansdowne, Penna.; Far Eastern Division, Board of Economic Warfare, Commerce Dept. Bldg., Washington, D.C. (District 2200, ext. 2827).  
 Boyes, Major Alfred (b. 1898, England), mining engineer. P.O. Box 234, Keewauk Minn. (4422); Butler Bros., Cooley, Minn. (Nashwauk 85).  
 Bredin, Mrs. Dee van B. (b. 1900, Sumatra), writer, photographer, and radio commentator. 2 Sutton Place South, New York, N.Y. (Plaza 8-2954).  
 Brevet, Jan (b. 1888, Java), salesman. 2146 Northwest 33rd Ave., Portland, Ore. (Be. 5796); By-Products Dept., Swift and Co., North Portland, Ore. (Un. 1156).  
 Brown, Lawrence E. (b. 1901, Ind.), oil worker. 315 Minner Ave., Oildale, Calif. (2-6176).  
 Bryan, Lieut. Park A. (b. 1905, Ind.), officer in U.S. Army.

## (b) Distribution of Smithsonian roster



Sources: Smithsonian Institution Archives. Figure (a) is a sample page of the Smithsonian roster and Figure (b) is a flow chart explaining how the roster was compiled and distributed.

Figure B.18: Sample records on YMCA chapters and SVM travel secretaries

(a) YMCA chapters

| <i>Institution.</i>    | <i>Location.</i> | <i>President.</i>  | <i>Corresponding Secretary.</i> |
|------------------------|------------------|--------------------|---------------------------------|
| <b>Kansas.</b>         |                  |                    |                                 |
| Midland Col.,          | Atchison,        | Robt. G. Shaffer,  | C. F. Leisenring.               |
| Baker Univ.,           | Baldwin,         | W. H. Howell,      | P. M. Pearson.                  |
| South'n Kan. Acad.,    | Eureka,          | J. H. Wiggins,     | J. R. Wells.                    |
| Normal Col.,           | Fort Scott,      | A. L. Shively,     | Wm. Schaumberg                  |
| Central Normal Col.,   | Great Bend,      | S. E. Kirkpatrick, | E. E. Wright.                   |
| Highland Univ.,        | Highland,        | Will P. James,     | Will McIntosh.                  |
| Campbell Normal Univ., | Holton,          |                    |                                 |
| State Univ.,           | Lawrence,        | C. L. Barrett,     | E. N. Johnson.                  |
| Haskell Institute,     | Lawrence,        | E. L. Ackley,      | C. P. Chapman.                  |
| Lane Univ.,            | Lecompton,       | Kiser Williams,    | S. Perry.                       |
| State Ag'l Col.,       | Manhattan,       | I. L. Oaks,        | H. L. Chambers.                 |
| Baptist Col.,          | Ottawa,          | W. H. Sanders,     | R. W. Newman.                   |
| Kan. Wesleyan Univ.,   | Salina,          | J. F. Crawford,    | N. F. Graham.                   |
| Washburn Col.,         | Topeka,          | J. C. Short,       | R. C. Postlethwaite.            |
| Garfield Univ.,        | Wichita,         | S. W. Naylor,      | H. B. Mills.                    |
| Southwest'n Kan. Col., | Winfield,        | Ira Mason,         | Sherman Ploughe.                |
|                        |                  | F. L. Guthrie,     | W. F. Tomlinson.                |
| <b>Kentucky.</b>       |                  |                    |                                 |
| Centre Col.,           | Danville,        | H. Brown,          | H. N. Cralk.                    |
| Georgetown Col.,       | Georgetown,      | H. P. Aulick,      | W. C. Lyle.                     |
| Ag. & Mech. Col.,      | Lexington,       | E. V. Muncy,       | C. C. Winston.                  |
| Kentucky Col.,         | Lexington,       | Robert L. Cave,    | Oscar Trent.                    |
| Ky. Wesleyan Col.,     | Millersburg,     | H. C. Woodyard,    | C. A. Tague.                    |
| Central Univ.,         | Richmond,        | D. Clay Lilly,     | Rob't Caldwell.                 |

(b) Travel secretaries' itineraries

|                                         |                    |     |                                         |    |
|-----------------------------------------|--------------------|-----|-----------------------------------------|----|
| Maine, {                                | Colby, . . . . .   | 7   | Roanoke, . . . . .                      | 5  |
|                                         | Bates, . . . . .   | 22  | Emory and Henry, . . . . .              | 1  |
|                                         | Bowdoin, . . . . . | 3   | Washington and Lee, . . . . .           | 12 |
| Dartmouth, . . . . .                    |                    | 6   | Pantops, . . . . .                      | 1  |
| Univ. of Vt., . . . . .                 |                    | 4   | Randolph-Macon, . . . . .               | 6  |
| Williams, . . . . .                     |                    | 19  | Va. Military Institute, . . . . .       | 1  |
| Amherst, . . . . .                      |                    | 25  | Lynchburg, . . . . .                    | 1  |
| Oberlin, (Students of College, 73; plus |                    |     | N. Y. Medical Conference, . . . . .     | 17 |
| Students of Alliance,).                 |                    | 143 | Union. Theo. Sem., . . . . .            | 15 |
| Syracuse, . . . . .                     |                    | 12  | Philadelphia Medical College, . . . . . | 11 |
| Madison, . . . . .                      |                    | 46  | Gen. Theo. Sem., N. Y., . . . . .       | 2  |
| Hamilton, . . . . .                     |                    | 15  | Mt. Vernon, . . . . .                   | 39 |
| Lafayette, . . . . .                    |                    | 9   | Univ. of Iowa, . . . . .                | 10 |
| Rutgers, . . . . .                      |                    | 23  | Mt. Pleasant, . . . . .                 | 12 |
| Lincoln, . . . . .                      |                    | 15  | Fairfield, . . . . .                    | 30 |
| Hackettstown, . . . . .                 |                    | 11  | Iowa State Y. M. and W. Con., . . . . . | 46 |
| Alexandria, . . . . .                   |                    | 12  | Grinnell College, . . . . .             | —  |
| Univ. of Va., . . . . .                 |                    | 9   | Toledo, . . . . .                       | —  |
|                                         |                    |     | Chicago, . . . . .                      | —  |

Sources: (a) *Intercollegian*, December 1889. (b) *Missionary Review of the World*, February 1887.

### B.3 Foreign country keywords

To identify publications about foreign countries, I searched for titles that included the following keywords, published between 1880 and 1945: africa, arab, arabia, argentina, asia, assam, brazil, burma, central america, ceylon, chile, china, corea, cuba, east indies, egypt, formosa, india, iran, iraq, japan, kenya, korea, laos, lebanon, malay, malaysia, manchuria, melanesia, mesopotamia, mexico, mongol, ottoman, palestine, persia, peru, philippine, philippines, polynesia, siam, singapore, south america, syria, tibet, turkey, turkish, west indies

### B.4 Wikipedia keywords

To identify individuals who served as intelligence or language experts during WWII, I ran an automated Python script for each Wikipedia biography. An individual is classified as a regional expert during WWII if the biography mentions both WWII-related terms and intelligence-related terms.

WWII-related words are ‘second world war’, ‘world war’, ‘world war two’. Intelligence-related words are as follows: office of strategic services, cryptograph, cryptanaly, army specialized, naval intelligence, military intelligence, intelligence officer, signal intelligence, coordinator of information, office of war information, board of economic warfare, foreign service, translat\*, interpre\*.

This approach allows for a comparison between missionaries and other influential individuals. It also enables us to examine the role of missionaries’ children. While it is theoretically possible to study children’s outcomes using census data, this method presents challenges. Missionary children who grew up in foreign countries do not appear in US census records until they return as adults, making it difficult to match them in the data. The Wikipedia data provides a shortcut for measuring the impact on missionary children, circumventing these challenges. It also allows for capturing other prominent missionaries outside the Student Volunteer Movement.



## References for Appendix

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